A Representation-Complete Reduction of the Riemann Hypothesis to a Native Seam Integer

Bilateral Recognition, Spectral Blindness, and a Finite Birman–Schwinger Obstruction

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Abstract

We develop a cut-derived recognition-seam framework and carry it from its geometric and index-theoretic form to a native multiplicative completed-Weil operator. The construction retains both cut sectors, target-relevant blind directions, and their cross seam; produces compact relative packets and the nonnegative seam integer

$$\mathfrak{k}_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta);$$

and identifies the same integer through one-sided spectral flow, threshold spectral zeta, and Morse spectral zeta. A complete-prime Arb active-band certificate, a covariant Euler–virial argument, and native Hardy-strip uniqueness establish positivity of the pure Gamma–prime source at the declared fixed shift. The remaining adverse odd boundary/Feshbach source is reduced exactly to the Hermitian Birman–Schwinger matrix

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta, \qquad \operatorname{rank} V_\eta \le 5,$$

with $\mathfrak{k}_\Sigma(\eta) = N_+(B_\eta - I)$.

We prove the quantitative shift transfer

$$S_{0.002,-} \ge 0, \qquad \boxed{S_{0.01,-} \ge 0.008I}, \qquad \boxed{\|S_{0.01,-}^{-1}\| \le 125}.$$

For an approximate native source solve $\widetilde{X}$, put

$$R = V_{0.01} - S_{0.01,-}\widetilde{X}, \qquad \widetilde{B}_{0.01} = \tfrac{1}{2}\big(V_{0.01}^*\widetilde{X} + \widetilde{X}^* V_{0.01}\big).$$

Then the final outward acceptance theorem is

$$\boxed{\lambda_{\max}(\widetilde{B}_{0.01}) + 125\,\|V_{0.01}\|\,\|R\| < 1 \Longrightarrow \mathfrak{k}_\Sigma(0.01) = 0.}$$

The actual outward value of the Hermitian matrix, a controlled cofinal sequence $\eta_j \downarrow 0$, and global seam vanishing remain open. No proof of the Riemann hypothesis is claimed.

Keywords. recognition-seam calculus; bilateral observer completion; target-relevant spectral blindness; native completed-Weil geometry; Birman–Schwinger inertia; source floor; spectral flow; Mellin transform; Riemann hypothesis.

Claim status. The representation-complete framework, pure-source positivity, quantitative source floor $S_{0.01,-} \ge 0.008I$, inverse bound $\|S_{0.01,-}^{-1}\| \le 125$, rank-at-most-five Hermitian reduction, and final outward acceptance theorem are proved in their declared settings. The actual outward five-by-five value and controlled $\eta \downarrow 0$ closure remain open; RH remains `ABSTAIN`.

The cut-derived framework, exact circle index calibration, diagonal-carrier no-go theorem, native multiplicative carrier and Euler-harmonic representation, native boundary-Gamma-prime Weil pairing, bounded self-adjoint Weil representative, parity/Feshbach reductions, compact relative packets, integer seam path, threshold and Morse spectral-zeta index formulas, Euler-virial criterion, complete-prime active-band source inequalities, Hardy compatible no-blindness, pure-source positivity, quantitative shift transfer, the explicit source floor 0.008 I at eta = 0.01, the inverse norm bound 125, universal target-faithful blind-quotient completion, bilateral four-block reconstruction, the rank-at-most-five Hermitian Birman-Schwinger reduction, and the final outward acceptance theorem are proved in their declared settings. The classical zero-sum identification and Weil criterion are cited inputs. The actual outward five-by-five value, a controlled decreasing-shift sequence, global seam vanishing, and the Riemann hypothesis remain open; RH remains **ABSTAIN**.

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Part I

Cut-Derived Recognition-Seam Calculus

1 Introduction

The Riemann hypothesis asks whether every nontrivial zero of the completed zeta function lies on the critical line. This paper does not announce a proof of that statement. It solves a prior representation problem: how to construct a native completed-Weil operator framework that preserves both sides of every cut, records target-relevant directions hidden from a chosen observer and reduces the remaining sign question to one exact nonnegative integer.

The governing proof discipline is

$$\boxed{\text{source structure} \longrightarrow \text{faithful representation} \longrightarrow \text{completion} \longrightarrow \text{terminal decision.}}$$

A reflection symmetry, finite positive matrix, regularized determinant or attractive geometric dictionary does not supply any missing arrow by itself. The historical route of this programme repeatedly exposed that distinction, which is why the developmental failures of promotion are recorded explicitly in Section 2.

1.1 From cut and recognition to a native completed carrier

The primitive layer starts with a cut, an idempotent recognition map and the resulting quotient. Labelled sheets are compared by reversible transport rather than automatically glued. The seam is the equalizer of the recognized cut and the recognized identity, while failure of equality remains as a typed residue. A recognition-compatible involution and its seam projections are derived only after this transport has been declared.

The exact circle family

$$D_a = -i\frac{\mathrm{d}}{\mathrm{d}\theta} + a$$

then supplies the calibration

$$\boxed{\mathrm{sf}(D_{\delta+tq}; 0) = \mathrm{Wind}(z^q) = \mathrm{ind}(T_{\overline{z^q}}) = q.}$$

This prevents the later seam count from being called an index by analogy alone.

The first reflected diagonal prime carrier is analysed rather than merely abandoned. Positive damping gives an entire trace-class family; at zero damping the exact Schatten classification and regularized determinants show that the licensed central carrier is zero-free. The route is therefore structurally zero-blind. Regularization can repair an operator ideal but cannot create a spectral event that the carrier never represented.

The replacement carrier is constructed on the multiplicative scale line. The native inversion and Euler flow are

$$(\mathsf{J}_\times F)(r) = F(r^{-1}), \qquad (\mathsf{U}_t F)(r) = F(e^{-t}r),$$

with self-adjoint generator

$$\mathsf{E} = \overline{-ir\frac{\mathrm{d}}{\mathrm{d}r}}.$$

The generalized characters $r^{i\xi}$ diagonalize this flow, inversion exchanges ξ and $-\xi$, and the resulting even and odd combinations are the cosine and sine seam sectors. The logarithmic variable $x = \log r$ gives the faithful Mellin/Fourier computation chart only after the multiplicative carrier has been defined.

The native two-channel energy completes the carrier to $\mathscr{X}_{\times,3}$, compactly embedded into multiplicative L^2. Its boundary, Gamma and Euler-dilation sectors define the completed-Weil pairing

$$\mathfrak{W}_\times = \mathfrak{B}_\times + \mathfrak{A}_{\Gamma,\times} - \mathfrak{P}_\times,$$

with bounded self-adjoint representative $W_\times$. The classical explicit formula is consumed as the shadow identifying this pairing with completed-zeta zero data, and the classical Weil criterion supplies the terminal equivalence. Neither classical input is re-claimed as a new derivation.

1.2 The seam integer

Parity and the certified positive odd block give an exact Feshbach reduction. For every $\eta > 0$, coercive even and odd references produce compact self-adjoint relative responses whose direct sum is

$$\mathbb{K}_\eta = K_{+,\eta} \oplus K^\sharp_{3,\eta}.$$

The terminal strict threshold count is

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \mathrm{rank}\,\mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).}$$

It is finite, additive across parity sectors and equal to minus the one-sided spectral flow of $I - t\mathbb{K}_\eta$. It is also the value at zero of a threshold spectral zeta and, after positive congruence, of a Morse spectral zeta. These are exact cancellation-free presentations of one obstruction, not proofs that it vanishes.

1.3 Blindness, source positivity and finite terminal reduction

Finite observations may miss target-relevant continuum directions. The paper therefore proves reference-whitened blind-kernel Schur transfer and compatible response-tail theorems. Older grids, standing-wave packets, Krylov ladders and visible spectral tops are retained as audits or independent enclosure designs unless their visible, blind and cross ledgers are closed on the same native reconstruction.

A second route acts on the unnormalized shifted sign operator. The Euler-scale conjugate observable gives a weighted virial identity. A covariant two-sheet connection removes the derivative of the prime phase and leaves the radial modulus derivative as the true arithmetic obstruction. At $\eta = 10^{-2}$, a complete-prime Arb ledger certifies the active-band source inequalities. The native exponential spatial energy places the Mellin transform in a Hardy strip, and positive-measure boundary uniqueness proves that no nonzero compatible state is blind on the active band. Hence the pure Gamma–prime source is positive.

The remaining adverse odd boundary and Feshbach source has rank at most five. Writing

$$F^{\times}_{3,\eta} = S_{\eta,-} - V_\eta V_\eta^*,$$

define

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta, \qquad \operatorname{rank} B_\eta \le 5.$$

At the declared shift,

$$\boxed{\mathfrak{k}_\Sigma(\eta) = N_+(B_\eta - I).}$$

Thus Birman–Schwinger inertia is the preferred finite terminal decision mechanism, while direct native operator enclosure remains an independent verification route.

1.4 Bilateral recognition closure

The repeated appearance of blind complements becomes the structural conclusion of the framework. For an observer $R : \mathscr{H} \to \mathscr{Y}$ and target $\Pi : \mathscr{H} \to \mathscr{Z}$, the target-relevant blind quotient is

$$\mathfrak{B}(R;\Pi) = \frac{\ker R}{\ker R \cap \ker \Pi}.$$

The universal completed observer appends precisely this quotient and has no remaining target-relevant blind direction. On the native compatible channel range,

$$\mathcal{C}_\times^* \mathcal{C}_\times = I, \qquad \mathcal{C}_\times \mathcal{C}_\times^* = P_{\mathrm{comp}}.$$

This is the exact analysis–synthesis meaning of “observer equals observed.” It is not a spectral sign theorem.

For a represented cut J and $P_\pm = (I \pm J)/2$, every bounded self-adjoint operator satisfies

$$\boxed{A = P_+AP_+ + P_+AP_- + P_-AP_+ + P_-AP_-.}$$

The uncut object therefore contains both sectors and their cross seam. This bilateral closure solves the representation and spectral-blindness problem without choosing the terminal outcome.

Theorem 1.1 (Main representation-complete seam-reduction theorem)**.** *On the declared architecture:*

(i) the cut, Euler flow, native energy, completed-Weil pairing and bounded self-adjoint representative are constructed before their classical chart;

(ii) parity/Feshbach reduction and compact relative responses produce the finite nonnegative seam integer $\mathfrak{k}_\Sigma(\eta)$ for every $\eta > 0$;

(iii) this integer equals the one-sided spectral-flow count and the value at zero of threshold and Morse spectral zetas;

(iv) every observer has a universal minimal target-faithful blind completion, native analysis and synthesis are inverse on the compatible range, and both cut sectors with their cross seam reconstruct the uncut operator;

(v) at $\eta = 10^{-2}$, pure-source positivity reduces the remaining adverse fixed-shift obstruction to a Hermitian Birman–Schwinger matrix of rank at most five, with

$$\mathfrak{k}_\Sigma(10^{-2}) = N_+(B_{10^{-2}} - I);$$

(vi) a complete native operator enclosure below threshold gives an independent certificate of the same zero-charge event.

The framework is therefore representation complete and the terminal sign problem is reduced exactly. The theorem does not assert an actual outward Birman–Schwinger certificate or a controlled zero-charge sequence.

Proof. The clauses assemble the cut and Euler representation theorems, native energy and Weil construction, parity/Feshbach and compact-response theorems, seam-path and spectral-zeta identities, bilateral recognition closure, complete-prime/Hardy source positivity and finite-rank Birman–Schwinger inertia reduction proved in the corresponding sections. □

1.5 Main contributions and claim boundary

The paper's principal contributions are:

C1. a cut-first recognition-seam and target-faithfulness architecture;
C2. an exact circle spectral-flow/index calibration;
C3. a structural zero-blindness theorem for the diagonal paired-prime route;
C4. a native multiplicative completed-Weil carrier and bounded self-adjoint operator realization;
C5. exact parity/Feshbach, compact-response and integer seam-path reductions;
C6. threshold, Morse-zeta and one-sided spectral-flow avatars of the seam integer;
C7. native blind-kernel Schur, compatible response-tail and Hardy no-blindness theorems;
C8. complete-prime covariant active-band source positivity at the declared shift;
C9. an exact rank-at-most-five Birman–Schwinger terminal reduction;
C10. universal bilateral observer completion and four-block reconstruction of the uncut operator;
C11. an honest developmental ledger and a seven-realization hierarchy assigning each route its exact role.

The present publication claims a representation-complete solution of the framework and blindness problem, an exact reduction to the native seam integer and a finite fixed-shift residual obstruction. It does not claim the actual outward 5×5 decision, an independent complete native enclosure along one common sequence, controlled global seam vanishing or the Riemann hypothesis. Accordingly, RH remains `ABSTAIN`.

2 Developmental route and discarded promotions

The logical order of a finished theorem is rarely the historical order in which its hypotheses were discovered. In the present programme that distinction is especially important. The work began as a geometric and index-theoretic search for a seam obstruction, passed through direct determinant and finite spectral models, and only later reached the native completed-Weil carrier. Each stage closed a genuine question, but each also exposed information that its chosen observer could not see. The resulting history is retained here because the successive failures of promotion are part of the framework theorem rather than embarrassing debris to be swept beneath a particularly ornate rug.

2.1 Five developmental stages

Stage I: cut, recognition and the integer obstruction. The initial route constructed the recognition quotient, labelled sheet transport, the represented involution, seam/anti-seam projections, defect currents and the exact circle calibration

$$\mathrm{sf} = \mathrm{Wind} = \mathrm{ind}\,.$$

The later completed-Weil fusion then reduced the terminal sign problem to the nonnegative integer

$$\mathfrak{k}_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).$$

This was a successful reduction, not a proof that the integer vanishes.

Stage II: native numerical convergence. Finite grids, standing-wave reconstructions, response-generated spaces, matched two-block bounds, Krylov accelerations and directed arithmetic tails were developed to estimate

$$\sup \sigma(\mathbb{K}_\eta).$$

These computations supplied valuable source closures, visible spectral tops, negative controls and design reserves. They also exposed three recurrent promotion errors: a finite compression may be blind to its complement, a coordinate Gram need not be the native Gram, and a directional residual is not a compatible global response tail. The numerical route is therefore retained as an independent enclosure mechanism, while its older finite packets are classified as audits or calibrations unless an explicit native transfer closes the omitted sector.

Stage III: threshold and Morse spectral zetas. The same seam integer was represented as the value at zero of an entire threshold spectral zeta and, after positive congruence, as the value at zero of a Morse spectral zeta. This established cancellation-free spectral and inertia avatars of the obstruction. It did not establish that either zeta vanishes. Positive congruence preserves the index at zero, not the complete spectral-zeta function away from zero.

Stage IV: Euler–virial, covariance and source positivity. The Euler-scale generator supplied a cut-compatible positive-commutator route. Moving the operator-specific calculation to the unnormalized shifted sign operator avoided unrecorded commutators of the whitening reference. A covariant two-sheet connection removed the derivative of the prime phase, leaving the gauge-invariant radial derivative of its modulus. Complete-prime Arb enclosures then certified the active-band source inequalities, and the native exponential energy supplied a Hardy-strip uniqueness theorem excluding compatible blind states. These results prove positivity of the pure Gamma–prime source at the declared shift. They do not evaluate the remaining finite boundary/Feshbach obstruction.

Stage V: bilateral recognition closure. The repeated appearance of blind complements led to the structural endpoint of the framework. A faithful observer may not choose the visible side and discard the rest. It must retain both cut sectors, their cross seam and every blind direction relevant to the target. This yields the universal blind-quotient completion, exact native analysis–synthesis identity on the compatible range, and four-block reconstruction of the uncut operator. The framework is therefore complete as a representation of the terminal problem, without prescribing the value of its obstruction.

2.2 Four publication classes

Every result used below belongs to exactly one of the following classes.

Class	Meaning
Proved theorem	Valid on the declared native or explicitly transported carrier under the hypotheses stated in the theorem.
Conditional theorem	A proved implication whose named operator, compactness, enclosure or sequence hypotheses remain to be supplied.
Calibration or audit	A synthetic, finite, floating-point or state-locked computation testing an identity, implementation or proof design without promotion to the actual continuum target.
Actual certificate	An outward enclosure evaluated on the declared native object, with visible, blind and cross terms controlled on the same reconstruction.

The word "proved" is never transferred from one class to another merely because a numerical value is attractive. In particular,

$$\text{calibration passes} \not\Longrightarrow \text{actual native certificate passes.}$$

2.3 Historical route versus final proof architecture

The historical route explains why the present objects were introduced. The final proof architecture is theorem-first:

$$\boxed{\begin{array}{c}\text{bilateral target-faithful vessel} \longrightarrow \text{native completed-Weil sign operator} \\ \longrightarrow \text{integer seam obstruction} \longrightarrow \text{finite Birman–Schwinger reduction} \\ \text{with an independent native enclosure route.}\end{array}}$$

The threshold zeta, Morse zeta, spectral flow, positive commutator, observability and determinant winding are not seven unrelated claims of completion. They are distinct realizations, proof engines or conditional avatars of this one obstruction. Their common vessel and exact hierarchy are made explicit in Section 28.

Developmental claim boundary. This section is an honest provenance and nonpromotion audit. It does not add a terminal sign estimate. The representation-complete framework is proved; the actual controlled seam-vanishing sequence and the Riemann hypothesis remain `OPEN/ABSTAIN`.

3 Primitive cut-recognition topology

This section gives the set-theoretic root of the recognition-seam calculus. Its purpose is not to imitate differential geometry before the required structure exists. It isolates the exact data from which recognition classes, invariant topology, seams, path defects and represented involutions may later be derived.

3.1 Cut-recognition data

Definition 3.1 (Primitive cut-recognition datum)**.** A primitive cut-recognition datum is a tuple

$$\mathfrak{G} = (\mathfrak{N}, X, \kappa, \chi, \mathcal{C})$$

with the following components.

(i) $\mathfrak{N}$ is a formal pre-distinction ground.
(ii) X is the appearance space generated by the admitted outputs of cuts and recognitions.
(iii) $\kappa : X \to X$ is a distinguished first-cut operation.
(iv) $\chi : X \to X$ is an idempotent recognition operation,

$$\chi^2 = \chi.$$

(v) $\mathcal{C}$ is a family of admissible cuts containing Id_X and κ and closed under the compositions declared by the model.

No algebraic structure on $\mathfrak{N}$ is required in the present paper.

Definition 3.2 (Recognition equivalence and Eye)**.** For $x, y \in X$, define

$$x \sim_R y \quad \iff \quad \chi(x) = \chi(y).$$

Let

$$\mathsf{E} : X \longrightarrow X/R, \qquad \mathsf{E}(x) = [x]_R,$$

be the quotient map, called the *Eye projection.*

Proposition 3.3 (Recognition quotient)**.** *The relation $\sim_R$ is an equivalence relation, and*

$$\mathsf{E}(x) = \mathsf{E}(y) \quad \iff \quad \chi(x) = \chi(y).$$

Moreover, the map

$$X/R \longrightarrow \operatorname{Ran}\chi, \qquad [x]_R \longmapsto \chi(x),$$

is a canonical bijection.

Proof. Reflexivity, symmetry and transitivity follow from equality in the codomain of χ. The displayed equivalence is the definition of the quotient. Idempotence gives $\chi(\chi(x)) = \chi(x)$, so every recognition class has the representative $\chi(x)$; equality of these representatives is precisely recognition equivalence. □

3.2 Invariant recognition topology

A recognition topology should not depend on distinctions that the declared cuts immediately undo. We therefore use recognition-saturated sets invariant under every admitted cut.

Definition 3.4 (Recognition-saturated cut-invariant set)**.** A subset $U \subseteq X$ is *recognition saturated* when

$$x \in U, \quad x \sim_R y \quad \Longrightarrow \quad y \in U.$$

It is $\mathcal{C}$*-invariant* when

$$C^{-1}(U) = U \qquad (C \in \mathcal{C}).$$

Let $\tau_{\mathcal{C},R}$ be the family of all recognition-saturated, $\mathcal{C}$-invariant subsets of X.

Theorem 3.5 (Invariant recognition topology)**.** *The family $\tau_{\mathcal{C},R}$ is a topology on X. In fact, it is closed under arbitrary unions, arbitrary intersections and complements. Consequently every member is clopen and $\tau_{\mathcal{C},R}$ is a complete Boolean algebra.*

The corresponding quotient topology is

$$\tau_{\mathsf{E}} = \{V \subseteq X/R : \mathsf{E}^{-1}(V) \in \tau_{\mathcal{C},R}\}.$$

Proof. Recognition saturation is preserved by arbitrary unions, arbitrary intersections and complements. For every $C \in \mathcal{C}$, inverse image commutes with all three operations, so the identity $C^{-1}(U) = U$ is preserved. The empty set and X satisfy both conditions. Since E is surjective, the displayed family is the quotient topology. □

Remark 3.6 (Strength and scope). The equality $C^{-1}(U) = U$ is deliberately strong. It produces a partition-like, zero-dimensional invariant topology rather than a general differential topology. Later realizations may instead use continuity of the induced cut, forward invariance, local invariance or measurable invariance, but each variant must state which closure properties survive.

3.3 Seams and cut costs

Definition 3.7 (Recognition seam)**.** For $C \in \mathcal{C}$, define

$$\mathsf{S}_C = \{x \in X : \chi(Cx) = \chi(x)\}.$$

Equivalently, S_C is the equalizer of $\chi \circ C$ and χ, or the fixed locus of the cut modulo recognition.

Thus a seam is not merely a boundary drawn in a picture. It is the locus on which applying the cut does not alter the recognized state.

Definition 3.8 (Abstract cut cost)**.** A cut cost for C is a function

$$\rho_C : X \longrightarrow [0, \infty]$$

whose zero set is prescribed to be the seam:

$$\rho_C(x) = 0 \quad \Longleftrightarrow \quad x \in \mathsf{S}_C.$$

This definition records a model-specific healing cost; the equality $\mathsf{S}_C = \rho_C^{-1}(0)$ is therefore definitional rather than an additional theorem.

A non-tautological construction is available whenever recognition is represented in a metric target.

Proposition 3.9 (Canonical metric seam cost)**.** *Let (Y, d_Y) be a metric space and let $E_Y : X \to Y$ satisfy*

$$E_Y(x) = E_Y(y) \quad \Longleftrightarrow \quad x \sim_R y.$$

Define

$$\rho_C^Y(x) = d_Y\big(E_Y(Cx), E_Y(x)\big).$$

Then

$$\mathsf{S}_C = (\rho_C^Y)^{-1}(0).$$

Proof. The metric separates points, so $\rho_C^Y(x) = 0$ if and only if $E_Y(Cx) = E_Y(x)$. By the hypothesis on E_Y, this is equivalent to $\chi(Cx) = \chi(x)$. $\square$

3.4 Recognition path defects

At the primitive set level there is not yet a connection, a tangent bundle or differential curvature. The lawful object is a comparison of declared parallel paths.

Definition 3.10 (Recognition path defect)**.** Let p and q be declared parallel transformation words with induced maps $F_p, F_q : X \to X$. Define

$$\Omega_R(p, q; x) = \begin{cases} 0, & \chi(F_p x) = \chi(F_q x), \\ 1, & \chi(F_p x) \neq \chi(F_q x). \end{cases}$$

A subset $A \subseteq X$ is *recognition-flat* for the declared path system if $\Omega_R(p, q; x) = 0$ for every declared parallel pair and every $x \in A$.

A later representation may send this binary defect to a normed residue, a cocycle, a holonomy discrepancy, a connection curvature or an index. The differential-geometric word *curvature* is used only after such a representation is supplied.

3.5 The derived quotient involution

The early RH programme often began with an involution J. The cut-first formulation identifies the exact hypotheses that produce one.

Definition 3.11 (Recognition-compatible involutive cut)**.** A map $C : X \to X$ is *recognition compatible* if

$$x \sim_R y \quad \Longrightarrow \quad Cx \sim_R Cy.$$

It is *involutive modulo recognition* if

$$C^2 x \sim_R x \qquad (x \in X).$$

Theorem 3.12 (Derived quotient involution)**.** *Assume that C is recognition compatible and involutive modulo recognition. Then*

$$\overline{C} : X/R \longrightarrow X/R, \qquad \overline{C}([x]_R) = [Cx]_R,$$

is a well-defined involution.

Proof. Recognition compatibility makes the definition independent of the representative. Moreover,

$$\overline{C}^2([x]_R) = [C^2 x]_R = [x]_R$$

by involutivity modulo recognition. $\square$

Corollary 3.13 (Canonical Hilbert representation)**.** *Let*

$$\mathcal{H}_R = \ell^2(X/R)$$

with canonical basis $\{e_{[x]} : [x] \in X/R\}$*. Define*

$$J_C e_{[x]} = e_{\overline{C}([x])}.$$

Then J_C *is a unitary self-adjoint involution:*

$$J_C^2 = I, \qquad J_C^* = J_C.$$

Its spectral projections

$$P_\pm = \frac{I \pm J_C}{2}$$

are the canonical matching and anti-matching projections of the quotient involution.

Proof. The map $\overline{C}$ is an involutive permutation of the quotient classes. Hence J_C permutes an orthonormal basis, so it is unitary. Since $J_C^{-1} = J_C$, it is self-adjoint. $\square$

Remark 3.14 (No primitive dagger promotion)*.* An intertwining relation such as

$$J_C \Phi = \Phi \chi C$$

does not by itself derive J_C if the involution is simply assumed on the represented side. A representation Φ must first factor through X/R and intertwine the canonical quotient involution above. In an RH realization a further theorem is still required to identify a represented fixed locus with the critical line; installing that equality as a definition would insert the desired conclusion into the representation.

4 Sheet rotation instead of quotient gluing

The primitive quotient records recognition classes, but the operative EMK geometry keeps two local sheets distinct and compares them by reversible transport. Agreement after transport defines a seam; disagreement survives as a typed residue or memory. The distinction matters whenever the target depends on sheet origin, orientation or crossing history.

4.1 Two-sheet carrier and typed transport

Let X_+ and X_- be labelled copies of a native state space and set

$$\widetilde{X} = X_+ \sqcup X_-.$$

Let

$$\mathsf{E}_+ : X_+ \to Y, \qquad \mathsf{E}_- : X_- \to Y$$

be recognition maps into a common target.

Definition 4.1 (Sheet transport)**.** A sheet transport is a bijection

$$U : X_+ \longrightarrow X_-$$

with inverse $U^{-1} : X_- \to X_+$.

A single map $X_+ \to X_-$ is not by itself a one-parameter action, because two such maps are generally not composable. The parameter law must live on a common carrier.

Definition 4.2 (One-parameter rotation system)**.** A one-parameter rotation system is an action

$$\widetilde{U}_\alpha : \widetilde{X} \longrightarrow \widetilde{X}, \qquad \widetilde{U}_0 = I, \qquad \widetilde{U}_{\alpha+\beta} = \widetilde{U}_\alpha \widetilde{U}_\beta.$$

For a distinguished seam parameter α_Σ, assume

$$\widetilde{U}_{\alpha_\Sigma}(X_+) = X_-.$$

The associated sheet transport is

$$U = \widetilde{U}_{\alpha_\Sigma}|_{X_+} : X_+ \to X_-.$$

The action may be a continuous rotation, phase transport, helical lift or finite sheet exchange. The paper consumes only the declared action law, invertibility and recognition comparison.

4.2 Rotational seam as an equalizer

Definition 4.3 (Rotational seam)**.** For a sheet transport U, define

$$\mathsf{S}_{\mathrm{rot}}(U) = \{x_+ \in X_+ : \mathsf{E}_-(Ux_+) = \mathsf{E}_+(x_+)\}.$$

Thus $\mathsf{S}_{\mathrm{rot}}(U)$ is the equalizer of E_+ and $\mathsf{E}_- \circ U$.

When Y is metric, the numerical residue is

$$r_{\mathrm{rot}}(x_+; U) = d_Y\big(\mathsf{E}_-(Ux_+), \mathsf{E}_+(x_+)\big),$$

and its zero set is the seam. When Y is linear, one may instead use the typed difference

$$\delta_{\mathrm{rot}}(x_+; U) = \mathsf{E}_-(Ux_+) - \mathsf{E}_+(x_+).$$

Without one of these structures, an ordered pair of recognized values should not be said to “vanish.”

4.3 Why this is not gluing

For comparison, a gluing map $g : X_+ \to X_-$ produces

$$X_g = (X_+ \sqcup X_-)/\sim_g, \qquad x_+ \sim_g g(x_+).$$

The quotient replaces two labelled states by one class. Sheet rotation retains the pair (x_+, Ux_+) and the inverse transport.

Proposition 4.4 (Information retained by sheet rotation)**.** *Let $q : \widetilde{X} \to X_g$ be the gluing quotient and let $U : X_+ \to X_-$ be invertible. Then:*

(i) q is noninjective whenever a nontrivial pair is glued;
(ii) the rotated carrier retains the sheet label and U^{-1};
(iii) a target depending on sheet origin or crossing history factors through q only if it is constant on every glued fibre;
(iv) no corresponding collapse is forced on the labelled rotated carrier.

Proof. The first statement follows from $q(x_+) = q(gx_+)$. The second is part of the labelled transport data. The third is the standard fibre criterion for factorization through a quotient, and the fourth follows because the labelled carrier retains both states. □

4.4 Rotation-induced involution and exact eigenspaces

Suppose the sheets are represented by Hilbert spaces $\mathcal{H}_+$ and $\mathcal{H}_-$ and

$$U : \mathcal{H}_+ \longrightarrow \mathcal{H}_-$$

is unitary. On $\mathcal{H}_+ \oplus \mathcal{H}_-$ define

$$J_U = \begin{pmatrix} 0 & U^* \\ U & 0 \end{pmatrix}.$$

Proposition 4.5 (Graph and anti-graph decomposition). *The operator J_U is a self-adjoint involution. Its eigenspaces are*

$$\operatorname{Ran} P_+ = \left\{ \frac{1}{\sqrt{2}}(u, Uu) : u \in \mathcal{H}_+ \right\},$$

$$\operatorname{Ran} P_- = \left\{ \frac{1}{\sqrt{2}}(u, -Uu) : u \in \mathcal{H}_+ \right\}, \qquad P_\pm = \frac{I \pm J_U}{2}.$$

Proof. Unitarity gives $U^* = U^{-1}$, hence $J_U^* = J_U$ and $J_U^2 = I$. Direct substitution shows that $(u, \pm Uu)$ are the ± 1 eigenvectors, and every vector splits uniquely into these graph and anti-graph parts. □

The $+1$ graph is a represented matching sector. It becomes a *recognized seam sector* only after the additional compatibility

$$\mathsf{E}_-(Uu) = \mathsf{E}_+(u)$$

is proved on the declared domain. Operator matching and recognition agreement are related but distinct statements.

4.5 Cayley commutator residue

For a bounded operator D and sufficiently small real h, define the Cayley step

$$C_h(D) = \left(I - \frac{h}{2}D\right)^{-1} \left(I + \frac{h}{2}D\right).$$

For bounded D_1, D_2, set

$$\mathcal{L}_h(D_1, D_2) = C_h(D_1)C_h(D_2)C_h(D_1)^{-1}C_h(D_2)^{-1}.$$

Theorem 4.6 (Cayley commutator expansion). *For bounded D_1, D_2,*

$$C_h(D) = I + hD + \frac{h^2}{2}D^2 + O(h^3)$$

in operator norm, and

$$\mathcal{L}_h(D_1, D_2) = I + h^2[D_1, D_2] + O(h^3).$$

Consequently,

$$\lim_{h \to 0} \frac{\mathcal{L}_h(D_1, D_2) - I}{h^2} = [D_1, D_2]$$

in operator norm.

Proof. Expand $(I - hD/2)^{-1}$ by the norm-convergent Neumann series and multiply to second order. Substitution into the four-factor commutator loop cancels the linear terms; the second-order residue is $D_1 D_2 - D_2 D_1$. □

Proposition 4.7 (Unitary Cayley steps)**.** *If $D^* = -D$, then $C_h(D)$ is unitary whenever it is defined.*

Proof. The factors $I - hD/2$ and $I + hD/2$ are adjoints of one another and commute because both are polynomials in D. Hence $C_h(D)^* C_h(D) = I$. □

Unbounded generators require a separate domain and resolvent theorem; they are not covered silently by this bounded expansion.

4.6 Operator-valued seam memory and UGD readout

The UGD coordinate presentation uses radial scale σ, recognition phase ϕ and seam memory k. Let

$$\nabla_{\mathrm{UGD}} = \mathrm{d} + A_\sigma\,\mathrm{d}\sigma + A_\phi\,\mathrm{d}\phi + A_k\,\mathrm{d}k, \qquad \mathcal{F} = \mathrm{d}A + A \wedge A.$$

The radial–tangential curvature sector is

$$F_{\sigma\phi} = [\nabla_\sigma, \nabla_\phi].$$

To integrate an operator-valued curvature over a surface, the fibres must first be compared by a declared trivialization or parallel transport. Denote the transported curvature by $\widetilde{F}_{\sigma\phi}$.

Definition 4.8 (Operator seam memory)**.** For an oriented surface Σ, define

$$\Delta K_\Sigma = \int_\Sigma \widetilde{F}_{\sigma\phi}\,\mathrm{d}\sigma \wedge \mathrm{d}\phi.$$

A scalar seam-memory increment requires a declared linear readout

$$\ell : \mathrm{End}(E_p) \to \mathbb{C}, \qquad \Delta k_\Sigma = \ell(\Delta K_\Sigma).$$

In polar coordinates, the proposed area–rotation form is

$$\omega_A = \frac{1}{A_q} r\,\mathrm{d}r \wedge \mathrm{d}\theta.$$

It may be used as the scalar memory density only after a model proves

$$\widetilde{F}_{\sigma\phi}\,\mathrm{d}\sigma \wedge \mathrm{d}\phi = B\,\omega_A, \qquad \ell(B) = 1.$$

Under these hypotheses,

$$\Delta k_\Sigma = \int_\Sigma \omega_A.$$

Thus the area formula is a coordinate realization of a declared curvature sector, not an independent universal law.

4.7 Decaying off-sheet calibration

A finite EMK packet may have the schematic form

$$\mathcal{T}(\theta, \phi; A) = \delta_{\theta\phi}\mathsf{E} + \mathbf{1}_{\{\phi=\theta^{\perp}\}}\varepsilon(\theta)e^{-\lambda A}.$$

The off-sheet channel satisfies

$$\frac{\mathrm{d}}{\mathrm{d}A}\left(\varepsilon(\theta)e^{-\lambda A}\right) = -\lambda\varepsilon(\theta)e^{-\lambda A},$$

so $\mathcal{T}(\theta, \phi; A)$ approaches its Eye component as $A \to \infty$. This is a finite recognition-closure calibration; it is not a completed no-leakage theorem.

4.8 RH interpretation

For the RH realization, the two analytic sheets are compared by the anti-holomorphic reflection

$$\rho(s) = 1 - \overline{s}.$$

Its fixed locus is

$$\rho(s) = s \quad \Longleftrightarrow \quad \Re(s) = \frac{1}{2}.$$

The programme retains the reflected pair and asks whether its represented recognition residue vanishes. A completed theorem must still prove that the relevant arithmetic states, or the completed sign carrier replacing them, are faithfully represented by this seam geometry. Reflection alone does not localize the zeros of ξ.

5 Target-faithfulness controls for sheet rotation

The distinction between rotation and gluing is target relative. Quotients are lawful when the target is constant on their fibres and destructive when the target varies inside a fibre. One factorization theorem governs all of the finite controls used here.

5.1 Universal quotient-factorization criterion

Theorem 5.1 (Target factorization through a quotient)**.** *Let $q : X \to Q$ be a surjective map and let $\Pi : X \to Z$ be a target. The following are equivalent:*

(i) there exists a map $\widehat{\Pi} : Q \to Z$ such that $\Pi = \widehat{\Pi} \circ q$;

(ii) Π is constant on every fibre of q;

(iii)

$$q(x) = q(y) \quad \Longrightarrow \quad \Pi(x) = \Pi(y).$$

When it exists, $\widehat{\Pi}$ is unique.

Proof. If $\Pi = \widehat{\Pi} \circ q$, equality of quotient images forces equality of target values. Conversely, if Π is constant on every fibre, define

$$\widehat{\Pi}(q(x)) = \Pi(x).$$

The fibre condition makes this well defined, and surjectivity gives uniqueness. □

Corollary 5.2 (Exact gluing blindness)**.** *Let the native carrier contain two labelled states p_+, p_- with*

$$q(p_+) = q(p_-).$$

If

$$\Pi_{\mathrm{or}}(p_+) = 1, \qquad \Pi_{\mathrm{or}}(p_-) = -1,$$

then Π_{or} does not factor through q.

Corollary 5.3 (Target-relative lawful gluing)**.** *If a target Π_{inv} takes the same value on every glued pair, then it factors uniquely through the gluing quotient.*

Thus the paper does not reject quotient constructions. It rejects a quotient only when its blind fibres contain distinctions relevant to the declared target.

5.2 Recognition equality without state identity

Let $Up_+ = p_-$ be a sheet transport and suppose

$$\mathsf{E}_+(p_+) = \mathsf{E}_-(p_-).$$

Then the rotational residue vanishes even though the labelled native states remain distinct.

Proposition 5.4 (Seam closure does not erase history)**.** *Recognition agreement after sheet transport implies equality of recognized images, not equality of labelled native states.*

Proof. The first statement is the displayed Eye equality. The native states remain distinct because their sheet labels differ. □

This is the distinction between a healed seam and an erased transition. Recognition may close while the route that produced it remains available to a target that needs it.

5.3 Minimal repair of pairwise gluing blindness

Suppose q identifies each two-point fibre

$$\{(x, +), (g(x), -)\}.$$

Introduce the sheet label

$$\ell(x, +) = 1, \qquad \ell(g(x), -) = -1$$

and the augmented representation

$$\widetilde{q} = (q, \ell).$$

Theorem 5.5 (Minimal binary repair)**.** *On every two-element glued fibre, $\widetilde{q}$ is injective. Any supplemental representation that restores every target on that fibre must take at least two distinct values there; equivalently, it must carry at least one binary degree of freedom.*

Proof. The quotient coordinate agrees on the pair while the labels are opposite, so $\widetilde{q}$ separates the states. Conversely, a supplemental map taking one value is constant on the fibre and cannot recover the orientation target. □

The binary label is the smallest finite repair for pairwise blindness. More complicated histories may require winding, path or seam-memory data rather than a single bit.

5.4 Reflected-point control for the RH realization

For a modeled reflected pair

$$s_+ = \frac{1}{2} + \varepsilon + it, \qquad s_- = \rho(s_+) = \frac{1}{2} - \varepsilon + it,$$

define, when $\varepsilon \neq 0$,

$$\Pi_{\text{side}}(s_+) = \operatorname{sgn}(\varepsilon), \qquad \Pi_{\text{side}}(s_-) = -\operatorname{sgn}(\varepsilon).$$

Gluing s_+ and s_- destroys this side target; the labelled reflected carrier retains it. At $\varepsilon = 0$ the side target becomes target-null because both descriptions lie on the reflection-fixed locus.

This is a target-faithfulness calibration only. No assertion is made that either modeled point is a zero of ξ, and no finite side label replaces the completed Weil-sign realization developed later.

6 Curvature eigenmodes and exponential seam-memory evolution

The early rotation-metric draft linked path-order discrepancy, curvature and exponential change. Its durable content survives after two corrections: scalar coordinate derivatives are replaced by covariant derivatives on a bundle, and a scalar exponential law is asserted only on an invariant curvature eigensector.

6.1 Connection-level path discrepancy

Let $E \to M$ be a complex vector bundle over an oriented smooth surface, equipped with a connection ∇. For vector fields X, Y, the curvature is

$$F_\nabla(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}.$$

For commuting coordinate vector fields,

$$F_\nabla(X, Y) = [\nabla_X, \nabla_Y].$$

This is the lawful replacement for a formula such as $[\partial_x, \partial_y] = \kappa$: ordinary coordinate derivatives commute on smooth scalar functions, while a connection may have nonzero curvature on bundle sections.

For a small oriented surface cell Σ_A based at p, smooth parallel transport has the local expansion

$$\operatorname{Hol}(\partial\Sigma_A) = I + A\mathcal{K}_p + o(A), \qquad \mathcal{K}_p = F_\nabla(X, Y)|_p,$$

for an area-normalized oriented pair (X, Y). This is a local small-loop statement; it is not automatically the exact finite-area evolution used below.

6.2 Bounded area evolution

Theorem 6.1 (Constant bounded generator)**.** *Let $\mathcal{K}$ be a bounded operator. The initial-value problem*

$$\frac{dW}{dA} = \mathcal{K}W, \qquad W(0) = I,$$

has the unique norm-differentiable solution

$$W(A) = e^{A\mathcal{K}}.$$

Under the opposite orientation convention, the generator is $-\mathcal{K}$.

Proof. The operator exponential converges in norm and satisfies the equation and initial condition. Uniqueness follows from the standard bounded linear evolution theorem. □

Theorem 6.2 (Bounded nonautonomous transport)**.** *Let $A \mapsto \mathcal{K}(A)$ be norm continuous on $[0, A_0]$. Then*

$$\frac{dW}{dA} = \mathcal{K}(A)W, \qquad W(0) = I,$$

has a unique norm-continuous propagator given by the Dyson series, conventionally written

$$W(A) = \mathcal{P}\exp\left(\int_0^A \mathcal{K}(a)\, da\right).$$

If $[\mathcal{K}(a), \mathcal{K}(b)] = 0$ for all a, b, path ordering may be omitted.

Proof. Picard iteration gives the uniformly convergent Dyson series on the compact interval. Termwise differentiation proves the equation. Pairwise commutation reduces the iterated integrals to the ordinary exponential of the integral. □

Unbounded generators require a separate common-domain and propagator theorem; no such theorem is hidden in the bounded notation.

6.3 Curvature eigenmode law

Theorem 6.3 (Curvature eigenmode exponential law)**.** *Suppose $v \neq 0$ satisfies*

$$\mathcal{K}v = \varkappa v$$

for a constant bounded generator $\mathcal{K}$. Then

$$W(A)v = e^{\varkappa A}v.$$

If the same mode is written as

$$\frac{d\psi}{dA} + \lambda_{\mathrm{mode}}\psi = 0,$$

then, under the present orientation convention,

$$\boxed{\lambda_{\mathrm{mode}} = -\varkappa.}$$

Proof. The relation $\mathcal{K}^n v = \varkappa^n v$ gives

$$e^{A\mathcal{K}}v = \sum_{n=0}^{\infty} \frac{A^n}{n!}\mathcal{K}^n v = e^{\varkappa A}v.$$

Comparison of the two scalar equations yields the sign relation. □

This is a scoped mode law, not a universal identity relating every decay constant to every curvature quantity. Variable curvature generally mixes modes unless an invariant one-dimensional bundle or another reduction theorem has been proved.

6.4 Dagger covariance defect

Let J be a fixed self-adjoint involution. Curvature does not alter $J^2 = I$; what may fail is covariance of the transport.

Definition 6.4 (Rotation-covariance defect)**.** For constant bounded $\mathcal{K}$, define

$$\mathfrak{D}_J(A) = Je^{A\mathcal{K}}J^{-1} - e^{A\mathcal{K}}.$$

Proposition 6.5 (Exact covariance criterion)**.** *The identity $\mathfrak{D}_J(A) = 0$ for all A in a neighbourhood of zero holds if and only if*

$$J\mathcal{K}J^{-1} = \mathcal{K}.$$

Proof. If the generators agree under conjugation, functional calculus gives

$$Je^{A\mathcal{K}}J^{-1} = e^{AJ\mathcal{K}J^{-1}} = e^{A\mathcal{K}}.$$

Conversely, differentiation at $A = 0$ gives the generator identity. □

Thus the measurable failure is a covariance defect, not a broken involution.

6.5 Rotation one-form versus scalar angle

If a smooth scalar angle θ defines $\theta_i = \partial_i\theta$, then

$$\partial_i\theta_j - \partial_j\theta_i = 0$$

away from singularities. Nonzero rotational curvature therefore requires an independent Abelian connection, a non-Abelian connection, a singular or multivalued phase, or a bundle-valued connection. The manuscript uses connection notation whenever nonzero curvature is intended.

6.6 Relation to UGD area rotation

In the radial–tangential sector,

$$\mathcal{K} = F_{\sigma\phi} = [\nabla_\sigma, \nabla_\phi].$$

If an invariant seam mode B satisfies

$$F_{\sigma\phi}B = \varkappa B,$$

then normalized swept-area evolution gives

$$B(A) = e^{\varkappa A}B(0).$$

The area–rotation density

$$\omega_A = \frac{1}{A_q} r\, dr \wedge d\theta$$

enters this scalar law only after the operator-valued curvature flux and the scalar readout conditions of the preceding section have been verified in the chosen model.

6.7 Staircase negative control

The staircase approximation of a Euclidean diagonal retains horizontal and vertical total variation, effectively an L^1 length, while the limiting diagonal has Euclidean L^2 length. Uniform convergence of curves does not imply convergence of their lengths. Accordingly,

$$\text{same endpoint limit} \not\Longrightarrow \text{same target functional},$$

and the staircase example is not used as evidence for nonzero coordinate curvature.

7 Derived dagger and seam geometry

We now construct the finite algebra that first made the seam comparison operational. The model distinguishes the Hilbert adjoint from a fundamental-symmetry dagger and connects the two-class recognition quotient to the familiar swap involution.

7.1 Two-class realization of the derived cut

Assume the recognized quotient consists of two classes a, b exchanged by the derived quotient involution:

$$\overline{C}(a) = b, \qquad \overline{C}(b) = a.$$

On $\ell^2(\{a, b\}) \cong \mathbb{C}^2$, the canonical representation from the derived-involution theorem is

$$J_C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Proposition 7.1 (Two-class cut representation)**.** *The matrix above is a self-adjoint involution and is unitarily equivalent to every canonical Hilbert representation of a two-class quotient involution.*

Proof. It permutes the two orthonormal basis vectors, so $J_C^* = J_C$ and $J_C^2 = I$. Any other ordered orthonormal basis of the two quotient classes differs by a unitary change of basis. □

Thus the finite swap is not inserted as an unexplained primitive. It is the two-class realization of the recognition-compatible cut constructed in Section 2.

7.2 Rotation, fundamental symmetry and cut involution

On $\mathbb{C}^2$ define

$$R_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad K = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then

$$R_0^2 = -I, \qquad K^2 = I, \qquad R_0 K = -K R_0.$$

The mixed cut channel is

$$J_{\text{cut}} = R_0 K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which is exactly the preceding two-class representation.

Proposition 7.2 (Cut involution)**.** *The operator J_{cut} is an ordinary self-adjoint involution:*

$$J_{\mathrm{cut}}^* = J_{\mathrm{cut}}, \qquad J_{\mathrm{cut}}^2 = I.$$

Moreover,

$$R_0 J_{\mathrm{cut}} = -J_{\mathrm{cut}} R_0.$$

Proof. The assertions follow by direct multiplication, using $R_0^2 = -I$, $K^2 = I$ and $R_0 K = -K R_0$. $\square$

The eigenspaces of J_{cut} give

$$\mathbb{C}^2 = \mathcal{H}_+ \oplus \mathcal{H}_-, \qquad P_\pm = \frac{I \pm J_{\mathrm{cut}}}{2}.$$

The names seam and anti-seam are representation relative: a concrete application must prove that the represented split is induced by its primitive recognition seam.

7.3 Fundamental-symmetry dagger

The matrix K defines the indefinite Hermitian form

$$[u, v]_K = \langle Ku, v \rangle.$$

For $A \in M_2(\mathbb{C})$ define

$$A^\sharp = K A^* K.$$

Theorem 7.3 (Dagger algebra)**.** *For all $A, B \in M_2(\mathbb{C})$,*

$$(A^\sharp)^\sharp = A, \qquad (AB)^\sharp = B^\sharp A^\sharp,$$

$$(\alpha A + \beta B)^\sharp = \overline{\alpha} A^\sharp + \overline{\beta} B^\sharp.$$

Thus $\sharp$ is a conjugate-linear involutive anti-automorphism.

Proof. Since $K^* = K$ and $K^2 = I$,

$$(A^\sharp)^\sharp = K(KA^*K)^*K = KKAKK = A.$$

The product and conjugate-linearity identities follow from the corresponding properties of the Hilbert adjoint. $\square$

Proposition 7.4 (Dagger versus Hilbert adjoint)**.** *The rotation generator satisfies*

$$R_0^* = -R_0, \qquad R_0^\sharp = R_0.$$

Hence dagger self-adjointness is not ordinary Hilbert self-adjointness.

Proof. The first identity follows because R_0 is real and skew-symmetric. Anticommutation gives $KR_0K = -R_0$, and therefore

$$R_0^\sharp = KR_0^*K = -KR_0K = R_0.$$

$\square$

A completed spectral argument must state which adjoint it consumes. Krein self-adjointness does not silently supply the Hilbert self-adjointness required by the spectral theorem.

7.4 Leakage across the represented seam

Let $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ be a Hilbert space with self-adjoint involution J and projections $P_\pm = (I \pm J)/2$.

Definition 7.5 (Seam leakage)**.** For a bounded operator A, define

$$\mathcal{L}_{-+}(A) = P_- A P_+, \qquad \mathcal{L}_{+-}(A) = P_+ A P_-.$$

Proposition 7.6 (Block preservation criterion)**.** *The following are equivalent:*
(i) $[A, J] = 0$*;*
(ii) both leakage blocks vanish;
(iii) A *is block diagonal with respect to* $\mathcal{H}_+ \oplus \mathcal{H}_-$*.*

Proof. In the seam decomposition,

$$A = \begin{pmatrix} P_+ A P_+ & P_+ A P_- \\ P_- A P_+ & P_- A P_- \end{pmatrix}, \qquad J = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}.$$

A direct block calculation proves the equivalence. □

Remark 7.7 (Leakage is not yet an RH obstruction)*.* A nonzero finite leakage block proves only failure to preserve the declared finite split. To become an RH obstruction, the split must belong to a completed faithful carrier and the target sign or zero theorem must connect its anti-seam sector to the number-theoretic statement.

7.5 Simultaneous covariance

If a unitary U transports the whole comparison frame,

$$A \mapsto UAU^*, \qquad J \mapsto UJU^*, \qquad P_\pm \mapsto UP_\pm U^*,$$

then

$$\mathcal{L}_{-+}(UAU^*) = U\mathcal{L}_{-+}(A)U^*.$$

Thus leakage is covariant under simultaneous frame transport. Conjugating only A while holding the projections fixed describes a different comparison and need not preserve the leakage value.

Part II
Exact Index and Carrier Calibrations

8 Circle spectral flow, winding and Fredholm index

The finite dagger algebra becomes analytically meaningful only after a densely defined operator with a declared domain is supplied. The circle family provides an exact unbounded calibration in which topology, spectrum and Fredholm index carry the same integer.

8.1 The shifted circle operator

Let

$$\mathcal{H} = L^2(S^1), \qquad \mathrm{Dom}(D_a) = H^1(S^1), \qquad a \in \mathbb{R},$$

with periodic boundary conditions, and define

$$D_a = -i\frac{d}{d\theta} + a.$$

For the normalized Fourier basis

$$e_n(\theta) = (2\pi)^{-1/2} e^{in\theta}, \qquad n \in \mathbb{Z},$$

one has

$$D_a e_n = (n + a)e_n.$$

Theorem 8.1 (Self-adjoint circle carrier and compact resolvent)**.** *For every real a, the operator D_a is densely defined, closed and self-adjoint. Its spectrum is*

$$\mathrm{Spec}(D_a) = \{n + a : n \in \mathbb{Z}\},$$

with simple eigenvalues, and $(D_a - z)^{-1}$ is compact whenever $z \notin \mathrm{Spec}(D_a)$.

Proof. The Fourier transform sends D_a to the maximal real multiplication operator

$$(M_a c)_n = (n + a)c_n$$

on

$$\left\{ c \in \ell^2(\mathbb{Z}) : \sum_{n\in\mathbb{Z}} (1 + n^2)|c_n|^2 < \infty \right\}.$$

This proves self-adjointness and the spectrum. The resolvent is diagonal with coefficients $(n + a - z)^{-1} \to 0$, so finite-coordinate truncations converge to it in operator norm. □

8.2 Compatibility with the finite dagger

On

$$\widetilde{\mathcal{H}} = L^2(S^1) \otimes \mathbb{C}^2$$

set

$$\widetilde{D}_a = D_a \otimes I_2, \qquad \mathrm{Dom}(\widetilde{D}_a) = H^1(S^1) \otimes \mathbb{C}^2.$$

Let $\widetilde{K} = I \otimes K$ and define $A^\sharp = \widetilde{K} A^* \widetilde{K}$.

Proposition 8.2. *The doubled circle operator satisfies*

$$\widetilde{D}_a^* = \widetilde{D}_a, \qquad \widetilde{D}_a^\sharp = \widetilde{D}_a.$$

Proof. Ordinary self-adjointness follows from the tensor product with the finite identity. Since $\widetilde{D}_a$ commutes with $\widetilde{K}$, the dagger identity follows. □

8.3 Endpoint gauge relation

For $q \in \mathbb{Z}$, let $U_q = M_{z^q}$ on $L^2(S^1)$. Then U_q is unitary and

$$\boxed{U_q^* D_\delta U_q = D_{\delta+q}.}$$

Indeed, $U_q e_n = e_{n+q}$. Thus the endpoint shift and the winding loop belong to one gauge-covariant construction rather than merely producing equal integers by separate calculations.

8.4 Exact spectral flow

Fix

$$0 < \delta < 1, \qquad q \in \mathbb{Z},$$

and define

$$D_t = D_{\delta+tq}, \qquad 0 \le t \le 1.$$

The endpoints are invertible and the eigenvalue branches are

$$\lambda_n(t) = n + \delta + tq.$$

Theorem 8.3 (Integer spectral flow)**.** *With upward crossings counted positively,*

$$\mathrm{sf}(D_t; 0) = q.$$

Proof. If $q > 0$, exactly the modes $n = -q, -q+1, \ldots, -1$ cross zero, each upward. If $q = -r < 0$, exactly the modes $n = 0, 1, \ldots, r-1$ cross downward. The signed count is q in both cases. No finite truncation is used. $\square$

8.5 Winding representative

Let

$$u_q : S^1 \to U(1), \qquad u_q(z) = z^q.$$

Then

$$\mathrm{Wind}(\nu_q) = \frac{1}{2\pi i} \oint_{S^1} \nu_q^{-1}\, d\nu_q = q.$$

8.6 Toeplitz Fredholm realization

Let $H^2(S^1)$ be the Hardy space with basis $1, z, z^2, \ldots$, and let Π_H denote the Hardy projection. Define

$$C_q = \Pi_H M_{z^{-q}} \Pi_H \big|_{H^2(S^1)}.$$

The notation Π_H is kept distinct from the seam projections $P_\pm$.

Theorem 8.4 (Toeplitz index)**.** *For every $q \in \mathbb{Z}$, the operator C_q is Fredholm and*

$$\mathrm{ind}(C_q) = q.$$

Proof. For $q \ge 0$,

$$C_q z^n = \begin{cases} z^{n-q}, & n \ge q, \\ 0, & 0 \le n < q, \end{cases}$$

so $\dim \mathrm{Ker}\, C_q = q$ and the cokernel is zero. For $q = -r < 0$, C_q is multiplication by z^r; it is injective and its cokernel has basis $1, z, \ldots, z^{r-1}$, so the index is $-r = q$. $\square$

Theorem 8.5 (Unified circle calibration)**.** *For $0 < \delta < 1$ and $q \in \mathbb{Z}$,*

$$\boxed{\mathrm{sf}(D_{\delta+tq}; 0) = \mathrm{Wind}(z^q) = \mathrm{ind}(\Pi_H M_{z^{-q}} \Pi_H) = q.}$$

The endpoints also satisfy $U_q^* D_\delta U_q = D_{\delta+q}$.

Proof. The spectral-flow, winding and Toeplitz calculations above give the same integer, and the endpoint gauge identity was proved directly on the Fourier basis. □

Remark 8.6 (Relation to general index theory)*.* The general equality between spectral flow and index belongs to classical index theory; see [APS75, Phi96]. The present model is retained because every statement can be verified explicitly from Fourier and Hardy-space bases.

Remark 8.7 (Arithmetic nonpromotion)*.* Nothing in this calibration identifies the integer q with a prime, a circle crossing with a zero of ξ, or the Toeplitz index with an RH obstruction. Those identifications require separate transfer, faithfulness and completion theorems.

9 Reflected carriers, trace ideals and the damped divisor

The circle theorem supplies an exact integer, but not a carrier for the completed zeta function. We next audit a sequence of reflected carriers. The result is deliberately two-sided: the finite and damped constructions are exact, while their exact zero geometry proves that the diagonal paired-prime model cannot carry the nontrivial zero divisor of ξ.

9.1 Three analytic symmetries

We distinguish

$$\tau(s) = 1 - s, \qquad c(s) = \overline{s}, \qquad \rho(s) = 1 - \overline{s}.$$

The map τ is the holomorphic functional-equation transformation, c is conjugation, and $\rho = \tau \circ c$ is reflection across the critical line. Their fixed loci are

$$\mathrm{Fix}(\tau) = \left\{\frac{1}{2}\right\}, \qquad \mathrm{Fix}(\rho) = \left\{s : \Re(s) = \frac{1}{2}\right\}.$$

Confusing $1 - s$ with $1 - \overline{s}$ turns a line into a point, which is a rather severe typographical economy.

9.2 Finite and weighted reflection carriers

Let

$$\mathcal{H}_N = \mathcal{H}_{N,+} \oplus \mathcal{H}_{N,-}$$

be finite dimensional, with self-adjoint involution J_N acting by $\pm I$ on the two summands and projections

$$P_{N,\pm} = \frac{I \pm J_N}{2}.$$

For a bounded operator A, the directed leakage is

$$\mathcal{L}_N(A) = P_{N,-} A P_{N,+}.$$

Proposition 9.1 (Finite leakage criterion)**.** *The following are equivalent:*

(i) $[A, J_N] = 0$*;*

(ii) both off-diagonal leakage blocks vanish;
(iii) A is block diagonal in the ± 1 decomposition.
A visible seam projection $P_{N,+}v$ does not determine whether $P_{N,-}v$ vanishes.

Proof. The first three statements follow from the block form of A relative to the eigenspaces of J_N. The final statement follows from the independent decomposition $v = P_{N,+}v + P_{N,-}v$. □

For sampled reflected points

$$s_k^+ = \frac{1}{2} + \varepsilon + it_k, \qquad s_k^- = \rho(s_k^+) = \frac{1}{2} - \varepsilon + it_k,$$

the pair-swap involution has normalized eigenspaces spanned by

$$u_k = \frac{e_{k,+} + e_{k,-}}{\sqrt{2}}, \qquad v_k = \frac{e_{k,+} - e_{k,-}}{\sqrt{2}}.$$

This realizes the reflection exactly on the declared sample and nowhere else.

Let the paired sequence space carry weights $w_{k,+}, w_{k,-} > 0$.

Theorem 9.2 (Reflection-compatible completion)**.** *The pair-swap is isometric if and only if*

$$w_{k,+} = w_{k,-} \qquad (k \geq 1).$$

Under this condition it extends to a unitary self-adjoint involution and its spectral projections have norm one.

Proof. On one pair, swapping changes

$$w_{k,+}|x_+|^2 + w_{k,-}|x_-|^2$$

to

$$w_{k,+}|x_-|^2 + w_{k,-}|x_+|^2.$$

Equality for every pair of scalars is equivalent to equality of the weights. □

9.3 Complete leakage ideal ladder

On a symmetrically weighted paired carrier, let u_k, v_k be the normalized seam and anti-seam basis and define

$$A_a u_k = a_k v_k, \qquad A_a v_k = 0.$$

Then $A_a = P_- A_a P_+$ is pure seam-to-anti-seam leakage.

Theorem 9.3 (Leakage profile and Schatten ideals)**.** *For the operator A_a:*
(i) A_a is bounded if and only if $a \in \ell^\infty$, and then $\|A_a\| = \|a\|_\infty$;
(ii) A_a is compact if and only if $a_k \to 0$;
(iii) for $0 < q < \infty$, one has $A_a \in \mathcal{S}_q$ if and only if $a \in \ell^q$;
(iv) in particular, Hilbert–Schmidt leakage is equivalent to $a \in \ell^2$, and trace-class leakage to $a \in \ell^1$.

Proof. Relative to the orthonormal seam and anti-seam bases, the nonzero singular values of A_a are precisely $|a_k|$. Every assertion follows from the corresponding singular-value criterion. □

As a calibration, the diagonal profile $(k+1)^{-\beta}$ is compact exactly for $\beta > 0$, Hilbert–Schmidt exactly for $\beta > 1/2$, and trace class exactly for $\beta > 1$.

9.4 Exact and avoided transitions

An oriented scalar branch

$$d_j(\eta) = \epsilon_j(\eta - c_j), \qquad \epsilon_j \in \{-1, 1\},$$

contributes ϵ_j when it crosses zero. By contrast,

$$D_{c,\mu}(\eta) = \begin{pmatrix} \eta - c & \mu \\ \mu & -(\eta - c) \end{pmatrix}$$

has eigenvalues $\pm\sqrt{(\eta - c)^2 + \mu^2}$ and no crossing for $\mu > 0$. Index change requires an actual spectral crossing, not merely close levels.

9.5 The paired prime-sector family

Let $\mathcal{P}$ be the primes and

$$\mathcal{H}_{\mathcal{P}} = \ell^2(\mathcal{P} \times \{R, L\}).$$

Let J exchange the two sectors. For $\lambda \geq 0$ and $s \in \mathbb{C}$, define

$$K_\lambda(s)e_{p,R} = e^{-\lambda(\log p)^2} p^{-s} e_{p,R},$$

$$K_\lambda(s)e_{p,L} = e^{-\lambda(\log p)^2} p^{s-1} e_{p,L}.$$

Theorem 9.4 (Complete covariance package)**.** *Whenever the operators are bounded,*

$$JK_\lambda(s)J = K_\lambda(1 - s),$$

$$K_\lambda(s)^* = K_\lambda(\overline{s}),$$

and therefore

$$\boxed{JK_\lambda(s)^*J = K_\lambda(1 - \overline{s}).}$$

Proof. The first identity swaps the two diagonal coefficients, the second conjugates them, and the third is their composition. □

Theorem 9.5 (Entire trace-class damped family)**.** *For every $\lambda > 0$, the map*

$$s \longmapsto K_\lambda(s)$$

is entire with values in the trace class, locally uniformly in trace norm. Hence

$$D_\lambda(s) = \det(I - K_\lambda(s))$$

is entire.

Proof. On a compact set, choose M with $|\Re s| \leq M$. The trace norm is bounded by

$$\sum_{p \in \mathcal{P}} e^{-\lambda(\log p)^2} (p^M + p^{1+M}).$$

For every $A > 1$, the summand is eventually bounded by a constant multiple of p^{-A}. The prime sum therefore converges locally uniformly, including all complex derivatives, and the standard analytic Fredholm-determinant theorem applies. □

9.6 Exact damped determinant and zero divisor

Because the operator is diagonal and trace class,

$$\boxed{D_\lambda(s) = \prod_{p\in\mathcal{P}} \left(1 - e^{-\lambda(\log p)^2} p^{-s}\right)\left(1 - e^{-\lambda(\log p)^2} p^{s-1}\right).}$$

Theorem 9.6 (Damped divisor theorem)**.** *For $\lambda > 0$, the zeros of D_λ are exactly*

$$s = -\lambda \log p - \frac{2\pi i n}{\log p},$$

and

$$s = 1 + \lambda \log p + \frac{2\pi i n}{\log p}, \qquad p \in \mathcal{P}, \quad n \in \mathbb{Z}.$$

Every zero is simple. In particular,

$$\boxed{D_\lambda(s) \neq 0 \quad \textit{for} \quad 0 \leq \Re(s) \leq 1.}$$

On the critical line,

$$\boxed{D_\lambda\left(\frac{1}{2} + it\right) > 0.}$$

Proof. The first Euler factor vanishes exactly when

$$-\lambda(\log p)^2 - s\log p = 2\pi i n,$$

and the second exactly when

$$-\lambda(\log p)^2 + (s-1)\log p = 2\pi i n.$$

This gives the displayed points. Their real parts are respectively negative and greater than one. The real part determines the prime, and the imaginary part then determines n, so factors from different primes cannot create multiplicity; each scalar factor has a simple zero.

For $s = 1/2 + it$, the two factors for each prime are complex conjugates. Their product is

$$\left|1 - e^{-\lambda(\log p)^2} p^{-1/2-it}\right|^2 > 0.$$

The absolutely convergent product is therefore strictly positive. □

Corollary 9.7 (Damped symmetries)**.** *The determinant satisfies*

$$D_\lambda(1-s) = D_\lambda(s), \qquad D_\lambda(\overline{s}) = \overline{D_\lambda(s)},$$

and hence

$$D_\lambda(1-\overline{s}) = \overline{D_\lambda(s)}.$$

Proof. The first identity swaps the two Euler factors; the second follows by conjugation of the absolutely convergent product. □

9.7 Meaning of the damped result

The damping does more than improve convergence. It moves the complete determinant divisor outside the closed critical strip. Therefore the damped determinant is not an incomplete approximation whose interior zeros have merely not yet been located. Its zero set is known exactly and is incompatible with the nontrivial divisor of ξ.

For $\lambda_0 > \lambda > 0$, ordinary relative determinants remain lawful away from reference zeros:

$$\det\Big[(I - K_\lambda(s))(I - K_{\lambda_0}(s))^{-1}\Big] = \frac{D_\lambda(s)}{D_{\lambda_0}(s)}.$$

The singular endpoint $\lambda \downarrow 0$ requires a different ideal and is treated next. Neither finite products nor reflection covariance can change the exact zero-blindness proved above.

10 Undamped Schatten windows and the diagonal no-go theorem

The damped family is trace class for every positive damping parameter, but its divisor lies outside the critical strip. At the undamped endpoint the ordinary Fredholm determinant disappears, yet higher regularized determinants become available on central substrips. Their exact formula shows that regularization repairs the operator ideal while leaving the carrier completely zero-blind.

10.1 Complete undamped classification

On

$$\mathcal{H}_{\mathcal{P}} = \ell^2(\mathcal{P} \times \{R, L\}),$$

define

$$K_0(s)e_{p,R} = p^{-s}e_{p,R}, \qquad K_0(s)e_{p,L} = p^{s-1}e_{p,L}.$$

Write $\sigma = \Re(s)$.

Theorem 10.1 (Boundedness, compactness and norm)**.** *The operator $K_0(s)$ is bounded if and only if*

$$0 \le \sigma \le 1.$$

Within that closed strip,

$$\boxed{\|K_0(s)\| = \max\{2^{-\sigma}, 2^{\sigma-1}\}.}$$

It is compact if and only if

$$0 < \sigma < 1.$$

Consequently,

$$\boxed{\|K_0(s)\| < 1 \quad \textit{throughout the open critical strip.}}$$

Proof. The moduli of the diagonal entries are $p^{-\sigma}$ and $p^{\sigma-1}$. Both are bounded in p exactly when $\sigma \ge 0$ and $\sigma \le 1$, respectively. In that range both sequences are decreasing in p, so the largest entries occur at $p = 2$, giving the norm formula. A bounded diagonal operator is compact exactly when all diagonal entries tend to zero, which requires both strict inequalities. □

Corollary 10.2 (Every interior resolvent is regular)**.** *For $0 < \Re(s) < 1$, the operator $I - K_0(s)$ is boundedly invertible and*

$$(I - K_0(s))^{-1} = \sum_{n=0}^{\infty} K_0(s)^n$$

in operator norm.

Proof. The norm is strictly below one, so the Neumann series converges. $\square$

10.2 Full Schatten ladder

Theorem 10.3 (Undamped Schatten criterion)**.** *Let $0 < q < \infty$. Then*

$$\boxed{K_0(s) \in \mathcal{S}_q \quad \Longleftrightarrow \quad q\sigma > 1 \text{ and } q(1-\sigma) > 1.}$$

Equivalently,

$$K_0(s) \in \mathcal{S}_q \quad \Longleftrightarrow \quad \frac{1}{q} < \Re(s) < 1 - \frac{1}{q}.$$

In particular,

$$K_0\left(\frac{1}{2} + it\right) \in \mathcal{S}_q \quad \Longleftrightarrow \quad q > 2.$$

Thus the bounded undamped family is nowhere Hilbert–Schmidt and nowhere trace class.

Proof. The q-power sum of the singular values is

$$\sum_{p \in \mathcal{P}} \left(p^{-q\sigma} + p^{-q(1-\sigma)}\right).$$

The prime sum $\sum_p p^{-a}$ converges exactly for $a > 1$. Applying this criterion to both sectors gives the result. $\square$

The ordinary determinant $\det(I - K_0(s))$ and the Hilbert–Schmidt determinant $\det_2(I - K_0(s))$ are therefore unavailable everywhere. For each integer $m \geq 3$, however, define the central window

$$\Omega_m = \left\{ s \in \mathbb{C} : \frac{1}{m} < \Re(s) < 1 - \frac{1}{m} \right\}.$$

Then $K_0(s) \in \mathcal{S}_m$ on Ω_m.

10.3 Holomorphic regularized endpoint

Theorem 10.4 (Schatten convergence as damping is removed)**.** *Fix an integer $m \geq 3$. The map*

$$s \longmapsto K_0(s)$$

is holomorphic with values in $\mathcal{S}_m$ on Ω_m. Moreover,

$$K_\lambda(s) \longrightarrow K_0(s)$$

locally uniformly in the $\mathcal{S}_m$ norm on Ω_m as $\lambda \downarrow 0$. Consequently,

$$\det_m(I - K_\lambda(s)) \longrightarrow \det_m(I - K_0(s))$$

locally uniformly on Ω_m.

Proof. On a compact subset of Ω_m, choose $\epsilon > 0$ such that

$$m\Re(s) \geq 1 + \epsilon, \qquad m(1 - \Re(s)) \geq 1 + \epsilon.$$

The $\mathcal{S}_m$ sums for $K_0(s)$ and its complex derivatives are then dominated by convergent prime sums with exponent greater than one. For the difference, the factor

$$|1 - e^{-\lambda(\log p)^2}|^m$$

is bounded by one and converges pointwise to zero. Dominated convergence, uniformly on the compact set, gives the Schatten convergence. Continuity of the regularized determinant in the $\mathcal{S}_m$ topology gives the final assertion. $\square$

10.4 Exact regularized determinant

Let

$$P_{\mathcal{P}}(z) = \sum_{p \in \mathcal{P}} p^{-z}, \qquad \Re(z) > 1,$$

be the prime zeta function in its absolute-convergence half-plane.

Theorem 10.5 (Explicit regularized determinant)**.** *For every integer $m \geq 3$ and every $s \in \Omega_m$,*

$$\boxed{\det_m(I - K_0(s)) = \exp\left[-\sum_{n=m}^{\infty} \frac{P_{\mathcal{P}}(ns) + P_{\mathcal{P}}(n(1-s))}{n}\right].}$$

The series converges locally uniformly on Ω_m. Hence

$$\boxed{\det_m(I - K_0(s)) \neq 0 \quad (s \in \Omega_m).}$$

Proof. Throughout Ω_m, one has $\|K_0(s)\| < 1$. The standard logarithm of the mth regularized determinant is therefore

$$\log \det_m(I - K_0(s)) = -\sum_{n=m}^{\infty} \frac{\operatorname{Tr}(K_0(s)^n)}{n}.$$

The diagonal trace is

$$\operatorname{Tr}(K_0(s)^n) = P_{\mathcal{P}}(ns) + P_{\mathcal{P}}(n(1-s)).$$

For $n \geq m$, both prime-zeta arguments have real part greater than one. Compact-subset margins give local uniform convergence. Exponentials of holomorphic functions never vanish. $\square$

For $m = 3$, this gives a lawful determinant throughout

$$\frac{1}{3} < \Re(s) < \frac{2}{3},$$

which contains the entire critical line. The determinant exists there and is zero-free there. Regularization has repaired the ideal, not the missing spectral event.

10.5 Relative endpoint with a fixed damped reference

Fix $\lambda_0 > 0$ and $0 < \Re(s) < 1$. Since both $I - K_0(s)$ and $I - K_{\lambda_0}(s)$ are invertible, define

$$\mathcal{R}_{0,\lambda_0}(s) = (I - K_0(s))(I - K_{\lambda_0}(s))^{-1}.$$

Proposition 10.6 (Relative ideal classification)**.** *For $0 < q < \infty$,*

$$\mathcal{R}_{0,\lambda_0}(s) - I \in \mathcal{S}_q \quad \Longleftrightarrow \quad \frac{1}{q} < \Re(s) < 1 - \frac{1}{q}.$$

In particular, the endpoint relative perturbation is nowhere trace class and nowhere Hilbert–Schmidt in the open critical strip. Higher regularized relative determinants exist on the corresponding central windows and are zero-free.

Proof. One has

$$\mathcal{R}_{0,\lambda_0} - I = (K_{\lambda_0} - K_0)(I - K_{\lambda_0})^{-1}.$$

The inverse is bounded and diagonal. The difference multiplier contains the factor

$$e^{-\lambda_0 (\log p)^2} - 1,$$

which tends to -1. It therefore has exactly the same $\mathcal{S}_q$ membership as $K_0(s)$. The relative operator is invertible, so every licensed regularized determinant is nonzero. $\square$

10.6 One-sided Euler anchors

On $\ell^2(\mathcal{P})$ define

$$K_R(s)e_p = p^{-s}e_p, \qquad K_L(s)e_p = p^{s-1}e_p.$$

Proposition 10.7 (One-sided reciprocal Euler determinants)**.** *For* $\Re(s) > 1$,

$$\det(I - K_R(s)) = \prod_p (1 - p^{-s}) = \frac{1}{\zeta(s)}.$$

For $\Re(s) < 0$,

$$\det(I - K_L(s)) = \prod_p (1 - p^{s-1}) = \frac{1}{\zeta(1-s)}.$$

The two trace-class domains are disjoint.

Proof. The trace norms are the corresponding absolutely convergent prime sums, and the determinant products are Euler's products. □

10.7 Archimedean completion and its limitation

Define the completion multiplier

$$C_\infty(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right).$$

In the half-plane $\Re(s) > 1$,

$$\boxed{\xi(s)\det(I - K_R(s)) = C_\infty(s).}$$

Equivalently, if $D_R(s) = 1/\zeta(s)$, then

$$\frac{\xi'(s)}{\xi(s)} + \frac{D_R'(s)}{D_R(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi\left(\frac{s}{2}\right).$$

Proposition 10.8 (Completion factors do not create interior zeros)**.** *The function* C_∞ *is holomorphic and nowhere zero in the open critical strip. Therefore multiplying the zero-free regularized determinant* $\det_m(I - K_0(s))$ *by* $C_\infty(s)$ *cannot create the nontrivial zeros of* ξ *in any central window* Ω_m*.*

Proof. For $0 < \Re(s) < 1$, neither s nor $s - 1$ vanishes, the Gamma function is holomorphic and nonzero, and the power of π is nonzero. The product of two nowhere-zero holomorphic functions is nowhere zero. □

Thus the obstruction is not merely a forgotten Gamma factor. The diagonal carrier has no interior noninvertibility mechanism.

10.8 Diagonal-carrier exclusion theorem

Theorem 10.9 (Zero-blindness of the paired diagonal route)**.** *Throughout the open critical strip,*

$$I - K_0(s)$$

is invertible. Every ordinary or higher regularized determinant lawfully attached to this noninvertibility event is therefore zero-free on its domain. In particular, on a domain containing a nontrivial zero of ξ*, there is no identity*

$$\det_m(I - K_0(s)) = g(s)\xi(s)$$

with g *holomorphic and nowhere zero.*

Proof. Invertibility follows from $\|K_0(s)\| < 1$. The explicit regularized formula proves zero-freeness. If the displayed identity held and ξ vanished at s_0 while $g(s_0) \neq 0$, then the left side would vanish, contradicting the preceding theorem. □

The correct conclusion is structural replacement, not stronger estimates on the same diagonal pair. A successful completed route needs a carrier with genuine prime interaction, trace-formula coupling, completed form geometry or another interior spectral event. The native X_3 Weil carrier developed later in this paper is precisely such a change of topology and mechanism.

11 Arithmetic recognition, prime-shell transfer and finite defect ledgers

The arithmetic layer is finite and exact. A logarithmic clock supplies a Dirichlet polynomial, M"obius inversion extracts the von Mangoldt stream, radical recognition aggregates prime-support fibres, and a two-channel transfer ledger separates symmetric route error from genuinely anti-seam error. None of these finite results identifies an actual zero of ξ; their role is to establish the source and defect calculus consumed by a later completed carrier.

11.1 Finite logarithmic clock and pressure extraction

For $N \geq 1$, let

$$I_N = \{1, \ldots, N\}, \qquad \mathcal{H}_N = \ell^2(I_N),$$

with basis e_n, and define

$$H_N e_n = \log(n) e_n.$$

Then

$$Z_N(s) = \mathrm{Tr}(e^{-sH_N}) = \sum_{n=1}^{N} n^{-s}$$

is an entire Dirichlet polynomial. It is not the analytic continuation of ζ.

For arithmetic functions, let $*$ denote Dirichlet convolution and let μ be the M"obius function.

Theorem 11.1 (M"obius pressure extractor)**.** *If $\ell(n) = \log n$, then*

$$\boxed{\ell * \mu = \Lambda.}$$

Consequently the extracted stream is supported exactly on prime powers.

Proof. This is M"obius inversion applied to the classical identity

$$\log n = \sum_{d|n} \Lambda(d).$$

Thus $(\ell * \mu)(p^k) = \log p$ and the value is zero otherwise. □

11.2 Radical Eye transfer as a finite operator

Let

$$\mathcal{Q}_N = \{q \le N : q \text{ is square-free}\}.$$

For $q \in \mathcal{Q}_N$, define the radical fibre and its size by

$$F_q = \{n \le N : \operatorname{rad}(n) = q\}, \qquad m_N(q) = |F_q|.$$

Set

$$\mathcal{R}_N = \ell^2(\mathcal{Q}_N).$$

Definition 11.2 (Radical pushforward)**.** Define

$$T_N : \mathcal{H}_N \longrightarrow \mathcal{R}_N, \qquad (T_N f)(q) = \sum_{n \in F_q} f(n).$$

This is the unnormalized arithmetic Eye transfer: it aggregates fibre mass rather than preserving Hilbert norm.

Theorem 11.3 (Complete radical-transfer structure)**.** *Let M_N be the diagonal operator*

$$M_N e_q = m_N(q) e_q.$$

Then:

(i) T_N is surjective;

(ii)

$$(T_N^* g)(n) = g(\operatorname{rad}(n));$$

(iii)

$$T_N T_N^* = M_N;$$

(iv)

$$\|T_N\| = \sqrt{\max_{q \in \mathcal{Q}_N} m_N(q)};$$

(v)

$$\operatorname{Ker} T_N = \left\{ f : \sum_{n \in F_q} f(n) = 0 \textit{ for every } q \in \mathcal{Q}_N \right\}.$$

Proof. For $f \in \mathcal{H}_N$ and $g \in \mathcal{R}_N$,

$$\langle T_N f, g \rangle = \sum_q \sum_{n \in F_q} f(n) \overline{g(q)} = \sum_n f(n) \overline{g(\operatorname{rad}(n))},$$

which gives the adjoint. Hence

$$(T_N T_N^* g)(q) = m_N(q) g(q).$$

Every square-free $q \le N$ belongs to its own fibre, so M_N is invertible and T_N is onto. The norm and kernel formulas follow. □

Definition 11.4 (Normalized Eye coisometry)**.** Define

$$E_N = M_N^{-1/2} T_N.$$

Corollary 11.5 (Recognition averaging projection)**.** *One has*

$$E_N E_N^* = I_{\mathcal{R}_N}.$$

Thus E_N is a coisometry and $E_N^* E_N$ is the orthogonal projection onto functions constant on radical fibres. Explicitly,

$$(E_N^* E_N f)(n) = \frac{1}{m_N(\mathrm{rad}(n))} \sum_{\substack{r \le N \\ \mathrm{rad}(r)=\mathrm{rad}(n)}} f(r).$$

The unnormalized T_N is appropriate for arithmetic mass aggregation. The normalized E_N is the object to use for Hilbert-space assertions such as coisometry and orthogonal projection.

11.3 Neutral, seam and anti-seam radical sectors

Decompose

$$\mathcal{Q}_{N,0} = \{1\},$$
$$\mathcal{Q}_{N,+} = \{p \le N : p \text{ prime}\},$$
$$\mathcal{Q}_{N,-} = \{q \le N : q \text{ square-free and has at least two prime divisors}\}.$$

Let Q_0, Q_+, Q_- be the corresponding orthogonal coordinate projections. The neutral class $q = 1$ must be retained for a general stream, although it vanishes for both ℓ and Λ.

Definition 11.6 (Arithmetic anti-seam current)**.** Define

$$\mathcal{J}_N^{\mathrm{ar}}(f) = Q_- T_N f, \qquad \mathcal{C}_N(f) = Q_+ T_N f.$$

Theorem 11.7 (Complete zero-current criterion)**.** *For an arbitrary complex stream f,*

$$\mathcal{J}_N^{\mathrm{ar}}(f) = 0$$

if and only if

$$\sum_{\substack{n \le N \\ \mathrm{rad}(n)=q}} f(n) = 0 \qquad (q \in \mathcal{Q}_{N,-}).$$

If $f(n) \ge 0$ for every n, then zero current is equivalent to

$$f(n) = 0$$

whenever n has at least two distinct prime divisors. Thus a nonnegative zero-current stream is supported on 1 and prime powers.

Proof. The first condition is equality of each composite-radical coordinate to zero. In the nonnegative case every coordinate is a sum of nonnegative terms and vanishes exactly when every term in its fibre vanishes. □

Theorem 11.8 (Exact radical transfer of the von Mangoldt stream)**.** *For every prime $p \le N$, put*

$$\nu_p(N) = \#\{k \ge 1 : p^k \le N\} = \left\lfloor \frac{\log N}{\log p} \right\rfloor.$$

Then

$$\boxed{T_N \Lambda = \sum_{p \le N} \nu_p(N) \log(p) e_p.}$$

In particular,

$$\mathcal{J}_N^{\mathrm{ar}}(\Lambda) = 0.$$

Proof. The value of Λ at every power p^k is $\log p$, and $\operatorname{rad}(p^k) = p$. Summing over the powers below N gives the coefficient. No prime power belongs to a composite-radical fibre. □

Proposition 11.9 (The raw logarithmic stream fills every composite fibre)**.** *For every $q \in \mathcal{Q}_{N,-}$,*

$$(T_N\ell)(q) = \sum_{n\in F_q} \log n \geq \log q > 0.$$

Consequently every available composite radical class appears in $\mathcal{J}_N^{\mathrm{ar}}(\ell)$.

Proof. The point q belongs to its own radical fibre and all summands are nonnegative. □

For signed or complex streams, composite-fibre contributions may cancel. For example, when $N \geq 12$,

$$T_N(\delta_6 - \delta_{12}) = 0$$

because $\operatorname{rad}(6) = \operatorname{rad}(12) = 6$. The current therefore detects net fibre mass, not every signed contamination.

11.4 Prime-shell realization of the seam sector

Let $\mathcal{P}_N$ be the primes not exceeding N and set

$$\mathcal{H}_{\mathrm{pr}} = \ell^2(\mathcal{P}_N), \qquad \mathcal{H}_{\mathrm{top}} = \ell^2(\mathcal{P}_N) \otimes \mathbb{C}^2.$$

Define

$$u_p = \frac{e_{p,R} + e_{p,L}}{\sqrt{2}}, \qquad v_p = \frac{e_{p,R} - e_{p,L}}{\sqrt{2}}.$$

Let J exchange R and L, with projections $P_\pm = (I \pm J)/2$, and define

$$Ae_p = u_p.$$

Theorem 11.10 (Unitary seam identification)**.** *The map A is unitary from $\mathcal{H}_{\mathrm{pr}}$ onto* $\operatorname{Ran} P_+$*. Equivalently,*

$$A^*A = I, \qquad AA^* = P_+, \qquad JA = A, \qquad P_-A = 0.$$

Proof. The vectors u_p form an orthonormal basis of the $+1$ eigenspace of J. □

Let $d_p(\lambda)$ be declared arithmetic damping weights and define

$$D_\lambda e_p = d_p(\lambda) e_p,$$

$$U_\lambda e_{p,R} = d_p(\lambda) e_{p,R}, \qquad U_\lambda e_{p,L} = d_p(\lambda) e_{p,L}.$$

Theorem 11.11 (Exact symmetric intertwining)**.** *One has*

$$U_\lambda A = AD_\lambda, \qquad [U_\lambda, J] = 0, \qquad U_\lambda|_{\operatorname{Ran} P_+} = AD_\lambda A^*.$$

If the weights are real, then $JU_\lambda^* J = U_\lambda$.

This is an exact calibration by construction: A is chosen to land in the symmetric sector and U_λ uses the same weight on both reflected coordinates. It does not establish a completed arithmetic-topology representation. Also, D_λ is a damping operator, not a projection unless its eigenvalues are idempotent.

11.5 Complete seam and anti-seam transfer defect

Allow right, left and arithmetic weights $r_p, l_p, d_p \in \mathbb{C}$ and define

$$U^{R,L}e_{p,R} = r_p e_{p,R}, \qquad U^{R,L}e_{p,L} = l_p e_{p,L}, \qquad De_p = d_p e_p.$$

Let

$$\Delta = U^{R,L}A - AD.$$

Theorem 11.12 (Exact orthogonal defect decomposition)**.** *For every prime shell,*

$$\boxed{\Delta e_p = \left(\frac{r_p + l_p}{2} - d_p\right) u_p + \frac{r_p - l_p}{2} v_p.}$$

Define

$$S_{\mathrm{def}} = P_+\Delta, \qquad J_{\mathrm{def}} = P_-\Delta.$$

Then

$$S_{\mathrm{def}} e_p = \left(\frac{r_p + l_p}{2} - d_p\right) u_p,$$

$$J_{\mathrm{def}} e_p = \frac{r_p - l_p}{2} v_p.$$

Consequently,

$$J_{\mathrm{def}} = 0 \quad \Longleftrightarrow \quad r_p = l_p \text{ for every } p,$$

$$S_{\mathrm{def}} = 0 \quad \Longleftrightarrow \quad d_p = \frac{r_p + l_p}{2} \text{ for every } p,$$

and

$$\Delta = 0 \quad \Longleftrightarrow \quad r_p = l_p = d_p \text{ for every } p.$$

Proof. Use

$$e_{p,R} = \frac{u_p + v_p}{\sqrt{2}}, \qquad e_{p,L} = \frac{u_p - v_p}{\sqrt{2}}$$

and substitute directly. □

Thus $J_{\mathrm{def}} = 0$ does not imply exact transfer. A common right-left error lies entirely in the seam residual and is invisible to the anti-seam projection.

Theorem 11.13 (Abstract two-channel identity)**.** *For arbitrary finite-dimensional carriers, let*

$$\Delta = UA - AD, \qquad S_{\mathrm{def}} = P_+\Delta, \qquad J_{\mathrm{def}} = P_-\Delta.$$

Then

$$\Delta = S_{\mathrm{def}} + J_{\mathrm{def}}, \qquad S_{\mathrm{def}}^* J_{\mathrm{def}} = 0,$$

and

$$\boxed{\Delta^*\Delta = S_{\mathrm{def}}^* S_{\mathrm{def}} + J_{\mathrm{def}}^* J_{\mathrm{def}}.}$$

In particular,

$$\|\Delta\|_{\mathcal{S}_2}^2 = \|S_{\mathrm{def}}\|_{\mathcal{S}_2}^2 + \|J_{\mathrm{def}}\|_{\mathcal{S}_2}^2.$$

Exact route covariance holds if and only if both channels vanish.

Proof. The decomposition follows from $P_+ + P_- = I$. Orthogonality $P_+P_- = 0$ gives the cross-term identity, and taking the Hilbert–Schmidt trace gives the norm formula. □

11.6 Oriented witnesses and their incompleteness

Let

$$\mathfrak{D}_- = \mathcal{S}_2(\mathcal{H}_{\mathrm{pr}}, \operatorname{Ran} P_-)$$

be regarded as a real Hilbert space with pairing

$$(X, Y)_{\mathbb{R}} = \Re \operatorname{Tr}(X^* Y).$$

For a declared test $C \in \mathfrak{D}_-$, define the oriented witness

$$\mathcal{O}_C(J) = (J, C)_{\mathbb{R}}.$$

Theorem 11.14 (One scalar witness is incomplete)**.** *If $C \neq 0$, then* $\operatorname{Ker} \mathcal{O}_C$ *is a real hyperplane in $\mathfrak{D}_-$. Whenever* $\dim_{\mathbb{R}} \mathfrak{D}_- > 1$*, there exists a nonzero current J with*

$$\mathcal{O}_C(J) = 0.$$

Thus one scalar witness certifies full current vanishing only after a separate theorem confines all admissible currents to a one-dimensional nondegenerate ray.

Proof. The map $J \mapsto (J, C)_{\mathbb{R}}$ is a nonzero real linear functional, so its kernel has codimension one. □

A complete nonnegative readout is

$$Q(J) = \|J\|_{\mathcal{S}_2}^2,$$

which vanishes exactly when $J = 0$.

Proposition 11.15 (Reflected-pair cancellation)**.** *Let $c \in \mathfrak{D}_-$ be nonzero and $g(\lambda) > 0$. For*

$$J_\varepsilon(\lambda) = \varepsilon g(\lambda) c, \qquad J_{-\varepsilon}(\lambda) = -J_\varepsilon(\lambda),$$

one has

$$\mathcal{O}_c(J_\varepsilon) + \mathcal{O}_c(J_{-\varepsilon}) = 0,$$

while

$$Q(J_\varepsilon) + Q(J_{-\varepsilon}) = 2\varepsilon^2 g(\lambda)^2 \|c\|^2 > 0$$

for $\varepsilon \neq 0$.

A signed local witness records orientation but can cancel across a reflected orbit. A cancellation-free terminal theorem must consume the full current, a complete family of tests, a nonnegative norm or the one-sided threshold charge developed in the completed Weil realization.

Remark 11.16 (Not a Fredholm index)*.* The scalar witness scales continuously:

$$\mathcal{O}_C(tJ) = t\mathcal{O}_C(J).$$

A Fredholm index is integer-valued and locally constant on Fredholm families. The witness becomes an index only after a separate spectral-flow or Fredholm theorem discretizes it. The circle theorem supplies such a calibration; the finite arithmetic witness does not.

11.7 Boundary of the finite arithmetic model

The finite chain proves

$$\ell \xrightarrow{*\mu} \Lambda \xrightarrow{T_N} \text{prime-radical seam mass},$$

and it provides exact diagnostics for failed finite transfer. It does not prove

$$\text{arithmetic defect current} = \text{defect created by an actual zero of } \xi.$$

That identification requires the completed analytic carrier, its faithful coordinate map, and the zero or sign theorem developed in the remaining parts of the paper.

12 The RH representation interface

The recognition-seam framework lives on state spaces, while the critical line lives in the complex parameter plane. A lawful bridge must therefore specify both the analytic symmetries and a coordinate map between the two levels. This section separates those structures before the native completed-Weil realization is constructed in Part III.

12.1 Functional, reality and critical-line symmetries

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s).$$

The function is entire and satisfies

$$\xi(s) = \xi(1-s), \qquad \xi(\overline{s}) = \overline{\xi(s)};$$

see [Tit86, Edw74]. Define

$$\tau(s) = 1-s, \qquad c(s) = \overline{s}, \qquad \rho(s) = 1-\overline{s}.$$

Then

$$\mathrm{Fix}(\tau) = \left\{\frac{1}{2}\right\}, \qquad \mathrm{Fix}(\rho) = \left\{s : \Re(s) = \frac{1}{2}\right\}.$$

The holomorphic functional transformation and the anti-holomorphic critical-line reflection are not the same map.

12.2 Exact operator covariance package

Theorem 12.1 (Functional, reality and dagger covariance). *Let $\Omega \subset \mathbb{C}$ be invariant under τ and c. Let J be a unitary involution and let $K(s)$ be an operator family satisfying*

$$JK(s)J = K(1-s), \qquad K(s)^* = K(\overline{s}).$$

Then

$$\boxed{JK(s)^*J = K(1-\overline{s}).}$$

Equivalently, for $A^\sharp = JA^*J$,

$$K(s)^\sharp = K(\rho(s)).$$

Proof. Apply the first covariance identity at $\overline{s}$ and use the adjoint identity:

$$JK(s)^*J = JK(\overline{s})J = K(1-\overline{s}).$$

□

A linear sheet swap implements the holomorphic exchange $s \leftrightarrow 1-s$ at the family level. Conjugation enters through the adjoint or through a separately declared antiunitary reality structure. Critical-line reflection is the combined dagger covariance.

12.3 Equivariant coordinate maps and fixed fibres

Let X be a represented state space with involution J_X and let

$$\pi : X \longrightarrow \mathbb{C}$$

be a spectral or analytic coordinate map.

Theorem 12.2 (Equivariance gives only a fixed-locus inclusion)**.** *Assume*

$$\pi(J_X x) = \rho(\pi(x)) \qquad (x \in X).$$

Then

$$\pi(\operatorname{Fix} J_X) \subseteq \operatorname{Fix}(\rho).$$

Moreover,

$$\pi(\operatorname{Fix} J_X) = \operatorname{Fix}(\rho) \cap \pi(X)$$

if and only if every represented critical-line parameter has at least one fixed state in its fibre. The stronger identity

$$\operatorname{Fix} J_X = \pi^{-1}(\operatorname{Fix} \rho)$$

requires every state over a critical-line parameter to be fixed.

Proof. If $J_X x = x$, then

$$\pi(x) = \pi(J_X x) = \rho(\pi(x)),$$

which proves the inclusion. The equality statements are precisely the corresponding fibre conditions.

□

Example 12.3 (Equivariance without a fixed state)**.** Let $X = \{a, b\}$, let J_X exchange the two states, and set

$$\pi(a) = \pi(b) = \frac{1}{2}.$$

Then $\pi(J_X x) = \rho(\pi(x))$ for both states, but $\operatorname{Fix} J_X$ is empty.

Thus parameter equivariance alone does not prove that the represented seam equals the critical line.

12.4 Completed representation obligations

A direct zero-localization representation must supply distinct theorems for:

R1. a completed analytic carrier and closed operator family;
R2. functional covariance under τ;
R3. reality covariance under c;
R4. dagger covariance under ρ;
R5. an equivariant spectral-coordinate map π;
R6. a fixed-fibre theorem identifying the represented seam with the critical-line locus;
R7. a licensed spectral or determinant divisor theorem for ξ;
R8. a completed anti-seam current and a no-leakage theorem.

No item may be installed by definition if it is later consumed as evidence for RH.

12.5 Two terminal analytic routes

Direct determinant or zero-localization route. One seeks a holomorphic Schatten or Fredholm family $K_\xi(s)$ such that a licensed zero object $F(s)$ obeys

$$F(s) = g(s)\xi(s)$$

with g holomorphic and nowhere zero. This requires a common or lawful varying domain, removal of regulators, archimedean completion, multiplicity preservation and exclusion of extra, missing or polluted zeros. The diagonal paired-prime carrier of the preceding sections is formally excluded because its critical-strip spectral and determinant zero sets are empty.

Completed Weil-sign route. The classical Weil criterion replaces zero-by-zero realization by nonnegativity of a completed Hermitian form. Part III constructs the native Hilbert envelope X_3, its cut-derived parity sectors and the bounded self-adjoint representative W_3. The terminal question then becomes a global sign problem rather than a determinant-divisor problem. This route consumes the classical explicit formula and Weil criterion, but it does not consume the failed diagonal determinant carrier.

The two routes share the same proof-governance demand: the chosen representation must be target faithful and stable under completion. They do not share the same terminal spectral event.

12.6 Cancellation-free direct-route theorem shape

Theorem 12.4 (Completed anti-seam exclusion implies RH)**.** *Suppose a completed direct representation assigns to every nontrivial zero ρ_0 of ξ a current*

$$J_\xi(\rho_0) \in \mathfrak{D}_-$$

such that:

(i) every nontrivial zero is represented;

(ii)

$$\Re(\rho_0) \neq \frac{1}{2} \quad \Longrightarrow \quad J_\xi(\rho_0) \neq 0;$$

(iii) completed admissibility gives

$$J_\xi(\rho_0) = 0$$

for every represented nontrivial zero.

Then every nontrivial zero lies on the critical line.

Proof. An off-critical zero would have both a nonzero and a zero current, a contradiction. □

The nonnegative quantity

$$Q_\xi(\rho_0) = \|J_\xi(\rho_0)\|^2$$

is a cancellation-free version of this obstruction. A single signed linear witness can replace it only after a completeness theorem prevents orthogonality and reflected-orbit cancellation.

Remark 12.5 (Why symmetry alone is insufficient). The functional equation and reality symmetry organize nontrivial zeros into reflected orbits. They do not force each member to be fixed by ρ. Covariance of the zero set is weaker than localization of every zero on the seam.

13 The zero-identification boundary

Modeled points, spectral characteristic values, determinant zeros and actual zeros of ξ are different mathematical objects. The arrows between them must be proved for the same completed carrier. This section closes the spectral-to-determinant arrow when analytic Schatten theory applies and records that, for the present diagonal paired-prime carrier, both sets are empty in the critical strip.

13.1 Four zero layers

Definition 13.1 (Modeled points)**.** Let Z_{model} be a finite or formal collection introduced to test reflection, seam projection, crossing or leakage rules. Membership is a modeling choice.

Definition 13.2 (Spectral characteristic values)**.** For a declared operator family $H(s)$, let Z_{spectral} be the parameters at which the declared spectral event occurs, such as noninvertibility, an eigenvalue crossing or loss of Fredholmness.

Definition 13.3 (Determinant zeros)**.** If a determinant F is analytically licensed on Ω, define

$$Z_{\det} = \{s \in \Omega : F(s) = 0\}.$$

The determinant type and normalization are part of the definition.

Definition 13.4 (Completed-zeta zeros)**.** Let

$$Z_\xi = \{s \in \mathbb{C} : \xi(s) = 0\},$$

with multiplicity.

A carrier-dependent route has the form

$$Z_{\mathrm{model}} \dashrightarrow Z_{\mathrm{spectral}} \dashrightarrow Z_{\det} \dashrightarrow Z_\xi.$$

The first arrow is a representation theorem, the second is an analytic determinant theorem, and the third is a divisor-identification theorem.

13.2 Analytic Schatten determinant bridge

Theorem 13.5 (Spectral event equals regularized determinant zero)**.** *Let Ω be a domain, let $m \geq 1$ be an integer, and let*

$$K : \Omega \longrightarrow \mathcal{S}_m$$

be holomorphic. Define

$$F_m(s) = \det_m(I - K(s)).$$

Then F_m is holomorphic and

$$\boxed{F_m(s) = 0 \quad \Longleftrightarrow \quad I - K(s) \text{ is not invertible} \quad \Longleftrightarrow \quad 1 \in \operatorname{Spec} K(s).}$$

Thus, for this declared event,

$$Z_{\text{spectral}} = Z_{\det}$$

as sets. Multiplicity requires the standard analytic characteristic-value convention on both sides.

Proof. The regularized determinant is holomorphic on analytic $\mathcal{S}_m$ families. Its zero set is exactly the set where $I - K(s)$ fails to be invertible, with algebraic multiplicity governed by analytic Fredholm theory. □

Corollary 13.6 (Strict contraction is zero-blind)**.** *If $\|K(s)\| < 1$ throughout a domain, then*

$$Z_{\text{spectral}} = Z_{\det} = \varnothing$$

there.

Proof. The Neumann series makes $I - K(s)$ invertible; apply the theorem. □

13.3 Finite paired determinant as a normalization test

Let $0 < \lambda_1 \leq \cdots \leq \lambda_m$ be declared levels and define

$$D_N(E) = \prod_{j=1}^{m} \left(1 - \frac{E^2}{\lambda_j^2}\right).$$

Then $D_N(0) = 1$, $D_N(E) = D_N(-E)$, its zeros are the prescribed signed levels, and

$$\frac{D_N'(E)}{D_N(E)} = \sum_{j=1}^{m} \frac{2E}{E^2 - \lambda_j^2}$$

away from them. If H_N has spectrum $\{\pm\lambda_j\}$, then its characteristic polynomial is the corresponding unnormalized product.

These are correct finite identities, but the zero agreement is built into the chosen spectrum. The scaffold tests normalization and multiplicity bookkeeping; it supplies no arithmetic correspondence unless the levels are derived independently from a completed carrier.

13.4 Nonvanishing-factor promotion

Lemma 13.7 (Zero and multiplicity transfer)**.** *Let F, g and ξ be holomorphic on a domain Ω. If*

$$F(s) = g(s)\xi(s)$$

and g is nowhere zero, then F and ξ have the same divisor on Ω.

Proof. At every point, g is a unit in the local ring of holomorphic germs, so multiplication by g preserves the order of vanishing. □

The lemma is easy because its hypothesis already contains the decisive zero theorem. Constructing a lawful F and proving the identity is the hard step.

13.5 Diagonal paired-prime carrier exclusion

The preceding determinant audit proves:

(i) for every $\lambda > 0$, the damped Fredholm determinant is zero-free on the closed critical strip;

(ii) for every integer $m \geq 3$, the undamped regularized determinant is zero-free on

$$\Omega_m = \left\{ s : \frac{1}{m} < \Re(s) < 1 - \frac{1}{m} \right\};$$

(iii) throughout the open critical strip,

$$\|K_0(s)\| = \max\{2^{-\Re(s)}, 2^{\Re(s)-1}\} < 1.$$

Theorem 13.8 (Carrier-exclusion theorem)**.** *For the declared diagonal paired-prime family,*

$$Z_{\text{spectral}} = Z_{\text{det}} = \varnothing$$

in every licensed critical-strip domain. Consequently there is no identity

$$\det_m(I - K_0(s)) = g(s)\xi(s)$$

on a domain containing a zero of ξ when g is holomorphic and nowhere zero. The same conclusion holds for the damped Fredholm determinant on the closed strip.

Proof. The norm estimate gives invertibility in the open strip, and the exact damped divisor lies outside the closed strip. The analytic Schatten bridge therefore gives empty determinant zero sets. A nowhere-zero multiple of ξ must vanish at every zero of ξ, which these determinants do not. □

The direct diagonal route is therefore not waiting for a more elaborate normalization. It lacks the required interior spectral event. The native completed-Weil form changes both topology and terminal mechanism rather than decorating this excluded carrier.

13.6 Zero-correspondence obstruction classes

A completed audit should distinguish:

Z1. extra determinant zeros;

Z2. missing target zeros;

Z3. multiplicity mismatch;

Z4. archimedean zeros or poles introduced by incomplete Γ and π treatment;

Z5. trivial-zero contamination;
Z6. spectral pollution in a limiting approximation;
Z7. domain or analytic-sheet mismatch;
Z8. regulator dependence of the divisor.
A clean zero theorem must eliminate every relevant class on the same completed carrier.

13.7 Guarded promotion theorem

Theorem 13.9 (Completed divisor promotion)**.** *Suppose a completed carrier provides a holomorphic Schatten family K and $F_m = \det_m(I - K)$ on Ω. Assume:*

(i) the determinant is licensed by the operator ideal and completion;
(ii) regulators have been removed without changing the divisor;
(iii) archimedean and trivial factors are accounted for;
(iv) spectral pollution and domain mismatch are absent;
(v) there is a holomorphic nowhere-zero g with

$$F_m = g\xi.$$

Then

$$Z_{\text{spectral}} = Z_{\det} = Z_\xi \cap \Omega$$

with the declared multiplicity convention.

Proof. The analytic Schatten bridge gives the first equality. The nonvanishing-factor lemma gives the equality of divisors, and the remaining assumptions ensure that the objects belong to the completed carrier rather than to a regulator or finite approximation. ☐

This is a promotion theorem and claim ledger. It states what a successful carrier would imply; it does not reduce the difficulty of constructing that carrier.

13.8 Forbidden direct identifications

Neither

$$Z_{\text{model}} \longrightarrow Z_\xi$$

nor

$$Z_{\text{spectral}} \longrightarrow Z_\xi$$

is lawful without the intervening representation and divisor theorems. A modeled point may calibrate a sign rule without being a zero. A self-adjoint operator may have real spectrum without representing the ordinates of zeta zeros.

Likewise,

$$\rho(s) = s \quad \Longleftrightarrow \quad \Re(s) = \frac{1}{2}$$

identifies a reflection-fixed locus, not a zero set. Seam membership and zero membership remain logically independent until a completed target-faithful theorem connects them.

Part III

Native Multiplicative and X_3 Completed-Weil Realization

14 Native multiplicative carrier, EMK cut and Euler-scale flow

The completed operator theory is most naturally organized on the multiplicative scale line before logarithmic or Fourier coordinates are introduced. The guiding order is

$$\boxed{\begin{array}{c}\text{native scale carrier} \longrightarrow \text{EMK cut and scale flow}\\ \longrightarrow \text{native energy completion} \longrightarrow \text{classical representation charts.}\end{array}}$$

The logarithmic, Mellin and Fourier descriptions used later are faithful computational representations of this source structure. They are not used to repair a carrier that is blind to the declared target.

Throughout this section, Hilbert inner products are linear in the first variable.

14.1 The multiplicative core and inversion cut

Let

$$\mathcal{D}_\times = C_c^\infty(0,\infty), \qquad \mathrm{d}^\times r = \frac{\mathrm{d}r}{r}.$$

The measure $\mathrm{d}^\times r$ is invariant under dilation and inversion.

Definition 14.1 (EMK inversion cut and scale flow)**.** For $F \in \mathcal{D}_\times$ and $t \in \mathbb{R}$, define

$$(\mathsf{J}_\times F)(r) = F(r^{-1}), \qquad (\mathsf{U}_t F)(r) = F(e^{-t}r).$$

The distinguished seam is the fixed scale $r = 1$; the two open sheets are $0 < r < 1$ and $r > 1$.

Theorem 14.2 (Native cut–flow law)**.** *On $L^2((0,\infty), \mathrm{d}^\times r)$, $\mathsf{J}_\times$ is a unitary self-adjoint involution, $(\mathsf{U}_t)_{t\in\mathbb{R}}$ is a strongly continuous unitary group, and*

$$\boxed{\mathsf{J}_\times \mathsf{U}_t \mathsf{J}_\times = \mathsf{U}_{-t}.}$$

Its self-adjoint generator is the closure E of

$$\mathsf{E}_0 = -ir\frac{\mathrm{d}}{\mathrm{d}r} \quad \text{on } \mathcal{D}_\times,$$

with

$$\mathsf{U}_t = e^{-it\mathsf{E}}, \qquad \boxed{\mathsf{J}_\times \mathsf{E} \mathsf{J}_\times = -\mathsf{E}.}$$

Proof. Invariance of $\mathrm{d}^\times r$ gives

$$\|\mathsf{J}_\times F\|_2 = \|F\|_2, \qquad \|\mathsf{U}_t F\|_2 = \|F\|_2.$$

The identities $\mathsf{J}_\times^2 = I$ and

$$(\mathsf{J}_\times \mathsf{U}_t \mathsf{J}_\times F)(r) = F(e^t r) = (\mathsf{U}_{-t}F)(r)$$

are direct. Strong continuity is first checked on $\mathcal{D}_\times$ and then extended by density. Stone's theorem gives a self-adjoint generator. Differentiation at $t = 0$ on the core gives $-ir\,\mathrm{d}/\mathrm{d}r$, and conjugating the unitary group identity gives the generator identity. $\square$

14.2 Euler–harmonic representation

Use the Mellin convention

$$(\mathcal{M}F)(\xi) = \int_0^\infty F(r)r^{-i\xi}\,\mathrm{d}^\times r.$$

Mellin–Plancherel gives

$$\|F\|^2_{L^2(\mathrm{d}^\times r)} = \frac{1}{2\pi}\int_{\mathbb{R}} |\mathcal{M}F(\xi)|^2\,\mathrm{d}\xi.$$

On the core,

$$\mathcal{M}(\mathsf{E}F)(\xi) = \xi\mathcal{M}F(\xi), \qquad \mathcal{M}(\mathsf{U}_tF)(\xi) = e^{-i\xi t}\mathcal{M}F(\xi),$$

and

$$\mathcal{M}(\mathsf{J}_\times F)(\xi) = \mathcal{M}F(-\xi).$$

For $\xi \in \mathbb{R}$, let

$$\chi_\xi(r) = r^{i\xi}$$

be the generalized multiplicative character, interpreted distributionally against $\mathcal{D}_\times$.

Theorem 14.3 (Euler–harmonic representation theorem)**.** *The native Euler-scale structure determines the classical harmonic characters as follows.*

(i) The generalized characters diagonalize the Euler generator and its flow:

$$\boxed{\mathsf{E}_0\chi_\xi = \xi\chi_\xi, \qquad \mathsf{U}_t\chi_\xi = e^{-i\xi t}\chi_\xi.}$$

Equivalently,

$$\boxed{\mathcal{M}\mathsf{E}\mathcal{M}^{-1} = M_\xi, \qquad \mathcal{M}\mathsf{U}_t\mathcal{M}^{-1} = M_{e^{-i\xi t}}.}$$

(ii) The EMK cut reverses the spectral character:

$$\boxed{\mathsf{J}_\times\chi_\xi = \chi_{-\xi}.}$$

Its even and odd combinations are

$$\boxed{\frac{\chi_\xi + \chi_{-\xi}}{2} = \cos(\xi\log r), \qquad \frac{\chi_\xi - \chi_{-\xi}}{2i} = \sin(\xi\log r).}$$

Thus cosine and sine are respectively the seam-even and seam-odd spectral channels of the native scale flow.

(iii) Under the logarithmic coordinate $x = \log r$,

$$\boxed{\chi_\xi(e^x) = e^{i\xi x},}$$

so the Mellin spectral representation becomes the ordinary Fourier representation of translation.

(iv) Let

$$\mathsf{H}_\times = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{H}_\times^2 = I.$$

For $a > 0$ and $x = \log r$,

$$\boxed{\exp\big((a\mathsf{H}_\times - i\xi I)x\big) = \begin{pmatrix} r^{-a-i\xi} & 0 \\ 0 & r^{a-i\xi} \end{pmatrix} = e^{-i\xi x}(\cosh(ax)I + \sinh(ax)\mathsf{H}_\times).}$$

The two entries are exactly the kernels of the weighted Mellin boundary channels in their declared order. Hyperbolic sheet separation and circular Euler phase therefore commute in one two-sheet representation.

Proof. The first character identity follows from

$$-ir\frac{\mathrm{d}}{\mathrm{d}r}r^{i\xi} = \xi r^{i\xi},$$

and the flow identity follows from

$$\chi_\xi(e^{-t}r) = e^{-i\xi t}\chi_\xi(r).$$

The Mellin multiplication formulas are the corresponding unitary spectral representation. Inversion gives $(r^{-1})^{i\xi} = r^{-i\xi}$, and the sine/cosine formulas are the symmetric and antisymmetric combinations. The logarithmic identity is immediate. Finally, $\mathsf{H}_\times$ commutes with iI and satisfies $\mathsf{H}_\times^2 = I$, so its exponential is $\cosh(ax)I + \sinh(ax)\mathsf{H}_\times$; multiplication by the circular phase gives the displayed matrix. □

Corollary 14.4 (Classical harmonic analysis as the Euler shadow)**.** *Within the declared completed-Weil realization, the chain*

$$\boxed{\begin{array}{c}\textit{Euler-scale flow} \longrightarrow \textit{self-adjoint generator} \longrightarrow \textit{Mellin characters}\\ \longrightarrow \textit{sine/cosine cut sectors} \longrightarrow \textit{Fourier translation chart}\end{array}}$$

is derived from the native carrier and its inversion cut. The additional weights $r^{\pm a}$ produce the commuting hyperbolic two-sheet factor. Classical harmonic analysis is therefore a faithful spectral shadow of the native scale dynamics in this realization, not an independent source axiom.

The corollary is deliberately scoped. It does not claim to derive the completed zeta explicit formula, the Weil criterion or the terminal no-crossing estimate; those are respectively a classical input and open sign obligations.

14.3 Native energy and completed cut

Let

$$G(\xi) = \operatorname{Re}\psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log\pi, \qquad w_\Gamma(\xi) = 1 + |G(\xi)|.$$

Definition 14.5 (Native scale energy)**.** For $F \in \mathcal{D}_\times$, set

$$\boxed{\begin{aligned}\|F\|_{\times,3}^2 = &\int_0^\infty \max(r^3, r^{-3})|F(r)|^2\,\mathrm{d}^\times r\\ &+ \frac{1}{2\pi}\int_{\mathbb{R}} w_\Gamma(\xi)|\mathcal{M}F(\xi)|^2\,\mathrm{d}\xi.\end{aligned}}$$

Let $\mathscr{X}_{\times,3}$ be the completion of $\mathcal{D}_\times$ in this norm.

The native Hilbert carrier is therefore produced by completing the declared scale and archimedean energy. The inversion cut is isometric because both weights are even under $r \mapsto r^{-1}$ and $\xi \mapsto -\xi$.

Theorem 14.6 (Completed EMK cut)**.** *The inversion cut extends uniquely to a unitary self-adjoint involution on $\mathscr{X}_{\times,3}$, again denoted $\mathsf{J}_\times$. Hence*

$$\boxed{\mathscr{X}_{\times,3} = \mathscr{X}_{\times,3}^+ \oplus \mathscr{X}_{\times,3}^-,}$$

where

$$\mathscr{X}_{\times,3}^\pm = \ker(\mathsf{J}_\times \mp I).$$

Proof. For core vectors, a change of variables gives equality of both terms in the native norm after inversion. The extension follows by density, and the orthogonal decomposition is the spectral decomposition of a self-adjoint involution. □

14.4 Direct native compactness

Let

$$\iota_\times : \mathscr{X}_{\times,3} \longrightarrow L^2((0,\infty), \mathrm{d}^\times r)$$

be the natural inclusion.

Theorem 14.7 (Native compact embedding)**.** *The inclusion $\iota_\times$ is compact.*

Proof. Let $\mathcal{K}$ be bounded in the native energy norm. For $R > 1$,

$$\int_{r>R} |F(r)|^2 \, \mathrm{d}^\times r + \int_{0<r<R^{-1}} |F(r)|^2 \, \mathrm{d}^\times r \leq R^{-3} \|F\|_{\times,3}^2,$$

so $\mathcal{K}$ has uniform multiplicative spatial tails. Put

$$m(\Omega) = \inf_{|\xi|>\Omega} w_\Gamma(\xi).$$

Since $m(\Omega) \to \infty$,

$$\int_{|\xi|>\Omega} |\mathcal{M}F(\xi)|^2 \, \mathrm{d}\xi \leq \frac{2\pi}{m(\Omega)} \|F\|_{\times,3}^2.$$

Finally,

$$\|\mathsf{U}_t F - F\|_2^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |e^{-i\xi t} - 1|^2 |\mathcal{M}F(\xi)|^2 \, \mathrm{d}\xi.$$

The high-frequency part is uniformly small by the preceding estimate, and the bounded-frequency part is uniformly small as $t \to 0$. The multiplicative Kolmogorov–Riesz criterion, equivalently the ordinary criterion after the Haar isomorphism $r = e^x$, gives relative compactness. □

14.5 The logarithmic representation theorem

Define

$$(\mathcal{L}F)(x) = F(e^x).$$

Since $\mathrm{d}^\times r = \mathrm{d}x$ and

$$\max(r^3, r^{-3}) = e^{3|x|},$$

this map carries the native spatial channel to the exponential spatial channel used in X_3.

Theorem 14.8 (Native-to-X_3 unitary chart)**.** *The logarithmic map extends to a unitary*

$$\boxed{\mathcal{L} : \mathscr{X}_{\times,3} \longrightarrow X_3.}$$

It intertwines the native and logarithmic structures:

$$\boxed{\mathcal{L}\mathsf{J}_\times\mathcal{L}^{-1} = J_3, \qquad \mathcal{L}\mathsf{E}\mathcal{L}^{-1} = -i\frac{\mathrm{d}}{\mathrm{d}x}, \qquad \mathcal{L}\mathsf{U}_t\mathcal{L}^{-1} = T_t.}$$

Consequently

$$\mathcal{L}\mathscr{X}_{\times,3}^\pm = X_3^\pm.$$

Proof. The norm identity follows from the change of variables $r = e^x$. With the Mellin convention above,

$$\mathcal{M}F(\xi) = \widehat{\mathcal{L}F}(-\xi).$$

The frequency weight is even, so the sign reversal does not change the norm. The intertwining identities are direct on the core and extend by continuity. □

This theorem fixes the proof direction: X_3 is a faithful logarithmic representation of the native scale carrier. It is not a separate source axiom.

14.6 Native completed-Weil form

For $F \in \mathcal{D}_\times$, define

$$\ell_+^\times(F) = \int_0^\infty F(r) r^{-1/2}\, \mathrm{d}^\times r, \qquad \ell_-^\times(F) = \int_0^\infty F(r) r^{1/2}\, \mathrm{d}^\times r.$$

Cauchy–Schwarz and the spatial energy give

$$|\ell_\pm^\times(F)|^2 \le \frac{3}{4}\|F\|_{\times,3}^2.$$

Define the native boundary, Gamma and prime forms by

$$\mathfrak{B}_\times(F,H) = \ell_+^\times(F)\overline{\ell_-^\times(H)} + \ell_-^\times(F)\overline{\ell_+^\times(H)},$$

$$\mathfrak{A}_{\Gamma,\times}(F,H) = \frac{1}{2\pi}\int_{\mathbb{R}} G(\xi)\mathcal{M}F(\xi)\overline{\mathcal{M}H(\xi)}\,\mathrm{d}\xi,$$

and

$$\mathfrak{P}_\times(F,H) = \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\left(\langle F, \mathsf{U}_{\log n}H\rangle + \langle F, \mathsf{U}_{-\log n}H\rangle\right).$$

The spatial channel gives the native correlation estimate

$$\boxed{|\langle F, \mathsf{U}_t H\rangle| \le e^{-3|t|/2}\|F\|_{\times,3}\|H\|_{\times,3}.}$$

Hence the prime form is absolutely bounded by

$$2\sum_{n\ge 2}\frac{\Lambda(n)}{n^2}\|F\|_{\times,3}\|H\|_{\times,3}.$$

Definition 14.9 (Native completed-Weil form)**.** Set

$$\boxed{\mathfrak{W}_\times(F,H) = \mathfrak{B}_\times(F,H) + \mathfrak{A}_{\Gamma,\times}(F,H) - \mathfrak{P}_\times(F,H).}$$

Theorem 14.10 (Native bounded Weil representative)**.** *The form $\mathfrak{W}_\times$ extends uniquely to a bounded Hermitian form on $\mathscr{X}_{\times,3}$. There is a unique bounded self-adjoint operator*

$$\boxed{W_\times \in \mathcal{B}(\mathscr{X}_{\times,3})}$$

such that

$$\mathfrak{W}_\times(F,H) = \langle F, W_\times H\rangle_{\times,3}.$$

Moreover,

$$\boxed{\mathcal{L}W_\times\mathcal{L}^{-1} = W_3, \qquad [W_\times, \mathsf{J}_\times] = 0.}$$

Proof. The three form estimates above prove boundedness. Hermiticity follows from the paired boundary terms, the real Gamma symbol and the adjoint pair of dilation operators. Riesz representation gives $W_\times$. Under $x = \log r$, the boundary, Gamma and prime terms become exactly the corresponding terms in $\mathfrak{W}_\xi$; uniqueness gives unitary equivalence. Inversion exchanges the two boundary channels, reflects the even Gamma symbol and interchanges the two prime dilations, so the form is inversion invariant and the representative commutes with the cut. ∎

14.7 Two-sheet prime seam operator

For $a > 0$, define the boundary map

$$\mathcal{B}_a^\times F(\xi) = \begin{pmatrix} \mathcal{M}(r^{-a}F)(\xi) \\ \mathcal{M}(r^{a}F)(\xi) \end{pmatrix}.$$

Mellin–Plancherel gives

$$\boxed{\frac{1}{2\pi}\|\mathcal{B}_a^\times F\|_2^2 = \int_0^\infty (r^{-2a} + r^{2a})|F(r)|^2 \, \mathrm{d}^\times r.}$$

At $a = 3/2$, this norm is equivalent, with constants 1 and 2, to the native spatial channel.

For $a > 1/2$, put

$$q_a(\xi) = \sum_{n\geq 2} \frac{\Lambda(n)}{n^{a+1/2}} e^{i\xi \log n} = -\frac{\zeta'}{\zeta}\left(a + \frac{1}{2} - i\xi\right)$$

and

$$\mathcal{Q}_a(\xi) = \begin{pmatrix} 0 & q_a(\xi) \\ \overline{q_a(\xi)} & 0 \end{pmatrix}.$$

This matrix is self-adjoint for real ξ.

Theorem 14.11 (Native prime seam operator)**.** *The prime operator has the two equivalent native forms*

$$\boxed{B_{\mathrm{prime}}^\times = \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \iota_\times^* (\mathsf{U}_{\log n} + \mathsf{U}_{-\log n})\iota_\times}$$

and

$$\boxed{B_{\mathrm{prime}}^\times = (\mathcal{B}_{3/2}^\times)^* M_{\mathcal{Q}_{3/2}} \mathcal{B}_{3/2}^\times.}$$

It is compact and self-adjoint on $\mathscr{X}_{\times,3}$, and

$$\|\mathcal{Q}_{3/2}\|_{L^\infty} \leq -\frac{\zeta'(2)}{\zeta(2)}.$$

Proof. Each dilation atom is compact after sandwiching by the compact inclusion $\iota_\times$. The correlation estimate gives

$$\left\|\iota_\times^*(\mathsf{U}_t + \mathsf{U}_{-t})\iota_\times\right\| \leq 2e^{-3|t|/2},$$

so the series converges absolutely in operator norm. The two-sheet formula follows from weighted Mellin–Plancherel. At $a = 3/2$, the seam factor converts $\Lambda(n)/\sqrt{n}$ into $\Lambda(n)/n^2$, permitting interchange of sum and integral. The norm bound follows from the absolutely convergent Dirichlet series. □

The first formula supplies native compactness; the second displays the hyperbolic two-sheet coupling and circular Euler spectrum. Neither formula requires a spatial grid interpolation.

15 The native X_3 completed-Weil carrier

The recognition-seam framework becomes analytically substantive only after a native completed carrier is constructed. In the MP development this carrier is not postulated as a Hilbert–Pólya spectrum. It is obtained by completing the explicit-formula test core in a two-channel norm that controls both spatial escape and archimedean frequency escape. The resulting completed Weil form has a bounded self-adjoint representative, admits a derived even/odd Hilbert decomposition, and supports the exact parity/Feshbach reduction used in the terminal recognition-seam geometry.

The purpose of this section is therefore foundational. The Riemann hypothesis enters only after the carrier theorem, through the classical Weil positivity criterion.

15.1 The two-channel Hilbert envelope

Fix the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{i\xi x}\,dx, \qquad \|f\|_{L^2}^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\xi)|^2\,d\xi.$$

Let

$$\mathcal{D} = C_c^\infty(\mathbb{R})$$

and put

$$G(\xi) = \operatorname{Re}\psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log\pi, \qquad w_\Gamma(\xi) = 1 + |G(\xi)|,$$

where $\psi = \Gamma'/\Gamma$ is the digamma function.

Definition 15.1 (Native X_3 envelope)**.** Define X_3 to be the completion of $\mathcal{D}$ in the norm

$$\boxed{\|f\|_{X_3}^2 = \int_{\mathbb{R}} e^{3|x|}|f(x)|^2\,dx + \frac{1}{2\pi}\int_{\mathbb{R}} w_\Gamma(\xi)|\widehat{f}(\xi)|^2\,d\xi.}$$

The first term is the spatial channel and the second is the completed archimedean frequency channel.

The definition makes X_3 a Hilbert space. Since $e^{3|x|} \geq 1$, the native inclusion

$$\iota : X_3 \longrightarrow L^2(\mathbb{R})$$

is bounded.

Lemma 15.2 (Weighted correlation decay)**.** *Let $T_t h(x) = h(x-t)$. For $f, h \in X_3$ and $t \in \mathbb{R}$,*

$$\boxed{|\langle f, T_t h\rangle_{L^2}| \leq e^{-3|t|/2}\|f\|_{X_3}\|h\|_{X_3}.}$$

Proof. For core vectors, insert the spatial weights and use $|x| + |x-t| \geq |t|$:

$$\begin{aligned} |\langle f, T_t h\rangle| &\leq \int e^{-\frac{3}{2}(|x|+|x-t|)}\big(e^{\frac{3}{2}|x|}|f(x)|\big)\big(e^{\frac{3}{2}|x-t|}|h(x-t)|\big)\,dx \\ &\leq e^{-3|t|/2}\|e^{3|x|/2}f\|_2\|e^{3|x|/2}h\|_2. \end{aligned}$$

Density gives the result on X_3. ∎

Theorem 15.3 (Native position-frequency compactness). *The inclusion*

$$\boxed{\iota : X_3 \hookrightarrow L^2(\mathbb{R})}$$

is compact.

Proof. Let (f_j) be bounded in X_3, with squared norm at most M. The spatial channel gives

$$\int_{|x|>R} |f_j(x)|^2\,dx \le Me^{-3R}.$$

Since $|G(\xi)| \to \infty$ as $|\xi| \to \infty$, if

$$m(\Omega) = \inf_{|\xi|>\Omega} w_\Gamma(\xi),$$

then $m(\Omega) \to \infty$ and

$$\int_{|\xi|>\Omega} |\widehat{f}_j(\xi)|^2\,d\xi \le \frac{2\pi M}{m(\Omega)}.$$

Finally, Plancherel gives

$$\|f_j(\cdot + h) - f_j\|_2^2 = \frac{1}{2\pi}\int |e^{-ih\xi} - 1|^2 |\widehat{f}_j(\xi)|^2\,d\xi.$$

After splitting at $|\xi| = \Omega$, the high-frequency part is uniformly small by the preceding tail and the low-frequency part is uniformly small for sufficiently small h. The Kolmogorov–Riesz criterion now gives relative compactness in $L^2(\mathbb{R})$. □

15.2 The completed critical Weil form

For $f \in \mathcal{D}$, define the completion-pole channels

$$\ell_+(f) = \widehat{f}(i/2) = \int_{\mathbb{R}} f(x)e^{-x/2}\,dx, \qquad \ell_-(f) = \widehat{f}(-i/2) = \int_{\mathbb{R}} f(x)e^{x/2}\,dx.$$

They satisfy

$$|\ell_\pm(f)|^2 \le \frac{3}{4}\|f\|_{X_3}^2.$$

Define

$$\begin{aligned}\mathfrak{B}(f,h) &= \ell_+(f)\overline{\ell_-(h)} + \ell_-(f)\overline{\ell_+(h)},\\ \mathfrak{A}_\Gamma(f,h) &= \frac{1}{2\pi}\int_{\mathbb{R}} G(\xi)\widehat{f}(\xi)\overline{\widehat{h}(\xi)}\,d\xi,\end{aligned}$$

and

$$\mathfrak{P}_{1/2}(f,h) = \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} \left(\langle f, T_{\log n}h\rangle + \langle f, T_{-\log n}h\rangle\right).$$

On $\mathcal{D}$ the prime sum is finite, because the correlation $f * h^\sharp$ has compact support. On X_3, Theorem 15.2 gives

$$|\mathfrak{P}_{1/2}(f,h)| \le 2\left(\sum_{n\ge 2}\frac{\Lambda(n)}{n^2}\right)\|f\|_{X_3}\|h\|_{X_3}.$$

The scalar series equals $-\zeta'(2)/\zeta(2)$ and is finite.

Definition 15.4 (Completed critical Weil form)**.** On $\mathcal{D}$, set

$$\boxed{\mathfrak{W}_\xi(f,h) = \mathfrak{B}(f,h) + \mathfrak{A}_\Gamma(f,h) - \mathfrak{P}_{1/2}(f,h).}$$

Theorem 15.5 (Bounded native Weil representative)**.** *The form $\mathfrak{W}_\xi$ extends uniquely to a bounded Hermitian form on X_3. Consequently there is a unique bounded self-adjoint operator*

$$\boxed{W_3 \in \mathcal{B}(X_3), \qquad \mathfrak{W}_\xi(f,h) = \langle f, W_3 h\rangle_{X_3}.}$$

Proof. The pole functionals are continuous by the displayed 3/4 estimate. The Gamma form is bounded because $|G| \le w_\Gamma$ and the frequency term is part of the native norm. The prime form is bounded by the preceding correlation estimate. Thus $\mathfrak{W}_\xi$ is bounded. It is Hermitian because the pole form is Hermitian, G is real, and the prime translations occur in adjoint pairs. The Riesz representation theorem gives W_3, and Hermiticity gives $W_3^* = W_3$. □

Remark 15.6 (Classical explicit-formula membrane)*.* The identification of $\mathfrak{W}_\xi$ with the symmetrically summed completed-zeta explicit formula, and the equivalence

$$\mathrm{RH} \iff \mathfrak{W}_\xi(f,f) \ge 0 \quad (f \text{ in the admissible Weil core}),$$

are classical analytic-number-theory inputs. The paper's new framework theorem begins after this membrane: it constructs the native Hilbert representation, its recognition-compatible parity split, and the reduced anti-seam obstruction. It does not claim a new proof of the explicit formula or of the Weil criterion.

15.3 The cut-to-X_3 completion theorem

On the explicit-formula core $\mathcal{D}$, define the reflection cut

$$(\kappa_0 f)(x) = f(-x).$$

For the faithful analytic realization used here, recognition on the core is equality. The primitive fixed and anti-fixed sectors are therefore

$$\mathcal{D}^+ = \{f \in \mathcal{D} : \kappa_0 f = f\}, \qquad \mathcal{D}^- = \{f \in \mathcal{D} : \kappa_0 f = -f\}.$$

Thus $\mathcal{D}^+$ is the core seam of the reflection cut and $\mathcal{D}^-$ is its oriented anti-fixed sector. The analytic target determines the two-channel metric; the cut does not manufacture the Gamma weight or the exponent 3. Its exact role is to require that the chosen completion preserve the declared reflection distinction.

Theorem 15.7 (Cut-to-X_3 completion)**.** *The reflection cut κ_0 is isometric for the native norm and extends uniquely to a unitary self-adjoint involution*

$$\boxed{J_3 : X_3 \longrightarrow X_3.}$$

Its fixed and anti-fixed Hilbert sectors are

$$\boxed{X_3^+ = \ker(J_3 - I) = \overline{\mathcal{D}^+}^{X_3}, \qquad X_3^- = \ker(J_3 + I) = \overline{\mathcal{D}^-}^{X_3},}$$

and

$$\boxed{X_3 = X_3^+ \oplus X_3^-.}$$

Proof. Both weights in Theorem 15.1 are even. Moreover,

$$\widehat{\kappa_0 f}(\xi) = \widehat{f}(-\xi).$$

A change of variables therefore gives

$$\|\kappa_0 f\|_{X_3} = \|f\|_{X_3} \qquad (f \in \mathcal{D}).$$

Hence κ_0 extends uniquely by density to a unitary J_3 on X_3. Since $\kappa_0^2 = I$ on the core, $J_3^2 = I$ on the completion; a unitary involution is self-adjoint. The projections

$$P_\pm = \frac{I \pm J_3}{2}$$

are complementary orthogonal projections. On the core, $P_\pm \mathcal{D} = \mathcal{D}^\pm$. Density of $\mathcal{D}$ and continuity of $P_\pm$ imply

$$\operatorname{Ran} P_\pm = \overline{\mathcal{D}^\pm}^{X_3}.$$

The spectral decomposition of the self-adjoint involution gives the direct sum. □

Theorem 15.8 (Cut-reduced Weil representative)**.** *The bounded Weil representative commutes with the completed cut,*

$$W_3 J_3 = J_3 W_3.$$

Consequently

$$\boxed{W_3 = W_3^+ \oplus W_3^-, \qquad W_3 \geq 0 \iff W_3^+ \geq 0 \text{ and } W_3^- \geq 0.}$$

Proof. Reflection interchanges ℓ_+ and ℓ_-, leaves the even Gamma multiplier invariant, and interchanges $T_{\log n}$ with $T_{-\log n}$. Therefore

$$\mathfrak{W}_\xi(J_3 f, J_3 h) = \mathfrak{W}_\xi(f, h).$$

By uniqueness of the Riesz representative, $W_3 J_3 = J_3 W_3$. Hence the two eigenspaces in Theorem 15.7 reduce W_3, and positivity is equivalent to positivity on both direct-sum sectors. □

The even and odd Hilbert spaces are therefore derived from the primitive reflection cut after completion. They are not independently postulated carriers selected after the sign problem is inspected.

Define the parity-resolved boundary functionals

$$\ell_{\text{ev}} = \frac{\ell_+ + \ell_-}{\sqrt{2}}, \qquad \ell_{\text{odd}} = \frac{\ell_- - \ell_+}{\sqrt{2}}.$$

Then

$$\ell_{\text{ev}}(f) = \sqrt{2}\int_{\mathbb{R}} f(x)\cosh(x/2)\,dx, \qquad \ell_{\text{odd}}(f) = \sqrt{2}\int_{\mathbb{R}} f(x)\sinh(x/2)\,dx,$$

and

$$\boxed{\mathfrak{B}(f,h) = \ell_{\text{ev}}(f)\overline{\ell_{\text{ev}}(h)} - \ell_{\text{odd}}(f)\overline{\ell_{\text{odd}}(h)}.}$$

Let $r_{\text{ev}} \in X_3^+$ and $r_{\text{odd}} \in X_3^-$ be their Riesz vectors. Direct dual estimates give

$$\boxed{\|r_{\text{ev}}\|_{X_3}^2 \leq \frac{17}{12}, \qquad \|r_{\text{odd}}\|_{X_3}^2 \leq \frac{1}{12}, \qquad \langle r_{\text{ev}}, r_{\text{odd}}\rangle_{X_3} = 0.}$$

The boundary sector thus carries a positive even rank-one channel and a negative odd rank-one channel. This signed parity split is structural, not numerical.

15.4 Compact prime and Gamma defects

The compact embedding in Theorem 15.3 converts the critical prime distribution into a compact native operator. For $a \in \mathbb{R}$, define

$$C_a = \iota^*(T_a + T_{-a})\iota.$$

Each C_a is compact and self-adjoint. Moreover, Theorem 15.2 gives

$$\|C_a\| \le 2e^{-3|a|/2}.$$

Consequently

$$\boxed{B_{\text{prime}} = \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} C_{\log n}}$$

converges absolutely in operator norm, is compact and self-adjoint, and represents $\mathfrak{P}_{1/2}$. Its directed tail satisfies, for $N \ge 2$,

$$\left\| B_{\text{prime}} - \sum_{2\le n\le N} \frac{\Lambda(n)}{\sqrt{n}} C_{\log n} \right\| \le 2 \sum_{n>N} \frac{\Lambda(n)}{n^2} \le \frac{2(\log N + 1)}{N}.$$

For the Gamma sector, define

$$C_\Gamma f = (2\pi)^{-1/2} \sqrt{w_\Gamma}\, \widehat{f}, \qquad s(\xi) = \frac{G(\xi)}{w_\Gamma(\xi)}.$$

Then

$$B_\Gamma = C_\Gamma^* M_s C_\Gamma$$

represents $\mathfrak{A}_\Gamma$. Put

$$A_\Gamma = C_\Gamma^* C_\Gamma, \qquad D_\Gamma = A_\Gamma - B_\Gamma.$$

The scalar defect

$$m_\Gamma(\xi) = w_\Gamma(\xi) - G(\xi) = 1 + |G(\xi)| - G(\xi)$$

is bounded, nonnegative and even. Hence

$$D_\Gamma = \iota^* \mathcal{F}^{-1} M_{m_\Gamma} \mathcal{F} \iota$$

is compact, positive and self-adjoint.

15.5 The exact even carrier and its shifted compact route

Let $B_{\text{ev}} = r_{\text{ev}} \otimes r_{\text{ev}}$ on X_3^+, where $(r \otimes r)h = \langle h, r\rangle r$. Restrict the prime and Gamma operators to X_3^+ and set

$$A_0 = A_{\Gamma,+} + B_{\text{ev}}, \qquad C_+ = D_{\Gamma,+} + B_{\text{prime},+}.$$

Theorem 15.9 (Exact critical even split)**.** *The even Weil representative satisfies*

$$\boxed{W_3^+ = A_0 - C_+.}$$

The operator A_0 is positive and injective, with

$$0 < A_0 \le \frac{29}{12} I,$$

while C_+ is compact and self-adjoint.

Proof. On the even sector the pole form is $+B_{\mathrm{ev}}$, while $B_{\Gamma,+} = A_{\Gamma,+} - D_{\Gamma,+}$. Substitution into

$$W_3^+ = B_{\mathrm{ev}} + B_{\Gamma,+} - B_{\mathrm{prime},+}$$

gives the split. The upper bound follows from $A_{\Gamma,+} \le I$ and $\|B_{\mathrm{ev}}\| \le 17/12$. The frequency channel is injective, so A_0 has trivial kernel. Compactness of C_+ follows from the two compact summands. $\square$

Injectivity is not coercivity. In fact the exact critical even carrier has an essential zero threshold.

Theorem 15.10 (Essential-zero threshold)**.** *There exists a singular sequence $(f_R) \subset X_3^+$ such that*

$$\|f_R\|_{X_3} = 1, \qquad f_R \rightharpoonup 0, \qquad \|W_3^+ f_R\|_{X_3} \longrightarrow 0.$$

Consequently

$$\boxed{0 \in \sigma_{\mathrm{ess}}(W_3^+).}$$

Proof. Choose nonzero even $\phi \in C_c^\infty((-1/4, 1/4))$ and set

$$u_R(x) = \phi(x - R) + \phi(x + R), \qquad f_R = \frac{u_R}{\|u_R\|_{X_3}}.$$

The spatial component of $\|u_R\|_{X_3}$ grows like $e^{3R/2}$, while its weighted Fourier component stays bounded. Hence $C_\Gamma f_R \to 0$. The even pole functional grows only like $e^{R/2}$ before normalization, so $\ell_{\mathrm{ev}}(f_R) \to 0$. Therefore $A_0 f_R \to 0$. The sequence is weakly null in the native two-channel realization, and compactness gives $C_+ f_R \to 0$. Thus $W_3^+ f_R \to 0$. Weyl's criterion gives the conclusion. $\square$

Corollary 15.11 (Unshifted coercive compact-reference no-go)**.** *There do not exist a bounded operator $A \ge aI$ with $a > 0$ and a compact self-adjoint K such that*

$$W_3^+ = A^{1/2}(I - K)A^{1/2}.$$

Proof. Such a factorization would make W_3^+ Fredholm, contradicting Theorem 15.10. $\square$

For $\eta > 0$, define

$$A_{+,\eta} = A_0 + \eta I, \qquad K_{+,\eta} = A_{+,\eta}^{-1/2} C_+ A_{+,\eta}^{-1/2}.$$

Theorem 15.12 (Shifted even Birman–Schwinger reduction)**.** *For every $\eta > 0$, $K_{+,\eta}$ is compact and self-adjoint and*

$$\boxed{W_3^+ + \eta I = A_{+,\eta}^{1/2}(I - K_{+,\eta})A_{+,\eta}^{1/2}.}$$

Therefore

$$\boxed{W_3^+ + \eta I \ge 0 \iff \sup \sigma(K_{+,\eta}) \le 1.}$$

Moreover,

$$W_3^+ \ge 0 \iff W_3^+ + \eta_j I \ge 0 \quad \textit{for every member of one sequence } \eta_j \downarrow 0.$$

15.6 The certified odd block and exact Feshbach complement

The parity reduction makes clear that an even certificate cannot decide the odd sector. The MP stack supplies a state-locked odd packet

$$V_3 = \operatorname{span}\{u, |x|u, |x|^2 u\}, \qquad E_3 = \operatorname{span}\{V_3, e_3\} \subset X_3^-.$$

For this packet, the certified Weil Gram G_3 on V_3 and the Schur scalar M_3 satisfy

$$\lambda_{\min}(G_3) = 2.0838177120437723 \times 10^{-13} > 0,$$

$$3.08520376681449 \times 10^{-13} < M_3 < 2.993597611951253 \times 10^{-12}.$$

Thus the bordered Weil Gram on E_3 is positive definite. These numerical inequalities belong to one hash-locked continuum packet; they are not promoted to a statement about the complement.

Let P_3 be the X_3-orthogonal projection onto E_3 and $Q_3 = I - P_3$ on X_3^-. Relative to

$$X_3^- = E_3 \oplus Q_3 X_3^-,$$

write

$$W_3^- = \begin{pmatrix} A_3 & C_3^* \\ C_3 & D_3 \end{pmatrix}.$$

The finite certificate gives $A_3 > 0$.

Definition 15.13 (Odd Feshbach complement)**.** Define

$$\boxed{F_3 = D_3 - C_3 A_3^{-1} C_3^*.}$$

Theorem 15.14 (Exact odd Feshbach reduction)**.** *The operator F_3 is bounded and self-adjoint on $Q_3 X_3^-$, and*

$$\boxed{W_3^- \geq 0 \iff F_3 \geq 0.}$$

Consequently

$$\boxed{W_3 \geq 0 \iff W_3^+ \geq 0 \text{ and } F_3 \geq 0.}$$

Proof. The exact factorization is

$$W_3^- = \begin{pmatrix} I & 0 \\ C_3 A_3^{-1} & I \end{pmatrix} \begin{pmatrix} A_3 & 0 \\ 0 & F_3 \end{pmatrix} \begin{pmatrix} I & A_3^{-1} C_3^* \\ 0 & I \end{pmatrix}.$$

The triangular factors are boundedly invertible. Since $A_3 > 0$ on the finite-dimensional space E_3, congruence gives the first equivalence. Combine it with Theorem 15.8. $\square$

For $\eta > 0$, set

$$A_{3,\eta} = A_3 + \eta I_{E_3}, \qquad D_{3,\eta} = D_3 + \eta I_{Q_3 X_3^-},$$

$$F_{3,\eta} = D_{3,\eta} - C_3 A_{3,\eta}^{-1} C_3^*.$$

Then

$$W_3^- + \eta I \geq 0 \iff F_{3,\eta} \geq 0, \qquad F_{3,\eta} \longrightarrow F_3 \quad \text{in norm as } \eta \downarrow 0.$$

15.7 Framework theorem before the RH specialization

Define the reduced Hilbert carrier

$$\mathcal{X}_{\mathrm{red}} = X_3^+ \oplus Q_3 X_3^-$$

and the reduced self-adjoint operator

$$\boxed{\mathbb{W}_0 = W_3^+ \oplus F_3.}$$

Theorem 15.15 (Native recognition-seam Weil carrier)**.** *The construction above has the following exact properties.*

(i) X_3 is a native two-channel Hilbert completion compactly embedded in $L^2(\mathbb{R})$.

(ii) The completed critical Weil form has a bounded self-adjoint representative W_3.

(iii) The primitive reflection cut extends to J_3 and derives the reducing Hilbert decomposition $X_3 = X_3^+ \oplus X_3^-$.

(iv) The certified positive odd block and Feshbach map give

$$W_3 \geq 0 \iff \mathbb{W}_0 \geq 0.$$

(v) The bounded nonnegative operator

$$\boxed{\mathfrak{A}_{\mathrm{SS}} = \Big[-\mathbb{W}_0(I + \mathbb{W}_0^2)^{-1/2}\Big]_+}$$

satisfies

$$\boxed{\mathfrak{A}_{\mathrm{SS}} = 0 \iff \mathbb{W}_0 \geq 0 \iff W_3 \geq 0.}$$

Its support projection is

$$\boxed{\mathbf{1}_{(0,\infty)}(\mathfrak{A}_{\mathrm{SS}}) = \mathbf{1}_{(-\infty,0)}(\mathbb{W}_0).}$$

Proof. Only item (v) remains. Apply continuous functional calculus to

$$\mathfrak{s}_-(x) = \max\left\{0, \frac{-x}{\sqrt{1+x^2}}\right\}.$$

This function vanishes exactly on $[0,\infty)$ and is strictly positive on $(-\infty,0)$. The zero and support statements follow from the spectral theorem. □

Corollary 15.16 (RH as a candidate specialization)**.** *Under the classical completed Weil criterion,*

$$\boxed{\mathrm{RH} \iff W_3 \geq 0 \iff \mathfrak{A}_{\mathrm{SS}} = 0.}$$

This corollary consumes the classical criterion. The framework theorem itself is the operator-geometric equivalence between completed-form positivity and vanishing of the native anti-seam operator.

16 Thermodynamic curvature provenance of the Lambda response

16.1 Purpose and claim boundary

The operator-valued Lambda response used below did not arise as notation detached from the earlier thermodynamic programme. Its structural ancestor is the four-direction response geometry generated by the oriented thermodynamic compass. This section records that provenance, repairs the previously uneven treatment of the s–t pair, and states the precise mathematical bridge that would be required to pass from thermodynamic curvature to the integer seam charge.

The thermodynamic material in this section is a provenance and model-calibration layer. It is not used as a substitute for the native boundary–Gamma–prime construction, and no identity between a thermodynamic scalar and the arithmetic seam integer is assumed.

16.2 The complete thermodynamic response tetrad

Let

$$\mathbf{C}_{\mathrm{th}} = (C_p, C_v, C_s, C_t)$$

be the caloric–mechanical constraint tetrad. Here C_p and C_v are the usual heat capacities, while the signed mechanical responses are

$$C_s = V\left(\frac{\partial P}{\partial V}\right)_S, \qquad C_t = V\left(\frac{\partial P}{\partial V}\right)_T.$$

For stable compressible matter these are negative. The conventional positive adiabatic and isothermal bulk moduli are therefore

$$K_S = -C_s, \qquad K_T = -C_t.$$

On a domain where all four responses are finite and nonzero, define the entropy-scaled thermodynamic Lambda coordinates

$$\boxed{\lambda_X^{\mathrm{th}} = -\frac{ST}{C_X}, \qquad X \in \{p, v, s, t\}.}$$

Thus

$$\boldsymbol{\lambda}_{\mathrm{th}} = \left(-\frac{ST}{C_p}, -\frac{ST}{C_v}, -\frac{ST}{C_s}, -\frac{ST}{C_t}\right)$$

is the completed first response tetrad.

Remark 16.1 (Historical lower-case mechanical coordinates). Earlier thermodynamic drafts also used derivative coordinates such as

$$\widehat{\lambda}_s = V\left(\frac{\partial P}{\partial V}\right)_S, \qquad \widehat{\lambda}_t = -P\left(\frac{\partial V}{\partial P}\right)_T.$$

These are lawful response quantities, but they do not have the same units or common scaling as $\lambda_X^{\mathrm{th}} = -ST/C_X$. The present paper therefore reserves $\boldsymbol{\lambda}_{\mathrm{th}}$ for the uniform entropy-scaled tetrad and does not silently identify it with the historical derivative pair.

16.3 Caloric and mechanical closure ratios

Define the caloric ratio

$$\Gamma_c = \frac{\lambda_p^{\mathrm{th}}}{\lambda_v^{\mathrm{th}}} = \frac{C_v}{C_p}$$

and orient the mechanical ratio by

$$\Gamma_m = \frac{\lambda_t^{\mathrm{th}}}{\lambda_s^{\mathrm{th}}} = \frac{C_s}{C_t} = \frac{K_S}{K_T} = \frac{\kappa_T}{\kappa_S},$$

where $\kappa_T = K_T^{-1}$ and $\kappa_S = K_S^{-1}$. The standard equilibrium identity

$$\frac{C_p}{C_v} = \frac{\kappa_T}{\kappa_S} = \frac{K_S}{K_T}$$

then gives

$$\boxed{\mathcal{I}_{\mathrm{th}} := \Gamma_c \Gamma_m = 1}$$

in the flat equilibrium regime. The chosen orientation of Γ_m is essential: reversing the mechanical quotient replaces Γ_m by its reciprocal and destroys this normalized closure statement.

For positive ratios define the dimensionless scalar departure

$$\delta_{\mathrm{th}} = \log \mathcal{I}_{\mathrm{th}}.$$

The equality $\delta_{\mathrm{th}} = 0$ is the scalar flat calibration. It is a sector projection of the response geometry, not by itself the full curvature tensor.

16.4 Thermodynamics as curvature

Let $x = (x_p, x_v, x_s, x_t) = (P, V, S, T)$ and define the response one-form

$$\omega_{\mathrm{th}} = \sum_{X \in \{p,v,s,t\}} \lambda_X^{\mathrm{th}} \, \mathrm{d}x_X.$$

Its curvature is

$$\boxed{\Omega_{\mathrm{th}} = \mathrm{d}\omega_{\mathrm{th}} = \frac{1}{2} \sum_{X,Y} \left(\partial_X \lambda_Y^{\mathrm{th}} - \partial_Y \lambda_X^{\mathrm{th}} \right) \mathrm{d}x_X \wedge \mathrm{d}x_Y.}$$

Flat response geometry means $\Omega_{\mathrm{th}} = 0$ locally. Nonzero Ω_{th} records the failure of response closure and is the correct geometric object for branch dependence, loop holonomy and memory-bearing transport. The scalar δ_{th} may be used as a constitutive projection of this curvature only after the relevant model proves such a projection law.

16.5 Why $\Gamma_c \Gamma_m$ is not the seam integer

The thermodynamic closure scalar and the seam charge have different mathematical types:

$$\mathcal{I}_{\mathrm{th}} \in \mathbb{R}_{>0}, \qquad \mathfrak{k}_\Sigma(\eta) \in \mathbb{Z}_{\geq 0}.$$

Proposition 16.2 (Type obstruction to a literal identification)**.** *On a connected thermodynamic domain on which $\mathcal{I}_{\mathrm{th}}$ varies continuously, there is no canonical identity*

$$\mathcal{I}_{\mathrm{th}} \equiv \mathfrak{k}_\Sigma(\eta).$$

Any continuous integer-valued function is locally constant, whereas $\mathcal{I}_{\mathrm{th}}$ may vary continuously under constitutive deformation. A relation between the two therefore requires an explicit quantization or index map.

Proof. The codomain $\mathbb{Z}_{\geq 0}$ is discrete. Hence the image of every connected set under a continuous integer-valued map is a singleton. A nonconstant continuous scalar response cannot equal such a map on the whole connected domain. $\square$

16.6 The lawful curvature-to-index bridge

The useful relation is not scalar equality but a response representation followed by threshold index.

Theorem 16.3 (Curvature-driven seam-index variation)**.** *Let $u \mapsto \mathbb{K}_\eta(u)$ be a norm-continuous family of compact self-adjoint operators on one Hilbert space, where u denotes admissible response or curvature data. Let u_s, $0 \le s \le 1$, be a continuous path and assume that 1 is not an eigenvalue of the endpoint operators. Define*

$$\mathbb{D}_s = I - \mathbb{K}_\eta(u_s), \qquad \mathfrak{k}_\Sigma(\eta; u) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta(u)).$$

Then

$$\boxed{\mathfrak{k}_\Sigma(\eta; u_1) - \mathfrak{k}_\Sigma(\eta; u_0) = -\operatorname{sf}(\mathbb{D}_s; 0).}$$

In particular, if u_0 is a flat zero-charge reference, then the terminal seam integer is the negative spectral flow generated along the represented curvature path.

Proof. Every eigenvalue of $\mathbb{K}_\eta(u_s)$ crossing upward through 1 increases the threshold rank by one. The corresponding eigenvalue of $I - \mathbb{K}_\eta(u_s)$ crosses downward through zero and contributes minus one to spectral flow. Summing the signed crossings gives the formula; compactness makes the threshold multiplicities finite. □

Corollary 16.4 (Quantitative no-crossing advantage)**.** *Assume that a flat reference satisfies*

$$\sup \sigma(\mathbb{K}_\eta(u_0)) \le 1 - m_\eta, \qquad m_\eta > 0,$$

and that an explicit curvature-to-operator estimate gives

$$\|\mathbb{K}_\eta(u) - \mathbb{K}_\eta(u_0)\| \le L_\eta \, |\delta_{\mathrm{th}}(u)|.$$

If

$$L_\eta \, |\delta_{\mathrm{th}}(u)| < m_\eta,$$

then

$$\mathfrak{k}_\Sigma(\eta; u) = 0.$$

Proof. The spectral supremum is 1-Lipschitz in operator norm. Therefore

$$\sup \sigma(\mathbb{K}_\eta(u)) < 1 - m_\eta + m_\eta = 1.$$

The threshold projection above one is zero. □

16.7 Specialization and remaining bridge obligation

The native arithmetic programme later constructs

$$\mathbb{K}_\eta = K_{+,\eta} \oplus K^\sharp_{3,\eta}$$

and the single integer charge

$$\mathfrak{k}_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).$$

The theorem above explains the exact form a thermodynamic-curvature interpretation would have to take. What is not yet proved is an intertwining map

$$\boxed{\mathfrak{R}_\eta : \Omega_{\mathrm{th}} \longmapsto \mathbb{K}_\eta}$$

that simultaneously preserves the native reference metric, identifies the flat calibration, and supplies a quantitative norm estimate. Until that map is constructed, the safe hierarchy is

$$\boxed{\text{response tetrad} \longrightarrow \text{connection} \longrightarrow \text{curvature} \longrightarrow \text{compact response} \longrightarrow \text{integer seam charge},}$$

not the unsupported equality $\Gamma_c \Gamma_m \equiv \mathfrak{k}_\Sigma$.

17 The native compact Lambda response on the odd complement

The preceding section constructs the completed-Weil operator directly on the multiplicative carrier and identifies X_3 as its logarithmic chart. We now close the first two odd-complement obligations that were previously left conditional: a positive self-adjoint reference and a compact self-adjoint relative operator.

17.1 Native odd Feshbach reduction

Let

$$E_3^\times = \mathcal{L}^{-1} E_3 \subset \mathscr{X}_{\times,3}^-$$

be the transported four-dimensional certified odd block, let $P_3^\times$ be the native-energy orthogonal projection onto $E_3^\times$, and set

$$Q_3^\times = I - P_3^\times.$$

Relative to

$$\mathscr{X}_{\times,3}^- = E_3^\times \oplus Q_3^\times \mathscr{X}_{\times,3}^-,$$

write

$$W_\times^- = \begin{pmatrix} A_3^\times & (C_3^\times)^* \\ C_3^\times & D_3^\times \end{pmatrix}.$$

Unitary transport of the state-locked interval certificate gives

$$A_3^\times > 0.$$

Define

$$F_3^\times = D_3^\times - C_3^\times (A_3^\times)^{-1} (C_3^\times)^*.$$

The triangular Feshbach congruence gives

$$\boxed{W_\times^- \geq 0 \iff F_3^\times \geq 0.}$$

For $\eta > 0$, put

$$A_{3,\eta}^\times = A_3^\times + \eta I_{E_3^\times},$$

$$F_{3,\eta}^\times = D_3^\times + \eta I - C_3^\times (A_{3,\eta}^\times)^{-1} (C_3^\times)^*.$$

Then

$$W_\times^- + \eta I \geq 0 \iff F_{3,\eta}^\times \geq 0, \qquad F_{3,\eta}^\times \to F_3^\times \quad \text{in norm.}$$

17.2 Positive Gamma reference and compact defect

Split the critical Gamma symbol as

$$G = G_+ - G_-, \qquad G_+ = \max\{G, 0\}, \qquad G_- = \max\{-G, 0\}.$$

Let

$$A_{\Gamma,+}^{\times,-} = (\iota_\times^-)^* \mathcal{M}^{-1} M_{G_+} \mathcal{M} \iota_\times^-$$

be the positive Gamma operator on the odd sector, and let

$$D_{\Gamma,-}^{\times,-} = (\iota_\times^-)^* \mathcal{M}^{-1} M_{G_-} \mathcal{M} \iota_\times^-$$

be the negative Gamma defect. The second operator is compact because G_- is bounded and $\iota_\times^-$ is compact.

Let

$$B_{\mathrm{odd}}^\times = r_{\mathrm{odd}}^\times \otimes r_{\mathrm{odd}}^\times$$

be the positive rank-one operator associated with the odd boundary functional; its contribution to the Weil form has negative sign. Define

$$\boxed{\mathcal{C}_-^\times = D_{\Gamma,-}^{\times,-} + B_{\mathrm{prime}}^{\times,-} + B_{\mathrm{odd}}^\times.}$$

Then $\mathcal{C}_-^\times$ is compact and self-adjoint and

$$\boxed{W_\times^- = A_{\Gamma,+}^{\times,-} - \mathcal{C}_-^\times.}$$

On the native odd complement

$$\mathscr{H}_3^\times = Q_3^\times \mathscr{X}_{\times,3}^-,$$

define

$$\boxed{R_{3,\eta}^{\times,\sharp} = Q_3^\times A_{\Gamma,+}^{\times,-} Q_3^\times + \eta I_{\mathscr{H}_3^\times}.}$$

Also define

$$\boxed{\begin{aligned} G_{3,\eta}^{\times,\sharp} &= Q_3^\times \mathcal{C}_-^\times Q_3^\times \\ &\qquad + C_3^\times (A_{3,\eta}^\times)^{-1} (C_3^\times)^*. \end{aligned}}$$

Theorem 17.1 (Self-adjoint complement reference)**.** *For every $\eta > 0$, $R_{3,\eta}^{\times,\sharp}$ is bounded, positive and self-adjoint on $\mathscr{H}_3^\times$, and*

$$\boxed{R_{3,\eta}^{\times,\sharp} \geq \eta I.}$$

Hence it is boundedly invertible and has a bounded positive inverse square root.

Proof. The compressed positive Gamma operator is bounded, positive and self-adjoint. Adding ηI gives the stated coercive lower bound. □

Theorem 17.2 (Compact odd relative operator)**.** *For every $\eta > 0$, $G_{3,\eta}^{\times,\sharp}$ is compact and self-adjoint, and*

$$\boxed{F_{3,\eta}^\times = R_{3,\eta}^{\times,\sharp} - G_{3,\eta}^{\times,\sharp}.}$$

Consequently

$$\boxed{\mathbf{\Lambda}_{3,\eta}^{(1)} = (R_{3,\eta}^{\times,\sharp})^{-1/2} G_{3,\eta}^{\times,\sharp} (R_{3,\eta}^{\times,\sharp})^{-1/2}}$$

is compact and self-adjoint and satisfies

$$\boxed{F_{3,\eta}^\times = (R_{3,\eta}^{\times,\sharp})^{1/2} (I - \mathbf{\Lambda}_{3,\eta}^{(1)}) (R_{3,\eta}^{\times,\sharp})^{1/2}.}$$

Thus

$$\boxed{F_{3,\eta}^\times \geq 0 \iff \sup \sigma(\mathbf{\Lambda}_{3,\eta}^{(1)}) \leq 1.}$$

Proof. The first summand of $G_{3,\eta}^{\times,\sharp}$ is a compression of a compact self-adjoint operator. The second has range contained in $\operatorname{Ran} C_3^\times$, which is finite dimensional because the source $E_3^\times$ is finite dimensional. Hence the second summand is finite rank. The exact identity follows by inserting

$$D_3^\times = Q_3^\times A_{\Gamma,+}^{\times,-} Q_3^\times - Q_3^\times \mathcal{C}_-^\times Q_3^\times$$

into the shifted Feshbach formula. Sandwiching the compact defect by the bounded inverse square root proves compactness of $\mathbf{\Lambda}_{3,\eta}^{(1)}$, and the sign equivalence follows by boundedly invertible congruence. □

No trace-class assertion is required for the sign theorem and none is made here.

17.3 Lambda response and the logarithmic chart

The notation $\mathbf{\Lambda}_{3,\eta}^{(1)}$ records the first normalized operator-valued response of the shifted odd seam. Its mathematical content is the exact identity

$$F_{3,\eta}^\times = R^{1/2}(I - \mathbf{\Lambda}_{3,\eta}^{(1)})R^{1/2};$$

the notation alone adds no positivity claim.

Let $K_{3,\eta}^\sharp$ denote the logarithmic-chart relative operator. The unitary representation theorem gives

$$\boxed{\mathcal{L}\mathbf{\Lambda}_{3,\eta}^{(1)}\mathcal{L}^{-1} = K_{3,\eta}^\sharp.}$$

Hence compactness, spectrum, threshold multiplicity and sign are identical in the native and logarithmic descriptions.

Proposition 17.3 (First response derivative)**.** *The shifted Feshbach complement is norm differentiable and*

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\eta} F_{3,\eta}^\times = I + C_3^\times (A_3^\times + \eta I)^{-2} (C_3^\times)^* \geq I.}$$

In particular, $F_{3,\eta}^\times$ is increasing in η.

Proof. Differentiate the resolvent term using

$$\frac{\mathrm{d}}{\mathrm{d}\eta}(A_3^\times + \eta I)^{-1} = -(A_3^\times + \eta I)^{-2}.$$

□

17.4 Threshold charge and a large-shift anchor

Because $\mathbf{\Lambda}_{3,\eta}^{(1)}$ is compact and self-adjoint, define the odd threshold charge for every $\eta > 0$ by

$$\boxed{\mathfrak{k}_3(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbf{\Lambda}_{3,\eta}^{(1)}).}$$

It is finite and

$$\mathfrak{k}_3(\eta) = 0 \iff F_{3,\eta}^\times \geq 0.$$

The following elementary norm bound provides a genuine zero-charge starting ray.

Theorem 17.4 (Large-shift odd no-crossing anchor)**.** *The operator bounds*

$$\|\mathcal{C}_-^\times\| \le \frac{901}{120}, \qquad \|C_3^\times\| \le \frac{1021}{120}$$

imply

$$\boxed{\|\mathbf{\Lambda}_{3,\eta}^{(1)}\| \le \beta(\eta) := \frac{901}{120\eta} + \frac{1021^2}{120^2\eta^2}.}$$

The positive solution of $\beta(\eta) = 1$ is

$$\eta_* = 13.053924914719259\ldots.$$

Therefore

$$\boxed{\eta \ge 14 \Longrightarrow \mathfrak{k}_3(\eta) = 0 \text{ and } F_{3,\eta}^\times > 0.}$$

At $\eta = 14$,

$$\beta(14) = \frac{2556121}{2822400} = 0.905655116213152\ldots,$$

so the threshold reserve exceeds 0.0943*.*

Proof. Since $R_{3,\eta}^{\times,\sharp} \ge \eta I$,

$$\|\mathbf{\Lambda}_{3,\eta}^{(1)}\| \le \eta^{-1}\|\mathcal{C}_-^\times\| + \eta^{-2}\|C_3^\times\|^2.$$

The displayed constants give the majorant. It is decreasing for $\eta > 0$, and direct evaluation gives the stated threshold and anchor. □

The difficult direction is therefore a controlled descent from the zero-charge ray toward $\eta = 0$, not the existence of the odd compact operator.

18 Spectral blindness, native reconstruction and the publication freeze

The compact response operator is now constructed. The remaining issue is not whether a finite calculation can produce a subcritical eigenvalue, but whether the finite representation sees every target-relevant native direction. This section makes that obligation explicit and records the exact point at which the present paper is frozen.

18.1 Target-relevant spectral blindness

Let $R : \mathscr{H} \to \mathscr{Y}$ be a representation and let $\Pi : \mathscr{H} \to \mathscr{Z}$ be the declared target map.

Definition 18.1 (Target-relevant blind quotient)**.** Define

$$\boxed{\mathfrak{B}(R;\Pi) = \frac{\ker R}{\ker R \cap \ker \Pi}.}$$

The representation is target faithful exactly when $\mathfrak{B}(R;\Pi) = 0$.

Theorem 18.2 (Unitary invariance of blindness)**.** *Let $U : \mathscr{H} \to \widetilde{\mathscr{H}}$ and $V : \mathscr{Y} \to \widetilde{\mathscr{Y}}$ be unitary, and transport*

$$\widetilde{R} = VRU^{-1}, \qquad \widetilde{\Pi} = \Pi U^{-1}.$$

Then U induces a canonical isomorphism

$$\boxed{\mathfrak{B}(R; \Pi) \cong \mathfrak{B}(\widetilde{R}; \widetilde{\Pi}).}$$

For a bounded operator A and its unitary chart $\widetilde{A} = UAU^{-1}$, kernel, spectrum, spectral multiplicities, positivity, compactness and Fredholm status are preserved.

Proof. The identities

$$\ker \widetilde{R} = U \ker R, \qquad \ker \widetilde{\Pi} = U \ker \Pi$$

give the quotient isomorphism. The operator statements are standard consequences of unitary equivalence. □

Three consequences are immediate.

(i) The excluded diagonal paired-prime carrier remains zero-blind in every unitary chart. A change of language cannot create the missing noninvertibility event.

(ii) The positive block $E_3^\times$ remains blind to $Q_3^\times \mathscr{X}_{\times,3}^-$. The Feshbach complement is the lawful lift closing that blind kernel.

(iii) The odd carrier remains blind to the independent even sector. No odd no-crossing theorem can replace $W_3^+ \geq 0$.

Thus the native framework is not justified by renaming classical coordinates. It is justified by constructing the carrier, cut, flow, energy and compact response before introducing the chart, while preserving every genuine theorem and every genuine obstruction under the explicit intertwiner.

18.2 Reference-whitened blind-kernel Schur transfer

Let

$$F = F^*, \qquad R = R^* \geq r_0 I > 0$$

act on a Hilbert space $\mathscr{H}$, and define the reference-whitened operator

$$\widehat{F} = R^{-1/2} F R^{-1/2}.$$

Let

$$J_N : \mathscr{H}_N \longrightarrow \mathscr{H}$$

be an isometry, and put

$$P_N = J_N J_N^*, \qquad Q_N = I - P_N.$$

The observation J_N^* has exact blind kernel

$$\ker J_N^* = Q_N \mathscr{H}.$$

Suppose a finite self-adjoint operator $\widehat{F}_N$ satisfies

$$\widehat{F}_N \geq \mu I_{\mathscr{H}_N}$$

for some $\mu > 0$. Define the visible transfer error

$$\varepsilon_{\mathrm{vis}} = \left\| P_N \widehat{F} P_N - J_N \widehat{F}_N J_N^* \right\|, \qquad a = \mu - \varepsilon_{\mathrm{vis}}.$$

Assume further that

$$Q_N \widehat{F} Q_N \geq \delta Q_N$$

with $\delta > 0$, and put

$$\beta = \|Q_N \widehat{F} P_N\|.$$

Theorem 18.3 (Native blind-kernel Schur closure)**.** *Under the preceding hypotheses,*

$$\boxed{F \geq \lambda_- R,}$$

where

$$\boxed{\lambda_- = \frac{a + \delta - \sqrt{(a-\delta)^2 + 4\beta^2}}{2}.}$$

In particular,

$$\boxed{\varepsilon_{\mathrm{vis}} + \frac{\beta^2}{\delta} < \mu \quad \Longrightarrow \quad F > 0.}$$

Proof. For $u = p + q$ with $p = P_N u$ and $q = Q_N u$, set

$$x = \|p\|, \qquad y = \|q\|.$$

The finite transfer, blind-complement lower bound and cross estimate give

$$\langle \widehat{F} u, u \rangle \geq a x^2 + \delta y^2 - 2\beta x y.$$

The least eigenvalue of

$$\begin{pmatrix} a & -\beta \\ -\beta & \delta \end{pmatrix}$$

is λ_-. Hence $\widehat{F} \geq \lambda_- I$, and congruence by $R^{1/2}$ gives $F \geq \lambda_- R$. The scalar inequality is the sufficient Schur condition $a - \beta^2/\delta > 0$. □

Every constant in this theorem is invariant under unitary transport. It may therefore be evaluated in a logarithmic or Fourier chart, but the final statement belongs to the native operator only after the intertwining and blind-complement estimates are included.

18.3 Visible matrix transfer

Let $S_N : \mathbb{C}^{m_N} \to \mathscr{H}$ be a finite synthesis map and define

$$R_N = S_N^* R S_N, \qquad F_N = S_N^* F S_N.$$

After deleting $\ker R_N$, set

$$\boxed{J_N = R^{1/2} S_N R_N^{-1/2}.}$$

Then $J_N^* J_N = I$ and

$$J_N^* (I - R^{-1/2} F R^{-1/2}) J_N = I - R_N^{-1/2} F_N R_N^{-1/2}.$$

Let $\widetilde{F}_N, \widetilde{R}_N$ be computed self-adjoint matrices and assume

$$\varepsilon_F = \left\| R_N^{-1/2} (F_N - \widetilde{F}_N) R_N^{-1/2} \right\|,$$

$$\varepsilon_R = \left\| R_N^{-1/2} (R_N - \widetilde{R}_N) R_N^{-1/2} \right\| < 1.$$

Put

$$\widetilde{\lambda}_N = \lambda_{\min}(\widetilde{F}_N, \widetilde{R}_N), \qquad \widetilde{a}_N = 1 - \widetilde{\lambda}_N.$$

Proposition 18.4 (Visible generalized-eigenvalue transfer)**.** *If* $\widetilde{\lambda}_N \geq 0$*, then*

$$\boxed{\lambda_{\max}\Big(J_N^*(I - R^{-1/2}FR^{-1/2})J_N\Big) \leq \widetilde{a}_N + (1-\widetilde{a}_N)\varepsilon_R + \varepsilon_F.}$$

Proof. The reference error gives

$$\widetilde{R}_N \geq (1-\varepsilon_R)R_N.$$

Since $\widetilde{F}_N \geq \widetilde{\lambda}_N \widetilde{R}_N$ and

$$F_N \geq \widetilde{F}_N - \varepsilon_F R_N,$$

we obtain

$$F_N \geq \big[\widetilde{\lambda}_N(1-\varepsilon_R) - \varepsilon_F\big]R_N.$$

Taking one minus the resulting relative lower bound proves the estimate. □

18.4 Native channel reconstruction

The native norm defines the isometric analysis map

$$\mathcal{C}_\times F = \left(\max(r^{3/2}, r^{-3/2})F, \frac{\sqrt{w_\Gamma}}{\sqrt{2\pi}}\mathcal{M}F\right)$$

from $\mathscr{X}_{\times,3}$ into

$$\mathcal{Y}_\times = L^2((0,\infty), \mathrm{d}^\times r) \oplus L^2(\mathbb{R}, \mathrm{d}\xi).$$

Thus

$$\mathcal{C}_\times^* \mathcal{C}_\times = I.$$

Restrict this map to the native odd complement $\mathscr{H}_3^\times$.

Let Π_N be any finite-rank orthogonal projection on the channel space and define

$$Z_N = Q_3^\times \mathcal{C}_\times^*|_{\operatorname{Ran}\Pi_N}, \qquad H_N = Z_N^* Z_N.$$

After restricting to $(\ker H_N)^\perp$, define

$$\boxed{J_N = Z_N H_N^{-1/2}.}$$

Then $J_N^* J_N = I$, and $P_N = J_N J_N^*$ is the native orthogonal projection onto the reconstructed range. Unitary rotation of the finite channel coordinates does not change P_N.

Let $\mathbf{\Lambda}$ be a compact self-adjoint native response operator and put

$$\mathbf{\Lambda}_N = J_N^* \mathbf{\Lambda} J_N.$$

Define the compatible channel tail

$$\boxed{\tau_N = \|(I-\Pi_N)\mathcal{C}_\times \mathbf{\Lambda}\|.}$$

Theorem 18.5 (Compatible response-tail gate)**.** *With* $Q_N = I - P_N$*,*

$$\boxed{\|Q_N \mathbf{\Lambda}\| \leq \tau_N.}$$

Consequently

$$\|Q_N \mathbf{\Lambda} P_N\| \leq \tau_N, \qquad \sup\sigma(Q_N \mathbf{\Lambda} Q_N) \leq \tau_N.$$

If a finite matrix $\widetilde{\Lambda}_N$ satisfies

$$\|\mathbf{\Lambda}_N - \widetilde{\Lambda}_N\| \le \varepsilon_{\mathrm{vis}},$$

and

$$a_N = \lambda_{\max}(\widetilde{\Lambda}_N) + \varepsilon_{\mathrm{vis}},$$

then

$$\boxed{\sup \sigma(\mathbf{\Lambda}) \le \mathcal{U}(a_N, \tau_N),}$$

where

$$\boxed{\mathcal{U}(a,\tau) = \frac{a + \tau + \sqrt{(a-\tau)^2 + 4\tau^2}}{2}.}$$

Proof. Since $\mathcal{C}_\times^* \mathcal{C}_\times = I$ and $Q_N Z_N = 0$,

$$Q_N \mathbf{\Lambda} = Q_N \mathcal{C}_\times^* (I - \Pi_N) \mathcal{C}_\times \mathbf{\Lambda}.$$

Both Q_N and $\mathcal{C}_\times^*$ are contractions, proving the tail bound. The spectral estimate is the largest eigenvalue of the 2×2 upper matrix with diagonal bounds a_N, τ_N and off-diagonal bound τ_N. □

This theorem forbids replacing a compatible operator-response tail by an unrelated coefficient tail.

18.5 Exact native standing-wave frame

Fix $L > 0$. For $k \ge 1$, define

$$\boxed{\Psi_k^\times(r) = \frac{1}{\sqrt{L}} \max(r^{3/2}, r^{-3/2})^{-1} \sin\left(\frac{k\pi \log r}{L}\right) \mathbf{1}_{\{|\log r| \le L\}}.}$$

These atoms are defined on the multiplicative carrier. Their logarithmic sine functions are representation coordinates.

Theorem 18.6 (Seam-interval standing-wave frame)**.** *For every $k, j \ge 1$,*

$$\Psi_k^\times(r^{-1}) = -\Psi_k^\times(r),$$

$$\boxed{\int_0^\infty \max(r^3, r^{-3}) \Psi_k^\times(r) \overline{\Psi_j^\times(r)}\, \mathrm{d}^\times r = \delta_{kj}.}$$

The family is complete in the inversion-odd spatial channel on the declared scale interval $|\log r| \le L$.

Proof. Put $x = \log r$. After cancellation of the native weight, the atoms become

$$L^{-1/2} \sin(k\pi x/L)$$

on $[-L, L]$. Oddness and orthonormality are the standard sine identities on the symmetric interval, and completeness is the completeness of the Dirichlet sine system in the odd subspace. □

In the logarithmic chart,

$$\psi_k(x) = \frac{1}{\sqrt{L}} e^{-3|x|/2} \sin(k\pi x/L) \mathbf{1}_{[-L,L]}(x).$$

Put

$$a = \frac{3}{2}, \qquad \omega_k = \frac{k\pi}{L},$$

and

$$J_L(d) = \frac{a + e^{-aL}\big[-a\cos(dL) + d\sin(dL)\big]}{a^2 + d^2}.$$

A direct half-line integration gives

$$\boxed{\widehat{\psi}_k(z) = \frac{i}{\sqrt{L}}\big[J_L(z-\omega_k) - J_L(z+\omega_k)\big].}$$

Thus the Gamma channel and the two strip-boundary prime channels have explicit continuum matrix elements. Because the atoms vanish at $\pm L$ and have exponentially suppressed derivative jumps there, two integrations by parts give a squared Gamma-frequency tail of order

$$O(\Omega^{-3}\log\Omega).$$

18.6 Directed finite-grid audit and standing-wave pilot

The first shifted odd calculation uses

$$\eta = 10^{-2}, \qquad P = 10^6, \qquad L = 40, \qquad N = 16384.$$

The state-locked floating-point Schur audit uses the matrix

$$M_N = F_{P,N} - 0.05R_{\eta,N}.$$

Its central block has computed minimum 0.002301110703920909, the outward working lower value is 0.002, the exterior working lower value is 0.009499998398583278, and the working cross bound is 0.00125357. The resulting directed Schur value is

$$0.0018345854726528853 > 0.$$

Together with the normalized omitted-prime and Gamma tail

$$\varepsilon_{\mathrm{op}} = 0.0029631022229691536,$$

the computation reports the finite-grid upper

$$0.9529631022229692 < 1.$$

The central matrix was checked by ordinary floating-point eigensolution and Cholesky reconstruction. A full outward interval enclosure of every assembled matrix entry and factorization is not supplied here. Accordingly, these values are recorded as a directed finite-grid numerical audit, not as a finite theorem and not as a continuum theorem.

The native standing-wave family removes the earlier box-tail and sampled-Gram defects. The state-locked shadow computation, retaining the exact certified G_3 block, gives

modes	accepted-complement rank	top response
40	40	0.7188463455004981
80	79	0.7497160611315330
160	159	0.8948401658035906
240	238	0.9234744083669618
320	318	0.9397231078506526

After final positive-reference re-orthonormalization,

$$\boxed{\widetilde{a}_{320} = 0.9397231078506517,}$$

with Rayleigh quotient 0.9397231078506516 and directional response residual

$$0.008432756180832257.$$

These values are recorded as a state-locked numerical shadow pilot. A directional residual is not the global compatible tail τ_{320}.

Let ε_R and ε_F be directed relative errors for the exact continuum reference and Feshbach matrices on the standing-wave synthesis. By Theorem 18.4,

$$\boxed{a_{320}^{\mathrm{cert}} \leq 0.9397231078506517 + 0.0602768921493483\,\varepsilon_R + \varepsilon_F.}$$

Therefore the first native continuum odd decision is reduced to

$$\boxed{0.0029631022229691536 + \mathcal{U}(0.9397231078506517 + 0.0602768921493483\,\varepsilon_R + \varepsilon_F, \tau_{\mathrm{prime},320}) < 1.}$$

If $\varepsilon_R = \varepsilon_F = 0$, the allowable compatible retained-prime tail is

$$\tau_{\mathrm{prime},320} < 0.21210269661515813.$$

18.7 Publication freeze and final stage

The present manuscript is frozen at the following boundary:

native multiplicative carrier and EMK cut	proved,
Euler-scale flow and Mellin representation	proved,
native energy completion and compact embedding	proved,
unitary logarithmic chart $\mathscr{X}_{\times,3} \simeq X_3$	proved,
native completed-Weil representative	proved,
native compact odd Lambda response for every $\eta > 0$	proved,
odd zero-charge anchor for $\eta \geq 14$	proved,
spectral-blindness invariance	proved,
blind-kernel Schur and compatible-tail gates	proved,
native standing-wave frame	proved,
pinned finite-grid Schur calculation	directed numerical audit,
state-locked standing-wave compression	computed; not promoted,
outward interval finite-matrix enclosure	open,
continuum $\eta = 10^{-2}$ odd no-crossing	open,
controlled descent $\eta \downarrow 0$	open,
global even no-crossing	open,
completed charge-vanishing/no-leakage theorem	open,
Riemann hypothesis	`ABSTAIN`.

The final analytic stage is therefore sharply defined: first enclose the finite matrices outward; then certify ε_R, ε_F and $\tau_{\mathrm{prime},320}$ on the native standing-wave reconstruction; close the odd continuum inequality at a decreasing sequence of shifts; independently close the even sector; and only then invoke the classical Weil criterion.

Part IV

Unified Terminal Obstruction and Bilateral Closure

19 Lifted seam charge on the reduced native carrier

The preceding section constructs the native completed-Weil carrier and proves the exact bounded anti-seam equivalence

$$\mathfrak{A}_{\mathrm{SS}} = 0 \iff W_3 \geq 0.$$

No compactness hypothesis is needed for that operator-valued formulation. Compactness enters only when the obstruction is lifted to a finite integer threshold count and represented by spectral flow.

19.1 The shifted reduced carrier

For $\eta > 0$, define

$$\boxed{\mathbb{W}_\eta = (W_3^+ + \eta I_{X_3^+}) \oplus F_{3,\eta}.}$$

Proposition 19.1 (Shifted global equivalence)**.** *For every $\eta > 0$,*

$$\boxed{W_3 + \eta I \geq 0 \iff \mathbb{W}_\eta \geq 0.}$$

Moreover,

$$\mathbb{W}_\eta \longrightarrow \mathbb{W}_0 \qquad \textit{in operator norm as } \eta \downarrow 0.$$

Proof. Parity reduces $W_3 + \eta I$ into its shifted even and odd sectors. The shifted Feshbach theorem replaces the odd condition by $F_{3,\eta} \geq 0$. Since $A_3 > 0$ on the finite-dimensional space E_3,

$$(A_3 + \eta I)^{-1} \longrightarrow A_3^{-1}$$

in norm, so $F_{3,\eta} \to F_3$ in norm. The even block converges trivially. □

Define

$$\mathfrak{A}_{\mathrm{SS},\eta} = \left[-\mathbb{W}_\eta (I + \mathbb{W}_\eta^2)^{-1/2}\right]_+.$$

Then

$$\mathfrak{A}_{\mathrm{SS},\eta} = 0 \iff W_3 + \eta I \geq 0.$$

Consequently,

$$\boxed{W_3 \geq 0 \iff \mathfrak{A}_{\mathrm{SS},\eta_j} = 0 \text{ for every member of one sequence } \eta_j \downarrow 0.}$$

The reverse implication follows directly from

$$\langle W_3 x, x\rangle \geq -\eta_j \|x\|^2$$

and passage to the limit. No additional abstract completion axiom is required for this exact operator statement. Finite numerical certificates still require their own uniform tail and projection controls.

19.2 Compact relative packets

The even sector already has the exact shifted factorization

$$W_3^+ + \eta I = A_{+,\eta}^{1/2}(I - K_{+,\eta})A_{+,\eta}^{1/2},$$

where

$$A_{+,\eta} = A_0 + \eta I \geq \eta I$$

and $K_{+,\eta}$ is compact and self-adjoint.

For the odd complement, assume that for the selected $\eta > 0$ there is a bounded self-adjoint reference

$$R_{3,\eta} \geq c_{3,\eta} I > 0$$

on $Q_3 X_3^-$ such that

$$K_{3,\eta} = R_{3,\eta}^{-1/2}(R_{3,\eta} - F_{3,\eta})R_{3,\eta}^{-1/2}$$

is compact and self-adjoint. This compact odd-reference theorem remains open for the actual native complement.

Set

$$\mathbb{A}_\eta = A_{+,\eta} \oplus R_{3,\eta}, \qquad \mathbb{K}_\eta = K_{+,\eta} \oplus K_{3,\eta}.$$

Then

$$\boxed{\mathbb{W}_\eta = \mathbb{A}_\eta^{1/2}(I - \mathbb{K}_\eta)\mathbb{A}_\eta^{1/2}.}$$

19.3 The lifted integer obstruction

Definition 19.2 (Strict lifted seam charge)**.** Under the compact relative packet, define

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).}$$

Its sector contributions are

$$\mathfrak{k}_+(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(K_{+,\eta}), \qquad \mathfrak{k}_3(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(K_{3,\eta}).$$

Because $\mathbb{K}_\eta$ is compact and self-adjoint, the spectral subspace above the fixed threshold 1 is finite dimensional.

Theorem 19.3 (Additive charge and exact shifted sign)**.** *Under the compact relative packet,*

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_+(\eta) + \mathfrak{k}_3(\eta).}$$

Furthermore,

$$\boxed{\mathfrak{k}_\Sigma(\eta) = 0 \iff \mathbb{W}_\eta \geq 0 \iff W_3 + \eta I \geq 0.}$$

In particular, the two sectors cannot cancel:

$$\mathfrak{k}_\Sigma(\eta) = 0 \iff \mathfrak{k}_+(\eta) = \mathfrak{k}_3(\eta) = 0.$$

Proof. Spectral projections respect direct sums, giving additivity. Since $\mathbb{A}_\eta^{1/2}$ is boundedly invertible,

$$\mathbb{W}_\eta \geq 0 \iff I - \mathbb{K}_\eta \geq 0 \iff \sigma(\mathbb{K}_\eta) \cap (1, \infty) = \varnothing.$$

This is exactly the zero-charge condition. The final equivalence is Theorem 19.1. □

19.4 One-sided spectral flow

Consider

$$\mathbb{D}_{\eta,t} = I - t\mathbb{K}_\eta, \qquad 0 \le t \le 1.$$

For an eigenvalue μ of $\mathbb{K}_\eta$, the associated branch is $1 - t\mu$. An interior crossing occurs exactly when $\mu > 1$, at $t = 1/\mu$, and every such crossing is downward because the derivative is $-\mu < 0$.

Theorem 19.4 (Charge as one-sided spectral flow)**.** *Assume $\mathbb{K}_\eta$ is compact and self-adjoint. If $1 \notin \sigma(\mathbb{K}_\eta)$, then*

$$\boxed{\operatorname{sf}(\mathbb{D}_{\eta,t}; 0) = -\mathfrak{k}_\Sigma(\eta).}$$

In general, for positive ε avoiding endpoint spectrum,

$$\operatorname{sf}((1+\varepsilon)I - t\mathbb{K}_\eta; 0) = -\operatorname{rank} \mathbf{1}_{(1+\varepsilon,\infty)}(\mathbb{K}_\eta),$$

and hence

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \lim_{\varepsilon \downarrow 0} -\operatorname{sf}((1+\varepsilon)I - t\mathbb{K}_\eta; 0).}$$

The one-sided orientation is essential. A zero spectral flow here means there are no threshold crossings; it cannot hide cancellation between upward and downward events.

19.5 Surya angle on the relative spectrum

For a relative eigenvalue μ, define

$$\delta_{\mathrm{SS}}(\mu) = \arctan(\mu - 1), \qquad c_{\mathrm{SS}}(\mu) = \sin \delta_{\mathrm{SS}}(\mu) = \frac{\mu - 1}{\sqrt{1 + (\mu-1)^2}}.$$

The operator-valued chart

$$\mathfrak{C}_{\mathrm{SS},\eta} = \sin(\arctan(\mathbb{K}_\eta - I))$$

satisfies

$$\boxed{\mathbf{1}_{(0,\infty)}(\mathfrak{C}_{\mathrm{SS},\eta}) = \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).}$$

Thus the relative Surya amplitude, the threshold anti-seam support, the lifted charge and the one-sided spectral flow are equivalent presentations of the same compact obstruction.

19.6 Conditional terminal theorem

Theorem 19.5 (Conditional lifted-charge specialization)**.** *Assume that a compact odd relative packet exists for every member of one sequence $\eta_j \downarrow 0$. Then*

$$\boxed{W_3 \ge 0 \iff \mathfrak{k}_\Sigma(\eta_j) = 0 \text{ for every } j.}$$

Under the classical completed Weil criterion, this becomes

$$\boxed{\mathrm{RH} \iff \mathfrak{k}_\Sigma(\eta_j) = 0 \text{ for every } j.}$$

Equivalently, the corresponding one-sided regularized spectral flows vanish for every j.

Proof. If $W_3 \ge 0$, then $W_3 + \eta_j I \ge 0$ for every j, so Theorem 19.3 gives zero charge. Conversely, zero charge gives $W_3 + \eta_j I \ge 0$. Taking quadratic forms and letting $j \to \infty$ gives $W_3 \ge 0$. The RH statement is the classical Weil-criterion specialization. □

The theorem is conditional only because the actual odd compact relative packet has not been constructed. It does not require a new determinant–ξ identity: it consumes the completed Weil form already represented by W_3.

19.7 Why principal holonomy is insufficient

A principal phase cannot replace the lifted charge. Since

$$e^{2\pi i m} = 1 \qquad (m \in \mathbb{Z}),$$

trivial $U(1)$ holonomy is compatible with any integer winding. The terminal theorem must therefore control

$$\mathfrak{k}_\Sigma(\eta)$$

or the full spectral-flow lift, not merely its exponential image.

19.8 Remaining analytic gates

The native framework and bounded anti-seam operator are established. The remaining compact-charge programme has four exact obligations:

G1. construct a boundedly coercive odd reference $R_{3,\eta}$ producing compact self-adjoint $K_{3,\eta}$;
G2. certify the even and odd threshold counts on the same shifted sequence;
G3. provide projection-consistent finite-to-infinite tail bounds for every numerical charge certificate;
G4. prove an EMK admissibility or no-leakage theorem forcing the global lifted charge to vanish.

The current repository proves the framework equivalences but not the final vanishing statement. The Riemann hypothesis therefore remains `ABSTAIN`.

20 Native threshold spectral–zeta correspondence

The word "spectral" in this section refers to the threshold spectrum of the native compact response $\mathbb{K}_\eta$. It does not mean that the ordinates of the nontrivial zeros are asserted to be eigenvalues of one fixed self-adjoint operator, and it does not introduce a spectral-zeta regularization $\mathrm{Tr}(A^{-s})$. The correspondence proved here is the exact chain linking native threshold vanishing, completed-Weil positivity and the absence of off-critical zeros.

Let $\mathscr{A}_\times$ denote the transported admissible Weil core in the native multiplicative completion. It is dense in $\mathscr{X}_{\times,3}$, and the bounded native Weil form extends from this core to the self-adjoint representative $W_\times$.

Definition 20.1 (Controlled seam-vanishing sequence)**.** A sequence $\eta_j \downarrow 0$ is called a controlled seam-vanishing sequence if the global compact packets $\mathbb{K}_{\eta_j}$ are defined on the same completed native architecture and

$$\mathfrak{k}_\Sigma(\eta_j) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_{\eta_j}) = 0 \qquad \text{for every } j.$$

Theorem 20.2 (Native threshold spectral–zeta correspondence)**.** *For every sequence $\eta_j \downarrow 0$, the following statements are equivalent:*

(i) $\mathfrak{k}_\Sigma(\eta_j) = 0$ *for every j;*
(ii) $W_\times \geq 0$*;*
(iii)

$$\mathfrak{W}_\times(F, F) \geq 0 \qquad (F \in \mathscr{A}_\times);$$

(iv) $\zeta(s)$ *has no nontrivial zero with* $\Re(s) > \frac{1}{2}$*;*
(v) the Riemann hypothesis holds.

Equivalently,

$$\boxed{\mathfrak{k}_\Sigma(\eta_j) = 0\ \forall j \iff W_\times \geq 0 \iff \zeta(s) \neq 0 \textit{ for every nontrivial } s \textit{ with } \Re(s) > \tfrac{1}{2} \iff \text{RH.}}$$

Proof. The equivalence of (i) and (ii) is the native Weil integer seam-path theorem, using norm convergence of the shifted global carrier as $\eta_j \downarrow 0$. The equivalence of (ii) and (iii) follows from the bounded Riesz representation of the native Weil form and density of the admissible core.

The classical completed explicit formula identifies the native form on $\mathscr{A}_\times$ with the completed Weil form. The classical Weil criterion therefore gives the equivalence of (iii) and (v); this is the classical-shadow membrane and is not re-claimed as a new proof of the explicit formula or of Weil's criterion.

It remains only to compare (iv) and (v). If RH holds, there is plainly no nontrivial zero to the right of the critical line. Conversely, suppose that there is no nontrivial zero with real part greater than 1/2. Every nontrivial zero lies in the critical strip. If one had real part less than 1/2, the functional equation $\xi(s) = \xi(1-s)$ would produce a reflected zero with real part greater than 1/2, a contradiction. Hence every nontrivial zero lies on the critical line. □

Corollary 20.3 (Seam vanishing implies the Riemann hypothesis)**.** *If a seam-vanishing theorem constructs one controlled sequence $\eta_j \downarrow 0$, then $\zeta(s)$ has no nontrivial zero with $\Re(s) > \frac{1}{2}$. By the functional equation, every nontrivial zero lies on $\Re(s) = \frac{1}{2}$, and therefore* RH *holds.*

Remark 20.4 (What is and is not being claimed). The theorem is a threshold correspondence through the completed-Weil sign problem. It does not assert a Hilbert–Pólya formula, an equality between zero ordinates and eigenvalues, or a multiplicity-preserving Fredholm divisor identity. A stronger direct spectral-divisor theorem would require a holomorphic operator pencil $K_\xi(s)$, a licensed regularized determinant and a nowhere-vanishing factor $g(s)$ such that

$$\det_p(I - K_\xi(s)) = g(s)\xi(s),$$

together with exclusion of extra, missing and polluted zeros. None of those stronger assertions is consumed here.

Claim status. The correspondence theorem is an exact packaging of the proved native realization, shifted sign equivalence, compact threshold charge and the classical Weil criterion. The premise of Theorem 20.3 is not yet established: construction of a controlled seam-vanishing sequence remains the active terminal obligation. Accordingly, the Riemann hypothesis remains `ABSTAIN`.

21 Threshold spectral zeta and the seam-index trinity

The preceding correspondence theorem shows that a controlled seam-vanishing sequence implies the Riemann hypothesis. We now give the seam integer a second, strictly spectral presentation. This does not by itself prove that the integer vanishes; it creates an independent analytic target whose vanishing is exactly the same terminal event.

Throughout this section fix $\eta > 0$ and write

$$K = \mathbb{K}_\eta, \qquad P_> = \mathbf{1}_{(1,\infty)}(K).$$

Because K is compact and self-adjoint, $P_>$ has finite rank. Put

$$\boxed{A_\eta = (K - I)P_>.}$$

On $P_>\mathscr{H}$, the operator A_η is positive and invertible. If $P_> = 0$, every spectral-zeta function below is defined to be identically zero.

Definition 21.1 (Threshold spectral zeta)**.** For $s \in \mathbb{C}$, define

$$\boxed{\zeta^{\mathrm{th}}_{\Sigma,\eta}(s) = \mathrm{Tr}_{P_> \mathscr{H}}(A_\eta^{-s}) = \sum_{\lambda \in \sigma(K),\, \lambda > 1} m(\lambda)(\lambda - 1)^{-s}.}$$

The positive real logarithm is used in the complex power.

Theorem 21.2 (Threshold spectral-zeta index formula)**.** *The function $\zeta^{\mathrm{th}}_{\Sigma,\eta}$ is entire and*

$$\boxed{\zeta^{\mathrm{th}}_{\Sigma,\eta}(0) = \operatorname{rank} P_> = \mathfrak{k}_\Sigma(\eta).}$$

For every real q,

$$\zeta^{\mathrm{th}}_{\Sigma,\eta}(q) \geq 0,$$

and the following are equivalent:

(i) $\mathfrak{k}_\Sigma(\eta) = 0$*;*
(ii) $\zeta^{\mathrm{th}}_{\Sigma,\eta}(0) = 0$*;*
(iii) $\zeta^{\mathrm{th}}_{\Sigma,\eta}(q_0) = 0$ *for one real* q_0*;*
(iv) $\zeta^{\mathrm{th}}_{\Sigma,\eta}(q) = 0$ *for every real* q*.*

Proof. The sum in Definition 21.1 is finite. Each term has the form $\exp(-s \log(\lambda - 1))$, so the sum is entire. At $s = 0$ every threshold eigenvalue contributes its algebraic multiplicity, giving the rank of $P_>$. For real q, every summand is strictly positive whenever $P_> \neq 0$. The stated equivalences follow. □

The threshold zeta therefore avoids any global trace-class assertion for K. Only the finite threshold sector is traced. In particular, no unproved regularized determinant has been imported in order to count the seam integer.

Theorem 21.3 (Seam-index trinity)**.** *Let*

$$D_\eta = I - \mathbb{K}_\eta.$$

If $1 \notin \sigma(\mathbb{K}_\eta)$, then

$$\boxed{\mathfrak{k}_\Sigma(\eta) = N_-(D_\eta) = -\operatorname{sf}(I - t\mathbb{K}_\eta; 0) = \zeta^{\mathrm{th}}_{\Sigma,\eta}(0).}$$

Here $N_-(D_\eta)$ is the Morse index of D_η. If 1 belongs to the spectrum, the spectral-flow term is understood through the one-sided endpoint regularization already used in the integer seam-path theorem, while the other three terms remain well defined with the strict threshold $(1, \infty)$*.*

Proof. An eigenvalue $\lambda > 1$ of $\mathbb{K}_\eta$ is exactly a negative eigenvalue $1 - \lambda$ of D_η, with the same multiplicity. The one-sided spectral-flow identity was proved in the native integer seam-path theorem, and Theorem 21.2 supplies the final equality. □

Thus the seam integer has three presentations:

$$\text{Morse index} \quad \longleftrightarrow \quad \text{one-sided spectral flow} \quad \longleftrightarrow \quad \text{threshold spectral zeta at } s = 0.$$

They are exact avatars of one obstruction, not three unrelated numerical statistics.

Corollary 21.4 (Spectral-zeta vanishing implies RH). *Let $\eta_j \downarrow 0$ be a controlled sequence on the same completed native architecture. Fix one real q. If*

$$\boxed{\zeta^{\mathrm{th}}_{\Sigma,\eta_j}(q) = 0 \qquad \textit{for every } j,}$$

then $\mathfrak{k}_\Sigma(\eta_j) = 0$ for every j. Hence the native threshold spectral–zeta correspondence gives

$$\zeta(s) \neq 0 \qquad \textit{for every nontrivial } s \textit{ with } \Re(s) > \frac{1}{2},$$

and the functional equation implies the Riemann hypothesis.

Proof. Theorem 21.2 converts the vanishing of one real zeta moment into vanishing of the seam charge. Apply Corollary 20.3. ☐

21.1 The double-attack architecture

The terminal problem may now be approached by two different proof engines.

Attack I: native operator enclosure. Use the seam-scale convergence theorem, exact finite-rank source augmentation, source exterior rates, same-level native reconstruction and outward arithmetic to prove

$$\sup \sigma(\mathbb{K}_{\eta_j}) < 1 \qquad (j \geq 0).$$

This directly gives $\mathfrak{k}_\Sigma(\eta_j) = 0$.

Attack II: threshold spectral-zeta vanishing. Prove for one real q that

$$\zeta^{\mathrm{th}}_{\Sigma,\eta_j}(q) = 0 \qquad (j \geq 0)$$

through a native trace, resolvent or divisor identity that does not consume the already desired no-crossing inequality. Positivity of every threshold-zeta summand makes this route cancellation-free.

The two attacks are logically equivalent at the terminal integer, but they need not use the same proof mechanism. To count this as a genuinely independent second attack, the proof of threshold-zeta vanishing must not assume completed Weil positivity, the zero-free half-plane, or the no-crossing estimate it is supposed to replace.

21.2 Conditional determinant avatar

A stronger direct divisor route becomes available after a native Schatten membership theorem. Suppose $\mathbb{K}_\eta \in \mathcal{S}_p$ and put

$$\Delta_{p,\eta}(z) = \det_p(I - z\mathbb{K}_\eta).$$

Its zeros are the reciprocal nonzero eigenvalues of $\mathbb{K}_\eta$, with algebraic multiplicity. Therefore, for a contour Γ enclosing precisely the real zeros in $(0,1)$ and no other zeros,

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \frac{1}{2\pi i}\int_\Gamma \frac{\Delta'_{p,\eta}(z)}{\Delta_{p,\eta}(z)}\,dz.}$$

This formula is conditional on the Schatten and contour hypotheses. A direct spectral-divisor theorem for the completed zeta function would additionally require a holomorphic operator pencil, a licensed determinant identity, a nowhere-vanishing comparison factor, and exclusion of extra, missing and polluted zeros. Those obligations are not supplied by the threshold-zeta index formula alone.

Claim status. The threshold spectral-zeta index formula, seam-index trinity and the corollary "threshold-zeta vanishing implies RH" are proved. An independent theorem forcing the threshold spectral zeta to vanish, native Schatten membership and a target-faithful determinant divisor theorem remain open. Accordingly the Riemann hypothesis remains `ABSTAIN`.

22 Native Euler–virial annihilation of the threshold spectral zeta

The threshold spectral zeta gives a cancellation-free presentation of the seam integer, but its value does not vanish merely because it has acquired a useful name. This section supplies a genuinely different proof engine: a native positive-commutator criterion generated by the Euler-scale carrier. It does not assume the desired no-crossing inequality, completed-Weil positivity, or a zero-free half-plane.

22.1 A cut-compatible native conjugate observable

Let

$$(\mathsf{X}_\times F)(r) = (\log r)F(r)$$

be the native seam-position observable and let

$$\mathsf{E} = -ir\frac{\mathrm{d}}{\mathrm{d}r}$$

be the Euler-scale generator. Choose a real odd function

$$v \in C^1(\mathbb{R}), \qquad v, v' \in L^\infty(\mathbb{R}),$$

and define on the multiplicative core

$$\boxed{\mathsf{A}_v = \frac{1}{2}(\mathsf{X}_\times v(\mathsf{E}) + v(\mathsf{E})\mathsf{X}_\times).}$$

The vector field v is bounded and Lipschitz, hence its flow on the Euler spectrum is complete. The corresponding first-order operator has a self-adjoint closure, again denoted A_v.

Both primitive factors reverse under the inversion cut:

$$\mathsf{J}_\times \mathsf{X}_\times \mathsf{J}_\times = -\mathsf{X}_\times, \qquad \mathsf{J}_\times \mathsf{E} \mathsf{J}_\times = -\mathsf{E}.$$

Since v is odd,

$$\boxed{[\mathsf{A}_v, \mathsf{J}_\times] = 0.}$$

Thus the conjugate observable preserves the cut parity sectors and does not mix the two terminal coordinates.

Under the Mellin chart,

$$\boxed{\mathcal{M}\mathsf{A}_v\mathcal{M}^{-1} = i\left(v(\xi)\frac{\mathrm{d}}{\mathrm{d}\xi} + \frac{1}{2}v'(\xi)\right).}$$

This differential expression is a shadow calculation of an operator already defined on the multiplicative carrier.

Proposition 22.1 (Native multiplier commutator)**.** *Let m be real, locally absolutely continuous and such that vm' is bounded. Then, as a bounded commutator form,*

$$\boxed{i[M_m, \mathcal{M}\mathsf{A}_v\mathcal{M}^{-1}] = M_{vm'}.}$$

The same identity holds entrywise for a Hermitian matrix-valued multiplier.

Proof. On a smooth compactly supported spectral core, the zeroth-order terms cancel and direct differentiation gives

$$[M_m, i(v\partial_\xi + v'/2)]f = -ivm'f.$$

Boundedness of vm' extends the identity. □

22.2 The weighted threshold virial trace

Fix $\eta > 0$ and abbreviate

$$K = \mathbb{K}_\eta, \qquad P_> = \mathbf{1}_{(1,\infty)}(K), \qquad T_\eta = (K - I)P_>.$$

The operator T_η is positive and invertible on the finite-dimensional space $P_>\mathscr{H}$. Recall

$$\zeta^{\mathrm{th}}_{\Sigma,\eta}(s) = \mathrm{Tr}_{P_>\mathscr{H}}(T_\eta^{-s}).$$

Let $\mathsf{A} = \mathsf{A}^*$ be any native conjugate observable for which $K \in C^1(\mathsf{A})$, so that

$$B_{\eta,\mathsf{A}} = i[K, \mathsf{A}]$$

extends to a bounded self-adjoint operator. For real q, define

$$\boxed{\mathcal{V}_{\eta,\mathsf{A}}(q) = \mathrm{Tr}\Big(T_\eta^{-q/2} P_> B_{\eta,\mathsf{A}} P_> T_\eta^{-q/2}\Big).}$$

If $P_> = 0$, this trace is defined to be zero.

Theorem 22.2 (Weighted threshold virial identity)**.** *For every real q,*

$$\boxed{\mathcal{V}_{\eta,\mathsf{A}}(q) = 0.}$$

Proof. For every eigenvalue λ of K, the bounded-commutator virial theorem gives

$$P_\lambda i[K, \mathsf{A}]P_\lambda = 0.$$

Choose an orthonormal eigenbasis of the finite threshold space. The factors $T_\eta^{-q/2}$ are diagonal in this basis, while every diagonal block of the commutator vanishes. The finite trace is therefore zero. □

22.3 Positive commutator forces spectral-zeta vanishing

Theorem 22.3 (Virial annihilation of the threshold spectral zeta)**.** *Assume that for some $c_\eta > 0$ and $\beta \in \mathbb{R}$ one has the threshold form estimate*

$$\boxed{P_> B_{\eta,\mathsf{A}} P_> \geq c_\eta T_\eta^\beta.}$$

Then

$$\boxed{P_> = 0, \qquad \mathfrak{k}_\Sigma(\eta) = 0, \qquad \zeta^{\mathrm{th}}_{\Sigma,\eta}(s) \equiv 0.}$$

Proof. Insert $q=\beta$ in Theorem 22.2. The assumed estimate gives

$$0=\mathcal{V}_{\eta,\mathsf{A}}(\beta)\geq c_\eta\operatorname{Tr}_{P_>\mathscr{H}}(I)=c_\eta\mathfrak{k}_\Sigma(\eta).$$

The charge is a nonnegative integer, so it vanishes. The threshold-zeta index formula then gives the remaining statements. □

More generally, the same argument with arbitrary q yields

$$0\geq c_\eta\zeta^{\mathrm{th}}_{\Sigma,\eta}(q-\beta).$$

Every real threshold-zeta moment is nonnegative, so any strict weighted commutator estimate is cancellation-free.

Corollary 22.4 (Euler–virial route to the Riemann hypothesis)**.** *Let $\eta_j\downarrow 0$ be controlled on the same completed native architecture. If for every j there is a cut-compatible native conjugate observable A_j satisfying the estimate of Theorem 22.3, then*

$$\zeta(s)\neq 0\qquad \textit{for every nontrivial } s \textit{ with } \Re(s)>\frac{1}{2},$$

and hence the Riemann hypothesis holds.

Proof. The virial theorem gives $\mathfrak{k}_\Sigma(\eta_j)=0$ for every j. Apply the native threshold spectral–zeta correspondence and the functional equation. □

22.4 A channelwise commutator budget

Write the native commutator as a sum of declared response channels,

$$B_{\eta,\mathsf{A}}=B_\Gamma+B_{\mathrm{prime}}+B_{\mathrm{fin}}+B_{\mathrm{rest}}.$$

The following form is useful because every term is evaluated on the same threshold projection.

Proposition 22.5 (Strict channel budget)**.** *Suppose for some $\beta\in\mathbb{R}$ that*

$$T_\eta^{-\beta/2}P_>B_\Gamma P_>T_\eta^{-\beta/2}\geq c_\Gamma I,$$

and

$$\left\|T_\eta^{-\beta/2}P_>(B_{\mathrm{prime}}+B_{\mathrm{fin}}+B_{\mathrm{rest}})P_>T_\eta^{-\beta/2}\right\|\leq\delta,\qquad \delta<c_\Gamma.$$

Then $\mathfrak{k}_\Sigma(\eta)=0$.

Proof. The full compressed commutator is at least $(c_\Gamma-\delta)T_\eta^\beta$. Apply Theorem 22.3. □

22.5 Native channel orientation

Let

$$g(\xi)=\operatorname{Re}\psi\left(\frac{1}{4}+\frac{i\xi}{2}\right)-\log\pi,\qquad g_-=\max\{-g,0\}.$$

The source-closure theorem proves that g is even and increasing for $\xi>0$. Choose the bounded odd vector field

$$\boxed{v_0(\xi)=-\frac{\xi}{1+\xi^2}.}$$

Then, almost everywhere,

$$\boxed{v_0(\xi)g_-'(\xi) \geq 0, \qquad v_0(\xi)(1+2g_-(\xi))' \geq 0.}$$

Thus the odd and even Gamma defect multipliers have the correct native virial orientation. Strictness on the unknown threshold subspace is an observability question, not a sign convention.

For the two-sheet prime symbol at $a = 3/2$,

$$q(\xi) = \sum_{n\geq 2} \frac{\Lambda(n)}{n^2} e^{i\xi \log n}, \qquad \mathcal{Q}(\xi) = \begin{pmatrix} 0 & q(\xi) \\ \overline{q(\xi)} & 0 \end{pmatrix},$$

the differentiated series converges absolutely and

$$\boxed{\|v_0 \mathcal{Q}'\|_\infty \leq \|v_0\|_\infty \sum_{n\geq 2} \frac{\Lambda(n)\log n}{n^2}.}$$

This supplies a finite prime-commutator budget. The boundary and Feshbach commutators are finite rank and are inserted through their exact source ranges.

The remaining operator-specific theorem is therefore sharply typed: prove a strict Gamma observability estimate on the threshold subspace and show that the prime plus finite-rank commutator budget is smaller than that strict margin. This is a native Euler–virial problem, not a request for another terminal mode cutoff.

Claim status. The cut-compatible conjugate family, multiplier commutator identity, weighted threshold virial trace, virial annihilation theorem, strict channel budget and the resulting conditional route to RH are proved. The operator-specific strict observability estimate and its directed prime/finite-rank budget remain open. The Riemann hypothesis remains `ABSTAIN`.

23 Congruence-safe Morse zeta and the covariant two-sheet virial reduction

The threshold-zeta virial theorem is abstractly correct on the normalized compact packet $\mathbb{K}_\eta$. An operator-specific channel proof, however, must not silently differentiate

$$\mathbb{K}_\eta = \mathbb{A}_\eta^{-1/2}(\mathbb{A}_\eta - \mathbb{W}_\eta)\mathbb{A}_\eta^{-1/2}$$

as though the reference factors commuted with the conjugate observable. Such a step would import unrecorded commutators of $\mathbb{A}_\eta^{-1/2}$ and could manufacture a false sign. We therefore move the channelwise second attack to the shifted native sign operator $\mathbb{W}_\eta$ itself. The seam integer is preserved by the exact positive congruence, while the boundary, Gamma and prime channels remain visible before normalization.

23.1 The congruence-safe Morse spectral zeta

Recall the exact shifted factorization

$$\boxed{\mathbb{W}_\eta = \mathbb{A}_\eta^{1/2}(I - \mathbb{K}_\eta)\mathbb{A}_\eta^{1/2}, \qquad \mathbb{A}_\eta \geq \eta I.}$$

Let

$$P_-^\eta = \mathbf{1}_{(-\infty,0)}(\mathbb{W}_\eta), \qquad H_\eta = (-\mathbb{W}_\eta)P_-^\eta.$$

The negative spectral subspace is finite dimensional because $I - \mathbb{K}_\eta$ is a self-adjoint Fredholm operator with finite Morse index and boundedly invertible congruence preserves inertia.

Definition 23.1 (Native Morse spectral zeta)**.** For $s \in \mathbb{C}$, define

$$\boxed{\zeta^{\mathrm{M}}_{\Sigma,\eta}(s) = \mathrm{Tr}_{P_-^\eta \mathscr{H}}(H_\eta^{-s}) = \sum_{\nu\in\sigma(\mathbb{W}_\eta),\, \nu<0} m(\nu)(-\nu)^{-s}.}$$

If $P_-^\eta = 0$, the function is defined to be identically zero.

Theorem 23.2 (Congruence-safe zeta index)**.** *The Morse spectral zeta is entire and*

$$\boxed{\zeta^{\mathrm{M}}_{\Sigma,\eta}(0) = N_-(\mathbb{W}_\eta) = \mathfrak{k}_\Sigma(\eta) = \zeta^{\mathrm{th}}_{\Sigma,\eta}(0).}$$

For every real q, the value $\zeta^{\mathrm{M}}_{\Sigma,\eta}(q)$ is nonnegative and vanishes if and only if the seam charge vanishes.

Proof. The defining sum is finite. At $s = 0$ it counts negative eigenvalues with multiplicity. Sylvester inertia under the displayed positive congruence gives

$$N_-(\mathbb{W}_\eta) = N_-(I - \mathbb{K}_\eta) = \mathrm{rank}\, \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).$$

Every summand is strictly positive on the real axis whenever the negative sector is nonempty. □

Remark 23.3 (Only the index is congruence invariant)*.* The two spectral-zeta functions need not agree away from $s = 0$. Positive congruence preserves inertia, not the magnitudes of negative eigenvalues. The identity consumed by the terminal proof is therefore

$$\zeta^{\mathrm{M}}_{\Sigma,\eta}(0) = \zeta^{\mathrm{th}}_{\Sigma,\eta}(0),$$

not an unlicensed equality of complete zeta functions.

23.2 Morse-weighted virial annihilation

Let $\mathsf{A} = \mathsf{A}^*$ be a native conjugate observable such that $\mathbb{W}_\eta \in C^1(\mathsf{A})$ and put

$$B^{\mathrm{M}}_{\eta,\mathsf{A}} = i[\mathbb{W}_\eta, \mathsf{A}].$$

For real q, define

$$\mathcal{V}^{\mathrm{M}}_{\eta,\mathsf{A}}(q) = \mathrm{Tr}\Big(H_\eta^{-q/2} P_-^\eta B^{\mathrm{M}}_{\eta,\mathsf{A}} P_-^\eta H_\eta^{-q/2}\Big).$$

Theorem 23.4 (Morse-weighted virial identity)**.** *For every real q,*

$$\boxed{\mathcal{V}^{\mathrm{M}}_{\eta,\mathsf{A}}(q) = 0.}$$

If for some $c_\eta > 0$ and $\beta \in \mathbb{R}$,

$$\boxed{P_-^\eta B^{\mathrm{M}}_{\eta,\mathsf{A}} P_-^\eta \geq c_\eta H_\eta^\beta,}$$

then

$$\boxed{P_-^\eta = 0, \qquad \mathfrak{k}_\Sigma(\eta) = 0, \qquad \zeta^{\mathrm{M}}_{\Sigma,\eta} \equiv 0.}$$

Proof. The diagonal commutator block vanishes on every eigenspace of $\mathbb{W}_\eta$. This proves the weighted trace identity. Taking $q = \beta$ in the strict estimate gives

$$0 \geq c_\eta \operatorname{rank} P_-^\eta.$$

The rank is nonnegative, so it is zero. $\square$

This formulation is the native channel route. It avoids differentiating the reference square root and is therefore immune to a false sign introduced by a normalizing coordinate.

23.3 The common two-sheet sign symbol

Let

$$g(\xi) = \operatorname{Re}\psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log\pi, \qquad w_\Gamma(\xi) = 1 + |g(\xi)|,$$

and let

$$q(\xi) = \sum_{n\geq 2} \frac{\Lambda(n)}{n^2} e^{i\xi\log n}, \qquad \mathcal{Q}(\xi) = \begin{pmatrix} 0 & q(\xi) \\ \overline{q(\xi)} & 0 \end{pmatrix}.$$

Before the exact finite-rank boundary and Feshbach augmentations are added, the frequency part of the shifted sign form has the same scalar Gamma coefficient in the direct even and direct odd sectors:

$$\boxed{\omega_\eta(\xi) = g(\xi) + \eta w_\Gamma(\xi).}$$

Indeed,

$$(1+\eta)(1+|g|) - (1+|g|-g) = \eta(1+|g|) + g$$

in the even sector, while

$$\eta(1+|g|) + g_+ - g_- = \eta(1+|g|) + g$$

in the odd sector. The common two-sheet source symbol is therefore

$$\boxed{\Omega_\eta(\xi) = \omega_\eta(\xi) I_2 - \mathcal{Q}(\xi).}$$

This identity is obtained before reference normalization.

23.4 Covariant removal of the prime phase

On an interval where $q(\xi) \neq 0$, write

$$q(\xi) = \rho(\xi)e^{i\phi(\xi)}, \qquad \widehat{\mathcal{Q}}(\xi) = \rho(\xi)^{-1}\mathcal{Q}(\xi).$$

Let v be real and odd. In the two-sheet spectral chart define the Hermitian connection

$$\mathcal{H}_{v,q}(\xi) = \frac{1}{2}v(\xi)\phi'(\xi)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and the covariant first-order observable

$$\mathsf{A}_{v,q}^{\mathrm{cov}} = i\left(v\partial_\xi + \frac{1}{2}v'\right) + M_{\mathcal{H}_{v,q}}.$$

Theorem 23.5 (Covariant two-sheet commutator). *On every interval where q is nonzero,*

$$\boxed{i[M_{\Omega_\eta}, \mathsf{A}^{\mathrm{cov}}_{v,q}] = M_{v(\omega'_\eta I_2 - \rho' \widehat{\mathcal{Q}})}.}$$

Consequently, wherever $v \geq 0$,

$$\boxed{\lambda_{\min}\left(v(\omega'_\eta I_2 - \rho' \widehat{\mathcal{Q}})\right) = v(\omega'_\eta - |\rho'|).}$$

Proof. The matrix multiplier commutator is

$$v\Omega'_\eta + i[\Omega_\eta, \mathcal{H}_{v,q}].$$

The diagonal connection removes the angular derivative of the off-diagonal prime symbol. Explicitly,

$$v\mathcal{Q}' + i[\mathcal{Q}, \mathcal{H}_{v,q}] = v\rho' \widehat{\mathcal{Q}}.$$

Substitution gives the first identity. Since $\widehat{\mathcal{Q}}$ is a Hermitian involution, the eigenvalue formula follows. □

The connection removes only the phase derivative. The radial derivative ρ' is gauge invariant and therefore represents the true prime obstruction to a strict source-symbol virial inequality.

23.5 The active-band criterion

Let $I_\eta = (-\Omega^*_\eta, \Omega^*_\eta)$ be a symmetric interval. The following theorem isolates the exact remaining obligations without importing a mode cutoff.

Theorem 23.6 (Covariant active-band seam-annihilation criterion). *Assume:*

(i) q has no zero on I_η;

(ii) the lower source-symbol branch satisfies

$$\omega_\eta(\xi) - |q(\xi)| \geq 0 \qquad (|\xi| \geq \Omega^*_\eta);$$

(iii) for an odd v with $v(\xi) > 0$ on $(0, \Omega^*_\eta)$,

$$\omega'_\eta(\xi) - |\rho'(\xi)| > 0 \qquad (0 < \xi < \Omega^*_\eta);$$

(iv) the compatible native channel reconstruction has no nonzero negative state blind to the open set on which the preceding gap is positive;

(v) the exact finite-rank and reconstruction commutator error is strictly smaller than the resulting finite-dimensional observability constant.

Then

$$\boxed{\mathfrak{k}_\Sigma(\eta) = 0.}$$

Proof. The source-symbol negative branch is confined to the active interval by (ii). The covariant commutator is nonnegative there by (iii) and strictly positive on every nonzero compatible negative state by (iv). Finite dimensionality of the negative spectral space upgrades strict positivity to a positive minimum. The budget in (v) preserves that minimum for the full native sign operator. Apply Theorem 23.4. □

Condition (iv) is a target-faithful observability statement. It cannot be replaced by the assertion that a sampled transform matrix has full rank. Condition (v) must use the exact boundary and Feshbach source augmentation and the same native projection.

Claim status. The Morse spectral-zeta index, congruence-safe virial identity, common native sign symbol and covariant removal of the prime phase are proved. The active band, the covariant derivative gap and the nonvanishing of q have strong numerical orientation at $\eta = 10^{-2}$ but still require outward Arb interval enclosures. Compatible no-blindness and the exact finite-rank commutator margin remain open. The Riemann hypothesis remains `ABSTAIN`.

24 Arb enclosure of the covariant active band

The covariant two-sheet virial theorem reduces the infinite source symbol to three scalar obligations: nonvanishing of the complete prime symbol on the active band, strict positivity of the radial derivative gap there, and nonnegativity of the lower source branch outside. We now formulate a directed certificate for all three obligations without a standing-wave terminal cutoff.

Put

$$q(\xi) = \sum_{n\geq 2} \frac{\Lambda(n)}{n^2} e^{i\xi \log n}, \qquad \omega_\eta(\xi) = g(\xi) + \eta(1 + |g(\xi)|).$$

The common two-sheet symbol has eigenvalue branches

$$\omega_\eta(\xi) \pm |q(\xi)|.$$

24.1 Complete prime moment enclosure

For an auxiliary integer P, define

$$q_P^{(j)}(\xi) = \sum_{n\leq P} \frac{\Lambda(n)(i\log n)^j}{n^2} e^{i\xi \log n}, \qquad j = 0, 1, 2.$$

Arb encloses every retained prime-power coefficient and one-step Euler phase. A geometric recurrence generates the uniform-grid packet with an explicit rounding ledger.

Let

$$R_j(P) = \sum_{n>P} \frac{\Lambda(n)(\log n)^j}{n^2}.$$

From $\psi(x) \leq x \log x$ and Stieltjes partial summation,

$$\boxed{R_j(P) \leq 2I_{j+1}(P) - jI_j(P),}$$

where

$$I_0(P) = P^{-1}, \qquad I_k(P) = \frac{(\log P)^k}{P} + kI_{k-1}(P).$$

Consequently the finite Arb packet plus $R_j(P)$ encloses the complete q, q', q''.

24.2 Interval lower bounds

On a grid interval $[x_k, x_{k+1}]$ of width h,

$$|q(\xi) - q_P(x_k)| \leq \varepsilon_{0,k} + R_0(P) + D_1^{\mathrm{up}} h,$$

where

$$D_j^{\mathrm{up}} \geq \sum_{n\geq 2} \frac{\Lambda(n)(\log n)^j}{n^2}.$$

This yields a directed lower bound for $|q|$. Once this lower bound is positive,

$$(|q|)' = \frac{\mathrm{Re}(\overline{q}q')}{|q|}$$

is enclosed using the certified disks for q and q' and the complete second moment D_2^{up}.

For $x \in [a, b] \subset (0, \infty)$, the positive real-digamma series gives

$$g'(x) \geq \sum_{m=0}^{M-1} \frac{(m + 1/4)a}{2\big((m + 1/4)^2 + b^2/4\big)^2}.$$

Since $\omega_\eta' \geq (1 - \eta)g'$, interval comparison proves the covariant radial derivative gap.

The interval adjacent to zero is treated after division by x. The exact identity $\mathrm{Re}(\overline{q}q')(0) = 0$ gives

$$|(|q|)'(x)| \leq x \frac{D_1^2 + D_0 D_2}{\inf_{[0,h]} |q|},$$

while the Gamma series supplies a positive lower coefficient for $\omega_\eta'(x)/x$.

Theorem 24.1 (Directed covariant active-band certificate)**.** *Fix $\eta = 10^{-2}$, $P = 10^6$, grid width $h = 10^{-2}$, active edge $\Omega_* = 7$, far edge $\Omega_\infty = 12$, and Arb precision 192 bits. Suppose the generated interval ledger satisfies*

$$\inf_{|\xi| \leq 7} |q(\xi)| > 0,$$

$$\inf_{0<\xi<7} \Big(\omega_\eta'(\xi) - |(|q|)'(\xi)|\Big) > 0,$$

and

$$\inf_{|\xi| \geq 7} (\omega_\eta(\xi) - |q(\xi)|) \geq 0.$$

Then assumptions (i)–(iii) of the covariant active-band seam-annihilation criterion hold for the complete Gamma–prime source symbol.

Proof. The interval ledger covers $[0, 7]$ and $[7, 12]$. For $\xi \geq 12$, monotonicity of g and the complete absolute moment bound give

$$\omega_\eta(\xi) - |q(\xi)| \geq \omega_\eta(12) - D_0^{\mathrm{up}}.$$

Reality symmetry supplies the negative half-line. The three displayed strict or nonnegative inequalities are exactly assumptions (i)–(iii) of Theorem 23.6. ☐

The auxiliary cutoff P sharpens scalar bounds only; its complement is included through R_0, R_1, R_2. It is not a definition of the native operator.

Claim status. The Arb driver and its regression controls are implemented. A passing local ledger certifies the complete-prime active-band source conditions. Compatible native no-blindness and the exact finite-rank/reconstruction commutator margin remain open, so the Riemann hypothesis remains `ABSTAIN`.

25 Native Hardy-channel no-blindness

The Arb active-band certificate proves a strict lower source-symbol gap on an open frequency band. The covariant virial theorem still requires a compatibility statement: no nonzero native negative state may be invisible on that band. This is not true for an arbitrary L^2 spectral packet. It is a consequence of the exponential spatial energy built into the native carrier.

25.1 The native Hardy strip

Let

$$f(x) = F(e^x), \qquad a = \frac{3}{2}.$$

The spatial part of the native energy gives

$$\int_{\mathbb{R}} e^{2a|x|}|f(x)|^2\,dx < \infty.$$

For $|\operatorname{Im} z| < a$, define

$$\Phi_F(z) = \int_{\mathbb{R}} f(x)e^{-izx}\,dx.$$

Theorem 25.1 (Native strip-Hardy embedding)**.** *The function Φ_F is analytic on the strip*

$$S_a = \{z \in \mathbb{C} : |\operatorname{Im} z| < a\}$$

and belongs to $H^2(S_a)$. More precisely, for every $|y| < a$,

$$\boxed{\frac{1}{2\pi}\|\Phi_F(\cdot + iy)\|^2_{L^2(\mathbb{R})} = \int_{\mathbb{R}} e^{2yx}|f(x)|^2\,dx \le \int_{\mathbb{R}} e^{2a|x|}|f(x)|^2\,dx.}$$

Its two L^2 boundary values are

$$\boxed{\Phi_F(\xi - ia) = \mathcal{M}(r^{-a}F)(\xi), \qquad \Phi_F(\xi + ia) = \mathcal{M}(r^{a}F)(\xi).}$$

Proof. If $|y| < a$, Cauchy–Schwarz with the weight $e^{2a|x|}$ gives absolute convergence locally uniformly in $z = \xi + iy$, hence analyticity. Plancherel applied to $e^{yx}f(x)$ gives the displayed identity. The endpoint functions $e^{\pm ax}f(x)$ belong to L^2, so the boundary limits exist in L^2 and are exactly the two Mellin channels. □

The logarithmic and Mellin transforms are therefore faithful charts of an analytic object already forced by the native energy.

25.2 Positive-measure boundary uniqueness

Lemma 25.2 (Hardy-strip boundary uniqueness)**.** *Let $\Phi \in H^2(S_a)$. If either one of its boundary values vanishes on a measurable set of positive Lebesgue measure, then $\Phi \equiv 0$.*

Proof. Map S_a conformally to the unit disk. The standard conformal Hardy-space transfer sends Φ to a disk Hardy function after multiplication by the nonvanishing square root of the boundary Jacobian. A nonzero disk Hardy function has nonzero radial boundary values almost everywhere: otherwise its boundary logarithm would equal $-\infty$ on a set of positive measure, which is incompatible with the Jensen–outer factor representation. Hence the pullback, and therefore Φ, is identically zero. □

Theorem 25.3 (Native active-band no-blindness)**.** *Let $E \subset \mathbb{R}$ have positive measure and let*

$$\mathcal{B}_a^{\times}F = \begin{pmatrix} \mathcal{M}(r^{-a}F) \\ \mathcal{M}(r^{a}F) \end{pmatrix}.$$

Then

$$\boxed{1_E\mathcal{B}_a^{\times}F = 0 \quad \Longrightarrow \quad F = 0.}$$

In fact, vanishing of either one of the two channel components on E already implies $F = 0$.

Proof. Apply Theorem 25.2 to the corresponding boundary value of Φ_F. The Mellin transform is injective on the native carrier. $\square$

The result is stable under restriction to either inversion parity sector and to the odd Feshbach complement.

25.3 Compatible reconstruction and strict observability

Let

$$J : \mathscr{S} \longrightarrow \mathscr{H}$$

be an isometric compatible reconstruction. This includes the canonical channel-Gram quotient

$$J_N = Z_N H_N^{-1/2}.$$

Define the active observation

$$\mathcal{O}_E = 1_E \mathcal{B}^{\times}_{3/2}.$$

Theorem 25.4 (Compatible reconstruction is not active-band blind)**.** *For every positive-measure band E,*

$$\boxed{\ker(\mathcal{O}_E J) = \{0\}.}$$

This property is invariant under every unitary change of coordinates in $\mathscr{S}$.

Proof. If $\mathcal{O}_E J u = 0$, then Theorem 25.3 gives $Ju = 0$. Since J is isometric, $u = 0$. Replacing J by JU with U unitary does not change its range or its kernel. $\square$

Let h be measurable and strictly positive almost everywhere on E. If $V \subset \mathscr{H}$ is finite dimensional and $J_V : \mathbb{C}^d \to V$ is an isometry, define

$$\boxed{\mathcal{G}_{E,h}(V) = J_V^* (\mathcal{B}^{\times}_{3/2})^* 1_E M_h \mathcal{B}^{\times}_{3/2} J_V.}$$

Theorem 25.5 (Finite-dimensional strict observability)**.** *For every nonzero finite-dimensional compatible subspace V,*

$$\boxed{\mathcal{G}_{E,h}(V) > 0.}$$

Consequently,

$$\boxed{c_{E,h}(V) := \lambda_{\min}(\mathcal{G}_{E,h}(V)) > 0.}$$

Proof. For $u \neq 0$,

$$\langle u, \mathcal{G}_{E,h}(V) u \rangle = \int_E h(\xi) \|\mathcal{B}^{\times}_{3/2} J_V u(\xi)\|^2 \, d\xi.$$

If this vanished, the channel would vanish almost everywhere on the set where $h > 0$, contradicting Theorem 25.4. Thus the quadratic form is strictly positive. Compactness of the unit sphere in finite dimension gives a positive minimum. $\square$

Apply the theorem to

$$V = P_-^{\eta} \mathscr{H},$$

the finite-dimensional negative spectral space of the shifted native sign operator. The Arb active-band certificate supplies the strictly positive covariant source gap h. Therefore every nonzero negative state is detected by the source commutator. Assumption (iv) of the covariant active-band criterion is closed.

The theorem proves existence of the exact observability constant on the actual negative space. It does not yet supply an outward numerical lower bound for that constant. Such a bound is required only to dominate the exact finite-rank and reconstruction commutator term.

Remark 25.6 (Why an arbitrary spectral packet is insufficient). A generic $L^2(\mathbb{R})$ function may be supported outside E and is then nonzero but perfectly blind on E. Thus sampled full rank or generic Fourier completeness cannot replace the chain

$$\boxed{\begin{array}{c}\text{native exponential energy} \longrightarrow H^2 \text{ Mellin strip} \\ \longrightarrow \text{boundary uniqueness} \longrightarrow \text{compatible no-blindness.}\end{array}}$$

Claim status. The native strip-Hardy embedding, positive-measure boundary uniqueness, compatible active-band no-blindness and existence of a strict observability constant on every finite obstruction space are proved. An outward lower bound for the actual negative-space observability constant and the exact finite-rank or reconstruction commutator margin remain open. The Riemann hypothesis remains `ABSTAIN`.

26 Virial–Birman–Schwinger reduction to the exact finite obstruction

The complete-prime active-band certificate and the native Hardy-strip theorem close the infinite Gamma–prime source obstruction at the fixed shift $\eta = 10^{-2}$. The only remaining terms are exact finite-rank sources: the odd boundary defect and the odd Feshbach correction. We now replace the unknown negative-space commutator margin by an exact Birman–Schwinger matrix of rank at most five.

26.1 Positivity of the pure source operator

Let $S_{\eta,\pm}$ denote the shifted direct even and odd Gamma–prime sign operators before the boundary and Feshbach terms are inserted. Their negative responses are compact relative to coercive references, and hence

$$\sigma_{\text{ess}}(S_{\eta,\pm}) \subset [\eta, \infty).$$

Theorem 26.1 (Virial positivity of the complete source operator)**.** *At* $\eta = 10^{-2}$,

$$\boxed{S_{\eta,+} > 0, \qquad S_{\eta,-} > 0.}$$

Proof. Suppose $S_{\eta,\pm}F = \lambda F$ with $\lambda \le 0$ and $F \neq 0$. The complete-prime Arb certificate confines every nonpositive source direction to the active band and gives a strict covariant derivative gap there. The native Hardy-channel theorem proves that no nonzero compatible state is blind to this band. Hence the covariant source commutator has strictly positive expectation on F. The eigenvector virial identity gives expectation zero, a contradiction. Thus there is no spectrum at or below zero. Since the essential spectrum begins at a positive value, strict positivity follows. □

The favorable even boundary term may be added without creating a negative direction. Only the odd finite defect remains.

26.2 The rank-at-most-five odd source

Let r_{odd} be the exact odd boundary source and put

$$V_{F,\eta} = C_3(A_3 + \eta I)^{-1/2}.$$

Since $\dim E_3 = 4$,

$$\operatorname{rank} V_{F,\eta} \leq 4.$$

Define

$$\boxed{V_\eta = \begin{bmatrix} r_{\mathrm{odd}} & V_{F,\eta} \end{bmatrix}, \qquad \operatorname{rank} V_\eta \leq 5.}$$

Then the shifted odd sign operator is

$$\boxed{F_{3,\eta}^{\times} = S_{\eta,-} - V_\eta V_\eta^*.}$$

Definition 26.2 (Finite native Birman–Schwinger matrix)**.** Define

$$\boxed{B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta \in M_r(\mathbb{C}), \qquad r \leq 5.}$$

Theorem 26.3 (Exact finite-rank inertia reduction)**.** *The following identities hold:*

$$\boxed{N_-(F_{3,\eta}^{\times}) = N_+(B_\eta - I),}$$

$$\boxed{\dim\ker F_{3,\eta}^{\times} = \dim\ker(B_\eta - I).}$$

Consequently,

$$\boxed{F_{3,\eta}^{\times} \geq 0 \iff B_\eta \leq I.}$$

Proof. Put

$$T_\eta = S_{\eta,-}^{-1/2} V_\eta.$$

Then

$$F_{3,\eta}^{\times} = S_{\eta,-}^{1/2}(I - T_\eta T_\eta^*) S_{\eta,-}^{1/2}.$$

Positive congruence preserves inertia. The nonzero spectra of $T_\eta T_\eta^*$ and

$$T_\eta^* T_\eta = B_\eta$$

agree with multiplicity. The negative, zero and positive threshold statements follow. □

Corollary 26.4 (Fixed-shift seam charge is a five-dimensional threshold count)**.** *At $\eta = 10^{-2}$,*

$$\boxed{\mathfrak{k}_\Sigma(\eta) = N_+(B_\eta - I).}$$

In particular,

$$\lambda_{\max}(B_\eta) < 1 \implies \mathfrak{k}_\Sigma(\eta) = 0.$$

Proof. The complete even source and its favorable boundary correction are positive by Theorem 26.1. The odd charge is the negative index of the shifted odd sign operator, which is given by Theorem 26.3. □

The finite obstruction zeta

$$\zeta_{B,\eta}(s) = \sum_{\mu \in \sigma(B_\eta),\, \mu > 1} m(\mu)(\mu - 1)^{-s}$$

therefore satisfies

$$\boxed{\zeta_{B,\eta}(0) = \mathfrak{k}_\Sigma(\eta).}$$

26.3 Outward resolvent enclosure

Let $\widetilde{X}$ approximate the five source-resolvent columns solving

$$S_{\eta,-}X = V_\eta,$$

and put

$$R = V_\eta - S_{\eta,-}\widetilde{X},$$

$$\widetilde{B}_\eta = \frac{1}{2}\left(V_\eta^*\widetilde{X} + \widetilde{X}^*V_\eta\right).$$

Proposition 26.5 (Residual enclosure of the finite matrix)**.** *If*

$$S_{\eta,-} \geq s_0 I, \qquad s_0 > 0,$$

then

$$\boxed{\|B_\eta - \widetilde{B}_\eta\| \leq \frac{\|V_\eta\|\,\|R\|}{s}.}$$

Quantitative specialization at eta = 0.01 (proved in the Revision Addendum):
S_(0.01,-) >= 0.008 I and ||S_(0.01,-)^(-1)|| <= 125. Hence
||B_(0.01) - B_tilde_(0.01)|| <= 125 ||V_(0.01)|| ||R||, and the strict criterion
lambda_max(B_tilde_(0.01)) + 125 ||V_(0.01)|| ||R|| < 1 implies k_Sigma(0.01) = 0.

$$X - \widetilde{X} = S_{\eta,-}^{-1}R,$$

we have

$$\left\|V_\eta^*(X - \widetilde{X})\right\| \leq \|V_\eta\|\,\|S_{\eta,-}^{-1}\|\,\|R\| \leq \frac{\|V_\eta\|\,\|R\|}{s_0}.$$

Symmetrization does not increase this bound. The eigenvalue conclusion follows from Weyl's inequality and Theorem 26.4. □

The fixed-shift second attack has therefore become:

$$\boxed{\text{one source floor} + \text{five native resolvent columns} + \text{one } 5 \times 5 \text{ outward eigenvalue gate.}}$$

The dimension is dictated by the exact boundary and Feshbach source ranges, not by a terminal mode cutoff.

Claim status (revised). The virial source theorem, shift transfer to theta = 0.002, source floor S_(0.01,-) >= 0.008 I, inverse bound 125, Hermitian rank-at-most-five reduction, residual enclosure, and final outward acceptance theorem are proved. The actual five source-resolvent columns and the outward value of the Hermitian matrix remain open. A controlled sequence eta_j -> 0 and the Riemann hypothesis remain OPEN / ABSTAIN.

27 Bilateral recognition closure: accepting both sides of the cut

The framework is closed here in a precise sense that is independent of the terminal Riemann-hypothesis claim. Closure does not mean declaring the seam charge to be zero. It means that every side created by a cut, every representation-visible direction, every target-relevant blind direction and every cross coupling is retained in the mathematical object. No preferred outcome is built into the architecture.

27.1 The universal target-faithful completion

Let

$$R : \mathscr{H} \longrightarrow \mathscr{Y}$$

be a bounded observer and let

$$\Pi : \mathscr{H} \longrightarrow \mathscr{Z}$$

be the declared target map. Put

$$\mathscr{N} = \ker R, \qquad \mathscr{N}_0 = \ker R \cap \ker \Pi.$$

Since $\mathscr{N}_0$ is closed in $\mathscr{N}$, the quotient

$$\mathfrak{B}(R;\Pi) = \mathscr{N}/\mathscr{N}_0$$

is a Hilbert space with its quotient norm. Let

$$q_\Pi : \mathscr{N} \longrightarrow \mathfrak{B}(R;\Pi)$$

be the quotient map and let $P_{\mathscr{N}}$ be the orthogonal projection onto $\mathscr{N}$.

Definition 27.1 (Bilateral completed observer)**.** Define

$$\boxed{R_\Pi^\sharp h = (Rh, q_\Pi P_{\mathscr{N}} h) \in \mathscr{Y} \oplus \mathfrak{B}(R;\Pi).}$$

The first component observes the visible side. The second records precisely the target-relevant part of the blind side.

Theorem 27.2 (Universal target-faithful completion)**.** *The completed observer satisfies*

$$\boxed{\ker R_\Pi^\sharp = \ker R \cap \ker \Pi.}$$

Consequently,

$$\boxed{\mathfrak{B}(R_\Pi^\sharp;\Pi) = 0.}$$

Moreover, let $L : \mathscr{H} \to \mathscr{W}$ be bounded, assume $L|_{\mathscr{N}_0} = 0$, and suppose that the augmented observer

$$h \longmapsto (Rh, Lh)$$

is target faithful. Then there is a unique bounded injective map

$$\ell : \mathfrak{B}(R;\Pi) \longrightarrow \mathscr{W}$$

such that

$$\boxed{L|_{\mathscr{N}} = \ell q_\Pi.}$$

Thus $R_\Pi^\sharp$ is the universal minimal completion of the observer by its target-relevant blind quotient.

Proof. If $R_\Pi^\sharp h = 0$, then $Rh = 0$, so $h \in \mathscr{N}$ and $P_{\mathscr{N}} h = h$. The second component gives $q_\Pi h = 0$, hence $h \in \mathscr{N}_0$. The reverse inclusion is immediate. The target-blind quotient of the completed observer therefore vanishes.

Because $L|_{\mathscr{N}_0} = 0$, the quotient universal property gives a unique bounded map ℓ with $L|_{\mathscr{N}} = \ell q_\Pi$. If $\ell q_\Pi n = 0$, then $Rn = 0$ and $Ln = 0$. Target faithfulness of (R, L) gives $\Pi n = 0$, so $n \in \mathscr{N}_0$ and $q_\Pi n = 0$. Hence ℓ is injective. ☐

The theorem is the operator-level form of the observer lifts and completion-stable recovery developed in the companion spectral-blindness papers. The blind side is not denied; it is either shown to be target irrelevant or is represented by an additional channel.

27.2 Observer and observed on the native compatible range

The native energy defines the isometric analysis map

$$\mathcal{C}_\times : \mathscr{X}_{\times,3} \longrightarrow \mathcal{Y}_\times$$

constructed earlier. Its compatible channel range is

$$\mathcal{Y}_{\mathrm{comp}} = \overline{\operatorname{Ran} \mathcal{C}_\times},$$

and its compatibility projection is

$$P_{\mathrm{comp}} = \mathcal{C}_\times \mathcal{C}_\times^*.$$

Theorem 27.3 (Native observer–observed identity)**.** *One has*

$$\boxed{\mathcal{C}_\times^* \mathcal{C}_\times = I_{\mathscr{X}_{\times,3}}, \qquad \mathcal{C}_\times \mathcal{C}_\times^* = P_{\mathrm{comp}}.}$$

Therefore

$$\mathcal{C}_\times : \mathscr{X}_{\times,3} \longrightarrow \mathcal{Y}_{\mathrm{comp}}$$

is unitary, with inverse $\mathcal{C}_\times^*|_{\mathcal{Y}_{\mathrm{comp}}}$. For every native state F,

$$\boxed{\mathcal{C}_\times^*(\mathcal{C}_\times F) = F.}$$

For arbitrary channel data y, one instead has

$$\mathcal{C}_\times(\mathcal{C}_\times^* y) = P_{\mathrm{comp}} y.$$

Thus "observer equals observed" means exact analysis–synthesis identity on the compatible range. It does not mean that every channel vector is compatible, that every threshold projection is empty, or that self-adjointness forces a desired spectral sign.

Proof. The first identity is the defining isometry of the native energy channel. The second is the standard range projection of an isometry. The remaining statements follow immediately. □

Let $\mathsf{J}_\times$ be the inversion cut and let $\mathsf{J}_\mathcal{Y}$ act by inversion on the spatial channel and reflection $\xi \mapsto -\xi$ on the Mellin channel. Directly from the definitions,

$$\mathcal{C}_\times \mathsf{J}_\times = \mathsf{J}_\mathcal{Y} \mathcal{C}_\times.$$

Hence the observer–observed identification preserves the two cut sectors.

27.3 The uncut operator is the accepted four-block object

Let $J = J^* = J^{-1}$ be any represented cut on a Hilbert space and put

$$P_\pm = \frac{I \pm J}{2}.$$

For a bounded self-adjoint operator A, define its diagonal cut part and its cross seam by

$$A_{\mathrm{diag}} = P_+ A P_+ + P_- A P_-,$$

$$\Sigma_J(A) = P_+ A P_- + P_- A P_+.$$

Theorem 27.4 (Bilateral cut reconstruction)**.** *For every bounded self-adjoint A,*

$$\boxed{A = P_+AP_+ + P_+AP_- + P_-AP_+ + P_-AP_-.}$$

Equivalently,

$$\boxed{A = A_{\mathrm{diag}} + \Sigma_J(A).}$$

Moreover,

$$\boxed{\Sigma_J(A) = 0 \iff [A, J] = 0.}$$

Under unitary transport of A and J, the two diagonal blocks, the cross seam and every target-relevant blind quotient are transported unitarily. No change of chart can erase one side of the cut or repair a blind direction.

Proof. Insert $I = P_+ + P_-$ on both sides of A. The commutator identity is the usual block-diagonal criterion for a represented involution. Unitary covariance follows by conjugating every projection and block, while blindness invariance is Theorem 18.2. □

This theorem gives the precise meaning of accepting both sides of the cut. The uncut operator is not obtained by choosing the preferred sector. It is reconstructed from both diagonal sectors and their seam coupling. Relative to a finite compatible observation P_N, the same rule reads

$$A = P_NAP_N + P_NAQ_N + Q_NAP_N + Q_NAQ_N, \qquad Q_N = I - P_N.$$

The visible block, blind block and cross block are therefore all part of the claim. The blind-kernel Schur theorem and compatible response-tail theorem are the quantitative forms of this bilateral acceptance.

27.4 Recognition closure theorem

Theorem 27.5 (Recognition closure of the native framework)**.** *The native recognition-seam programme is representation-complete in the following sense.*

(i) The carrier, cut, Euler flow, energy and target are defined before any classical chart.
(ii) Native analysis and synthesis are inverse on the compatible channel range.
(iii) Both cut sectors and their cross seam reconstruct the uncut operator.
(iv) Every projected observer has a universal minimal target-faithful completion by its blind quotient.
(v) Every finite spectral claim is promoted only after its visible, blind and cross blocks are controlled on the same native reconstruction.
(vi) The terminal Riemann-hypothesis question is represented by an exact nonnegative integer seam obstruction and its equivalent spectral-flow, threshold-zeta and Morse presentations. The framework does not assume that this obstruction vanishes.

Thus the framework is closed as a faithful representation of the problem even while the terminal sign question remains open.

Proof. Items (i)–(ii) are the native carrier and Theorem 27.3. Item (iii) is Theorem 27.4. Item (iv) is Theorem 27.2. Item (v) follows from the spectral-blindness invariance, blind-kernel Schur and compatible response-tail theorems. Item (vi) is the integer seam-path, threshold spectral-zeta and Morse-zeta construction. None of these results includes the implication that the integer is zero. □

27.5 Bilateral outcome discipline

At every actual certificate, the framework accepts three mathematically distinct outcomes:

$$\begin{array}{lcl} \text{strictly subcritical enclosure} & \Longrightarrow & \text{the declared seam charge vanishes,} \\ \text{strictly supercritical enclosure} & \Longrightarrow & \text{a genuine crossing is detected,} \\ \text{enclosure meeting the threshold} & \Longrightarrow & \text{the computation is inconclusive.} \end{array}$$

No outcome is removed by language, and no philosophical preference is allowed to choose among them. A controlled vanishing sequence would imply the Riemann hypothesis by the proved spectral–zeta correspondence. Until such a sequence is established, the correct claim remains `ABSTAIN`.

Remark 27.6 (The boundary between philosophy and theorem). The phrase "observer equals observed" is retained as an interpretation of

$$\mathcal{C}_\times^* \mathcal{C}_\times = I$$

on compatible native data. It is not used as a substitute for

$$\mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta) = 0.$$

Likewise, accepting both sides of the cut means retaining both sectors and the cross seam, not declaring their obstruction absent. The philosophy selects the architecture; the theorem ledger decides the claim.

Framework closure status. The bilateral observer completion, native observer–observed identity, exact four-block cut reconstruction and representation-complete closure theorem are proved. The framework publication may therefore close without claiming that the terminal obstruction vanishes. The controlled seam-vanishing sequence and the Riemann hypothesis remain `OPEN/ABSTAIN`.

28 Seven realizations of the terminal seam obstruction

The preceding construction produces one terminal nonnegative integer,

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).}$$

The programme encountered several ways of reading or attacking this integer. They are collected here in one hierarchy so that an avatar of the obstruction is not mistaken for its vanishing and a proof engine is not advertised as a second terminal theorem merely because it has acquired a more dramatic name.

28.1 The seven realizations

No.	Realization	Exact role	Status
1	Native operator enclosure	Prove directly that $\sup \sigma(\mathbb{K}_\eta) \le 1$ from one native compression, its compatible blind-complement tail and all directed source/rounding errors.	Independent certificate route; theorem architecture proved, actual controlled sequence open.
2	Euler–virial positive commutator	A strict compressed commutator on the threshold or negative Morse sector contradicts the weighted virial identity unless that sector is empty.	Annihilation criterion proved; source positivity established at the declared fixed shift.
3	Morse–zeta covariant index	Positive congruence transfers the threshold count to the negative inertia of the shifted native sign operator and hence to the value at zero of its Morse zeta.	Congruence-safe index identity proved.
4	Threshold spectral zeta	The entire finite sum over strict threshold gaps satisfies $\zeta^{\mathrm{th}}_{\Sigma,\eta}(0) = \mathfrak{k}_\Sigma(\eta)$ and $\zeta^{\mathrm{th}}_{\Sigma,\eta}(-1) = \mathrm{Tr}(\mathbb{K}_\eta - I)_+$.	Cancellation-free avatar proved; vanishing not automatic.
5	Birman–Schwinger inertia	After pure-source positivity, all remaining adverse finite sources are compressed to $B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta$, with rank at most five and $\mathfrak{k}_\Sigma(\eta) = N_+(B_\eta - I)$ at the declared shift.	Exact preferred terminal reduction proved; actual outward matrix certificate open.
6	Determinant winding	If a licensed Schatten determinant $\Delta_{p,\eta}(z) = \det_p(I - z\mathbb{K}_\eta)$ exists, each eigenvalue $\lambda > 1$ gives a zero $z = \lambda^{-1}$ in $(0, 1)$; a contour enclosing precisely these positive-real zeros counts the seam integer.	Conditional avatar; Schatten and divisor licensing remain open.
7	Spectral no-blindness	Hardy-strip boundary uniqueness proves that a compatible native state cannot vanish on a positive-measure active band. On a finite obstruction space the weighted observability Gram is strictly positive.	Faithfulness theorem proved; it supports source positivity rather than independently forcing the final finite correction below threshold.

These seven rows are not seven proofs of the Riemann hypothesis. They are one direct enclosure route, two index/zeta coordinate systems, one commutator proof engine, one finite terminal reduction, one conditional determinant avatar and one faithfulness theorem.

28.2 The bilateral vessel

The common vessel is the bilateral recognition closure of Section 27. For every represented cut,

$$A = P_+AP_+ + P_+AP_- + P_-AP_+ + P_-AP_-,$$

and for every observer–target pair the target-relevant blind quotient is added as an explicit channel. Thus none of the seven realizations is permitted to choose a visible sector and silently discard its blind or cross component.

On the native compatible channel range,

$$\mathcal{C}_\times^*\mathcal{C}_\times = I, \qquad \mathcal{C}_\times\mathcal{C}_\times^* = P_{\mathrm{comp}}.$$

This is the precise operator identity behind the phrase "observer equals observed." It licenses faithful analysis and synthesis; it does not select a subcritical spectral outcome.

28.3 Preferred decision hierarchy

The final architecture uses the realizations in the following order.

$$\boxed{\begin{array}{c}\text{bilateral recognition and spectral-blindness closure}\\ \Downarrow \quad \text{faithful vessel}\\ \text{Hardy no-blindness and covariant Euler–virial source positivity}\\ \Downarrow \quad \text{positive infinite source}\\ \text{rank-} \le 5 \text{ Birman–Schwinger matrix } B_\eta\\ \Downarrow \quad \text{finite terminal decision}\\ \mathfrak{k}_\Sigma(\eta) = N_+(B_\eta - I).\end{array}}$$

The native operator-enclosure route remains independent: it estimates the top of the complete response directly and therefore supplies a valuable check on the finite-source reduction. The threshold zeta, Morse zeta and spectral flow record the same integer in cancellation-free coordinates. The determinant winding is reserved for a future stage in which the required operator ideal and analytic divisor theorem are actually licensed.

Theorem 28.1 (Representation-complete finite-obstruction reduction)**.** *On the declared native completed-Weil architecture, the following statements hold.*

(i) The bilateral observer completion is target faithful, native analysis and synthesis are inverse on the compatible range, and both cut sectors together with their cross seam reconstruct the uncut operator.

(ii) For every $\eta > 0$, the completed sign question is represented by the finite nonnegative integer $\mathfrak{k}_\Sigma(\eta)$ and its one-sided spectral-flow, threshold-zeta and Morse-zeta avatars.

(iii) At $\eta = 10^{-2}$, the complete Gamma–prime source is positive and the remaining adverse odd boundary/Feshbach source has rank at most five. Consequently there is a Hermitian matrix

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta, \qquad \operatorname{rank} B_\eta \le 5,$$

for which

$$\boxed{\mathfrak{k}_\Sigma(\eta) = N_+(B_\eta - I).}$$

(iv) Independently, every lawful native enclosure satisfying

$$\sup \sigma(\mathbb{K}_\eta) < 1$$

forces the same integer to vanish.

Thus the framework and blindness problem are solved at the representation level, while the actual terminal sign is reduced to an outward finite Birman–Schwinger decision, with native operator enclosure as an independent verification route.

Proof. Part (i) is the universal target-faithful completion, native observer–observed identity and bilateral cut reconstruction. Part (ii) is the compact-packet, seam-path, spectral-flow and spectral-zeta construction. The complete-prime active-band certificate and Hardy-strip no-blindness give the pure-source positivity consumed in part (iii), while the finite-rank Birman–Schwinger inertia theorem gives the displayed equality. Part (iv) is the threshold definition of $\mathfrak{k}_\Sigma(\eta)$. □

Corollary 28.2 (Conditional terminal implication)**.** *Suppose one controlled sequence $\eta_j \downarrow 0$ is constructed on the same native architecture and, for every j, either*

$$\lambda_{\max}(B_{\eta_j}) < 1$$

is certified together with the corresponding even-sector closure, or a complete native operator enclosure proves

$$\sup \sigma(\mathbb{K}_{\eta_j}) < 1.$$

Then

$$\mathfrak{k}_\Sigma(\eta_j) = 0 \qquad (j \geq 1),$$

the completed native Weil operator is nonnegative and the classical Weil criterion yields the Riemann hypothesis.

Proof. Either certificate empties the strict threshold sector at each shift. The controlled-sequence correspondence gives unshifted completed-Weil positivity, and the terminal implication is the classical Weil criterion. □

28.4 Outcome discipline

An actual outward terminal enclosure has three possible conclusions:

$$\begin{array}{lcl}
\lambda_{\max}(B_\eta) < 1 & \Longrightarrow & \mathfrak{k}_\Sigma(\eta) = 0, \\
\lambda_{\max}(B_\eta) > 1 & \Longrightarrow & \text{a genuine seam crossing is present}, \\
1 \in \text{the outward enclosure} & \Longrightarrow & \text{the certificate is inconclusive.}
\end{array}$$

The framework accepts all three. This bilateral outcome discipline is not agnosticism disguised as a theorem; it is the condition that prevents the architecture from baking its desired answer into the observer.

Claim status. The seven-realization classification, their hierarchy and the representation-complete finite-obstruction reduction are proved from the preceding results. The actual native source-resolvent columns, outward 5×5 certificate, independent complete operator enclosure along a common sequence and the controlled limit $\eta_j \downarrow 0$ remain open. The Riemann hypothesis remains `ABSTAIN`.

29 Relation to neighbouring RH operator programmes

Several operator, dynamical and functional-analytic programmes use language overlapping with this manuscript. Similar vocabulary does not identify the carriers, spectra or terminal theorems. The comparisons below state the exact distinction.

29.1 Hilbert–Pólya-type spectral realization

The Hilbert–Pólya aspiration seeks a self-adjoint operator whose spectrum recovers the imaginary parts of the nontrivial zeros; see [Edw74, Tit86]. The circle operator

$$D_a = -i\frac{d}{d\theta} + a$$

is only an index calibration: its spectrum is $\{n + a\}$, not the zeta-zero ordinates.

The bounded native Weil representative W_3 is also not proposed as a Hilbert–Pólya operator. Its role is to represent the completed Weil form relative to the X_3 metric. The paper consumes the classical equivalence

$$\mathrm{RH} \iff W_3 \geq 0,$$

not an equality between $\mathrm{Spec}(W_3)$ and the zeta-zero set.

29.2 Berry–Keating and the $H = xp$ model

Berry and Keating proposed the Hamiltonian $H = xp$ as a semiclassical model related to the average counting of Riemann zeros [BK97]. The present sheet transport has a different purpose: it retains recognition history and derives a seam involution. The area–rotation density

$$\omega_A = \frac{1}{A_q} r\,\mathrm{d}r \wedge \mathrm{d}\theta$$

is a conditional readout of radial–tangential curvature, not a phase-space regularization of xp, and no semiclassical zero-counting formula is inferred from it.

29.3 Connes' noncommutative trace-formula programme

Connes' programme places the explicit formula on the noncommutative space of adele classes and interprets zeros through an absorption spectrum and resonances [Con99]. The finite arithmetic current and the native X_3 Weil operator are not that noncommutative carrier. An identification would require an explicit comparison of spaces and a theorem equating the relevant trace distributions; none is claimed.

29.4 de Branges spaces

De Branges' approach uses Hilbert spaces of entire functions, Hermite–Biehler structure and canonical systems; his conjectural framework would imply RH for Dirichlet zeta functions [dB94]. Neither the Eye quotient nor X_3 is presented as a de Branges space. No Hermite–Biehler function, de Branges ordering theorem or canonical system is constructed.

29.5 Deninger's cohomological and foliated route

Deninger develops analogies with foliated dynamical systems, leafwise cohomology and Lefschetz-type trace formulas [Den98]. Seam memory is not automatically a cohomology class, and the labelled sheet carrier is not a foliated arithmetic space. A comparison would require an actual cochain complex, flow generator and trace formula.

29.6 The present native Weil-sign route

The present paper uses a different terminal mechanism. The explicit formula defines a bounded Hermitian form on the two-channel Hilbert envelope X_3, producing a bounded self-adjoint representative W_3. Spatial reflection derives the reducing sectors $X_3^{\pm}$, a positive odd finite block licenses a Feshbach complement, and the global sign reduces to

$$W_3 \geq 0 \iff W_3^+ \geq 0 \text{ and } F_3 \geq 0.$$

The bounded Surya chart packages the negative spectral sector of this reduced form operator. This is a Weil-positivity realization, not a zero-spectrum realization, adelic trace formula, canonical system or cohomological flow.

29.7 Comparison table

Programme	Central carrier or mechanism	Relation to the present paper
Hilbert–Pólya	self-adjoint zero-spectrum operator	D_a is only an index calibration; W_3 is a form representative
Berry–Keating	semiclassical xp Hamiltonian	distinct from recognition sheet transport and X_3 form positivity
Connes	adele-class space and trace formula	neither finite defect current nor W_3 is the adelic absorption carrier
de Branges	entire-function Hilbert spaces and canonical systems	no Hermite–Biehler or canonical-system realization is constructed
Deninger	foliated flow, cohomology and Lefschetz trace	seam memory is not a cohomology class without the missing dynamical structure
Present route	native completed Weil form on X_3	reduces RH to a global form-sign obstruction, not a zero-spectrum identity

29.8 Shared terminal boundary

Every programme must prove that its carrier actually decides the intended number-theoretic target. For zero-spectrum routes this means a complete zero correspondence. For the present Weil route it means global nonnegativity of W_3, equivalently vanishing of the reduced anti-seam operator. Formal resemblance, finite samples and symmetry alone decide neither target.

30 Position relative to classical theory and originality boundary

The manuscript intersects mature areas of topology, operator theory, harmonic analysis and analytic number theory. Its claim is not that the constituent tools are individually new. The contribution is their guarded assembly into a representation-complete completed-Weil architecture, the exact diagnosis and repair of target-relevant spectral blindness, the reduction to one nonnegative seam integer and, at the declared fixed shift, the further reduction of the residual obstruction to a Hermitian Birman–Schwinger matrix of rank at most five.

30.1 The Weil provenance split

The word "Weil" refers here to three logically different statements, which must not be collapsed.

Native construction proved in this programme. On the declared multiplicative core the paper constructs the Hermitian boundary, Gamma and prime sectors

$$\mathfrak{B}_\times, \qquad \mathfrak{A}_{\Gamma,\times}, \qquad \mathfrak{P}_\times,$$

and their signed overlay

$$\boxed{\mathfrak{W}_\times = \mathfrak{B}_\times + \mathfrak{A}_{\Gamma,\times} - \mathfrak{P}_\times.}$$

The paper proves bounded extension to the native energy completion, the bounded self-adjoint representative $W_\times$, unitary equivalence with W_3, parity and Feshbach reduction, compact relative packets, the integer seam path and the subsequent blindness, virial and finite-rank reductions.

Classical analytic identification consumed through an import membrane. The analytic continuation and functional equation of the completed zeta function, together with the classical explicit formula identifying the boundary–Gamma–prime pairing with the symmetric nontrivial-zero sum, are imported; see [Rie59, Tit86, Edw74]. The paper fixes conventions, normalization, domains and sector ledgers and then proves the native bounded-operator consequences. It does not claim that the bare cut axioms independently derive the completed zero sum.

Classical terminal equivalence. The implication

$$\mathfrak{W}_\times(F, F) \geq 0 \text{ on the admissible test class} \quad \Longleftrightarrow \quad \text{RH}$$

is the classical Weil positivity criterion. The present paper imports that equivalence but derives the carrier and exact threshold obstruction to which it applies.

Thus the accurate provenance statement is

$$\boxed{\begin{array}{c}\text{classical explicit-formula identification} \\ +\ \text{native pairing/operator and blindness architecture} \\ +\ \text{classical terminal Weil criterion.}\end{array}}$$

Calling the whole construction a classical import would erase the central operator work; calling the zero-sum identity or RH equivalence a new derivation would overstate it.

30.2 Classical mechanisms and their paper-specific roles

Mellin, Fourier and logarithmic representations. Mellin–Plancherel, Fourier analysis, Stone's theorem and compactness criteria are classical. The paper-specific direction is native first: the multiplicative carrier, inversion cut, Euler-scale flow and energy are declared before their logarithmic and harmonic coordinates.

Connections, curvature and indefinite adjoints. Connection curvature, parallel transport, holonomy and fundamental-symmetry adjoints are classical. Their role is typed carefully; no universal scalar curvature law, Hilbert self-adjointness or terminal sign is inferred from the vocabulary alone.

Spectral flow, Toeplitz index, trace ideals and spectral zetas. These mechanisms are classical; see [APS75, Phi96, BBW93, Sim05]. The circle family is an exact calibration. Trace-ideal theory proves a structural no-go theorem for the diagonal paired-prime carrier. The threshold and Morse spectral zetas are finite entire sums attached to the compact threshold sector, not a Hilbert–Pólya spectrum of zeta zeros.

Schur, Feshbach and Birman–Schwinger mechanisms. These are classical operator methods; see [Kat95]. Their paper-specific role is to act on the declared native completed-Weil blocks, preserve the target-relevant blind sector and reduce the remaining fixed-shift adverse source to the actual boundary/Feshbach source range of rank at most five.

Hardy uniqueness and positive commutators. Hardy-space boundary uniqueness and virial or positive-commutator principles are classical. The paper derives the relevant Hardy strip from the native exponential energy, identifies the two native boundary channels and combines this with a complete-prime covariant active-band certificate to prove source no-blindness and positivity.

30.3 Published blindness and recovery calibrations

The finite information-loss theory is developed in the companion papers [Dab26b, Dab26a]. They separate ordered history, endpoint and spectral shadow, prove target-relative blindness indices and construct stable observer lifts. The present paper does not republish those finite classifications. It extends their discipline to an infinite-dimensional completed-Weil carrier and proves a universal minimal blind-quotient completion for an arbitrary observer–target pair.

30.4 Paper-specific theorem package

The principal paper-specific contributions are:

N1. a cut-first recognition, sheet-transport and represented-involution architecture with exact target-factorization controls;

N2. the exact circle identity $\mathrm{sf} = \mathrm{Wind} = \mathrm{ind} = q$;

N3. complete carrier and determinant diagnostics proving the diagonal paired-prime route zero-blind;

N4. finite arithmetic recognition and two-channel defect ledgers;

N5. a native multiplicative carrier with inversion cut, Euler flow, energy completion and faithful logarithmic/Mellin chart;

N6. the native boundary–Gamma–prime completed-Weil overlay and bounded self-adjoint representative;

N7. exact parity/Feshbach reduction, compact shifted responses and the additive integer seam path;

N8. one-sided spectral-flow, threshold-zeta and Morse-zeta presentations of the same nonnegative integer;

N9. a covariant Euler–virial source analysis, complete-prime Arb active-band certificate and native Hardy no-blindness theorem;

N10. positivity of the pure source and an exact rank-at-most-five Birman–Schwinger inertia reduction at the declared fixed shift;
N11. a universal target-faithful observer completion by the target-relevant blind quotient;
N12. the native observer–observed analysis/synthesis identity on the compatible range;
N13. bilateral four-block reconstruction of the uncut operator from both cut sectors and their cross seam;
N14. a representation-complete closure theorem separating framework closure from terminal sign closure;
N15. a seven-realization hierarchy identifying Birman–Schwinger inertia as the preferred finite decision mechanism and native operator enclosure as an independent verification route;
N16. an explicit developmental and nonpromotion ledger distinguishing proved theorems, conditional theorems, calibrations and actual certificates.

30.5 Conservative originality and submission statement

The originality claim is the representation-complete synthesis, native completed-Weil realization, blindness completion, integer obstruction atlas and finite terminal reduction. It is not ownership of the classical tools and not a proof of the terminal sign.

The paper should therefore be evaluated as a framework-establishment, carrier-exclusion, native-derivation, target-faithfulness and exact-reduction theorem package. It proves

$$\boxed{\begin{array}{c}\text{representation-complete bilateral framework} \longrightarrow \text{native seam integer}\\ \longrightarrow \text{rank-} \leq 5 \text{ finite obstruction at the declared shift.}\end{array}}$$

It does not claim the actual outward finite certificate, a common controlled zero-charge sequence or the Riemann hypothesis. Those remain future terminal results rather than hidden hypotheses of the present publication.

31 Open terminal obligations after framework closure

The native multiplicative carrier, recognition-compatible cut, Euler flow, energy completion, completed-Weil representative, compact relative packets, integer seam path, spectral-zeta and Morse presentations, spectral-blindness closure, bilateral observer completion and four-block cut reconstruction are now constructed. The framework is therefore representation complete.

The remaining questions are not missing definitions or missing carrier architecture. They concern the value of the actual terminal obstruction and its behavior as the coercive shift tends to zero.

31.1 Framework closure is not terminal sign closure

The distinction is

$$\boxed{\text{faithful representation of the problem} \neq \text{proof that the represented obstruction vanishes.}}$$

The bilateral recognition theorem proves that visible data, target-relevant blind data, both cut sectors and their cross seam are all retained. It does not imply

$$\mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta) = 0.$$

Likewise,

$$\mathcal{C}_\times^* \mathcal{C}_\times = I$$

is the native observer–observed identity on compatible data; it is not a threshold sign theorem.

31.2 The actual fixed-shift finite obstruction

At

$$\eta = 10^{-2},$$

the complete-prime Arb active-band certificate and Hardy-strip no-blindness give strict positivity of the pure Gamma–prime shifted source operator $S_{\eta,-}$. The remaining odd defect is exact finite rank:

$$F^{\times}_{3,\eta} = S_{\eta,-} - V_\eta V_\eta^*, \qquad \operatorname{rank} V_\eta \le 5.$$

Define

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta.$$

Then

$$\boxed{N_-(F^{\times}_{3,\eta}) = N_+(B_\eta - I),}$$

and

$$\boxed{F^{\times}_{3,\eta} \ge 0 \iff B_\eta \le I.}$$

```
The quantitative source-floor gate is now closed:

F1. PROVED: S_(0.01,-) >= 0.008 I and ||S_(0.01,-)^(-1)|| <= 125.
F2. OPEN / FINITE DATA: materialize the actual odd boundary and Feshbach source columns V_(0.01).
F3. OPEN / FINITE DATA: certify outward full-source residuals for S_(0.01,-) X = V_(0.01).
F4. OPEN / FINITE DATA: enclose the Hermitian matrix B_tilde_(0.01).

The final theorem is already proved: if
lambda_max(B_tilde_(0.01)) + 125 ||V_(0.01)|| ||V_(0.01) - S_(0.01,-) X_tilde|| < 1,
then B_(0.01) < I, F_(3,0.01)^times > 0, and k_Sigma(0.01) = 0.

A subcritical, supercritical, or inconclusive interval remains admissible; only the actual matrix data are unresolved at the fixed shift.
```

31.3 Controlled shift descent

A single fixed-shift result does not determine the unshifted sign. A terminal theorem must construct one sequence

$$\eta_j \downarrow 0$$

on a common completed native architecture and certify

$$\mathfrak{k}_\Sigma(\eta_j) = 0 \qquad (j \ge 0).$$

The same sequence must be used in both cut sectors. The necessary stability obligations include:

D1. control of the collapsing coercive reference as $\eta_j \downarrow 0$;
D2. compatible reconstruction and blind-complement control at every level;
D3. stability of the threshold projection at the endpoint;
D4. passage from shifted nonnegativity to $W_\times \ge 0$.

31.4 Even and odd bilateral closure

The cut gives

$$W_\times \geq 0 \iff W_\times^+ \geq 0 \text{ and } F_3^\times \geq 0.$$

Neither side is allowed to stand in for the other. The pure source active-band symbol is common to both parity sectors at the declared fixed shift, while the finite augmentations differ. A terminal proof must retain those differences and certify both sector charges on the same descent sequence.

Because

$$\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_+(\eta) + \mathfrak{k}_3(\eta)$$

with nonnegative integer summands, cancellation between the two sides of the cut is impossible.

31.5 Completed no-leakage and the zeta implication

The threshold spectral–zeta correspondence proves that one controlled seam-vanishing sequence is equivalent, under the declared completed-Weil identification, to completed positivity and hence to the absence of nontrivial zeta zeros in $\Re(s) > 1/2$. The functional equation then gives RH.

This implication is proved. Its seam-vanishing premise is not.

A direct determinant-divisor route would require additional Schatten, holomorphic-pencil, nowhere-vanishing-factor and no-pollution theorems. Those stronger obligations are not consumed by the present framework closure.

31.6 Claim discipline

Every future terminal certificate must state:

- the exact native operator and target;
- the compatible reconstruction and target-relevant blind quotient;
- the actual source columns rather than calibration columns;
- outward rounding status for every scalar, matrix and residual;
- visible, blind and cross estimates on the same projection;
- the shift level and its relation to the controlled descent sequence.

The following substitutions remain forbidden:

$$\begin{aligned}
\text{self-adjointness} &\implies \text{no threshold crossing},\\
\text{observer–observed identity} &\implies \text{seam charge } 0,\\
\text{calibration pass} &\implies \text{actual operator pass},\\
\text{framework closure} &\implies \text{RH},\\
\text{philosophical acceptance} &\implies \text{spectral sign}.
\end{aligned}$$

31.7 Terminal status (revised)

```
cut-derived recognition-seam framework ........................................ proved
native carrier, Euler flow, completed-Weil and parity/Feshbach reduction ...... proved
compact packets, seam integer, threshold/Morse avatars ......................... proved
complete-prime active-band and Hardy no-blindness ............................... proved
shift-transferred source positivity at theta = 0.002 ........................... proved
source floor S_(0.01,-) >= 0.008 I ............................................. proved
inverse bound ||S_(0.01,-)^(-1)|| <= 125 ....................................... proved
Hermitian rank-at-most-five obstruction theorem ................................ proved
final outward residual acceptance theorem ...................................... proved
actual outward five-by-five value ................................................ open
controlled cofinal eta_j -> 0 sequence ........................................... open
Riemann hypothesis .............................................................. ABSTAIN
```

The remaining fixed-shift work is finite data, not a missing reduction. A fixed-shift pass alone does not remove the global eta -> 0 obligation.

32 Conclusion

The programme began by asking what a cut creates, what recognition preserves and how failure of closure can become an operator-readable obstruction. Its first exact achievements were deliberately modest: labelled sheet transport rather than irreversible gluing, a derived dagger split, an exact circle spectral-flow/index calibration and finite arithmetic defect ledgers. The diagonal paired-prime determinant route was then resolved negatively. Its failure was not a shortage of numerical resolution but a structural blindness: no lawful regularization could manufacture the missing central spectral event.

The later native route changed the terminal carrier. On the multiplicative scale line the inversion cut, Euler-scale flow and native energy produce the Mellin characters and their even/odd seam sectors before the logarithmic Fourier chart is introduced. The boundary, Gamma and prime channels form a bounded completed-Weil pairing with self-adjoint representative $W_\times$. Parity and Feshbach reduction, followed by coercive shifted references, give one global compact packet

$$\mathbb{K}_\eta = K_{+,\eta} \oplus K^\sharp_{3,\eta}$$

and one nonnegative integer

$$\boxed{\mathfrak{k}_\Sigma(\eta) = \operatorname{rank} \mathbf{1}_{(1,\infty)}(\mathbb{K}_\eta).}$$

The same integer is minus a one-sided spectral flow, the threshold count of the shifted sign operator and the value at zero of threshold and Morse spectral zetas. These identities prevent cancellation and make the terminal obstruction explicit. They do not presume its value.

The numerical convergence route, spectral-zeta route and Euler–virial route were not discarded when they exposed new gates. Their proper roles are now separated. Native operator enclosure is an independent complete-response certificate mechanism. Threshold and Morse zetas are cancellation-free index avatars. The Euler–virial method, covariant prime gauge, complete-prime Arb active-band theorem and native Hardy uniqueness prove positivity of the pure source at the declared shift. The remaining adverse odd boundary/Feshbach source has rank at most five and produces the exact Birman–Schwinger matrix

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta, \qquad \operatorname{rank} B_\eta \le 5,$$

with

$$\boxed{\mathfrak{k}_\Sigma(10^{-2}) = N_+(B_{10^{-2}} - I).}$$

Thus the preferred fixed-shift decision is finite for a structural reason: its dimension is the exact source rank, not a selected terminal mode cutoff. The actual outward matrix value remains to be certified. The native operator enclosure remains a separate verification path rather than being merged into the same evidence ledger.

The deepest closure obtained here is representation-theoretic. For any observer R and target Π, the completed observer

$$R^\sharp_\Pi h = \big(Rh, q_\Pi P_{\ker R} h\big)$$

has no target-relevant blind quotient and is universal among bounded faithful extensions. On the native compatible range,

$$\mathcal{C}_\times^* \mathcal{C}_\times = I, \qquad \mathcal{C}_\times \mathcal{C}_\times^* = P_{\mathrm{comp}}.$$

This is the precise content of observer and observed as two faithful presentations of the same compatible state. For every represented cut,

$$\boxed{A = P_+AP_+ + P_+AP_- + P_-AP_+ + P_-AP_-.}$$

The uncut operator therefore contains both cut sectors and their cross seam. Recognition does not select the preferred side and call it the whole.

The paper has consequently solved one problem and reduced another:

$$\boxed{\begin{aligned} \text{representation and target-blindness problem}: &\ \text{solved}, \\ \text{terminal completed-Weil sign problem}: &\ \text{reduced to } \mathfrak{k}_\Sigma, \\ \text{preferred fixed-shift decision}: &\ \text{rank-} \le 5 \text{ Birman–Schwinger}, \\ \text{independent decision route}: &\ \text{native operator enclosure.} \end{aligned}}$$

The seven realizations of the seam integer are therefore an obstruction atlas, not seven claims of completion.

An actual terminal certificate is allowed to be subcritical, supercritical or inconclusive. This bilateral outcome discipline is the mathematical form of surrendering attachment to the desired answer while retaining full attachment to the proof standard. A controlled sequence $\eta_j \downarrow 0$ on which the global charge vanishes would imply completed-Weil positivity and then the Riemann hypothesis by the classical Weil criterion. Such a sequence is not proved here.

The final status is

$$\boxed{\begin{aligned} \text{representation-complete bilateral framework}: &\ \text{proved}, \\ \text{native seam-integer reduction}: &\ \text{proved}, \\ \text{rank-at-most-five fixed-shift reduction}: &\ \text{proved}, \\ \text{actual controlled global seam vanishing}: &\ \text{open}, \\ \text{Riemann hypothesis}: &\ \texttt{ABSTAIN}. \end{aligned}}$$

Recognition does not erase the cut. It accepts both sides, remembers their seam, and refuses to hide what either side contains.

A Formal claim ledger

This appendix records the exact claim boundary of the unified manuscript. “Proved” always means proved under the hypotheses and on the carrier stated in the corresponding section. Classical and separately published inputs are cited rather than relabelled as new results. Framework closure means faithful representation closure; it does not mean that the terminal seam obstruction has been proved to vanish.

Statement	Status
Recognition quotient and canonical representative image from idempotent recognition	proved
Invariant recognition topology is a complete Boolean algebra of clopen sets	proved in declared set model
Seam as the equalizer of recognized cut and identity	defined
Canonical metric seam cost when a separating metric representation exists	proved
Recognition-compatible involutive cut descends to X/R	proved
Canonical self-adjoint involution on $\ell^2(X/R)$	proved
Typed sheet action, rotational seam and target-factorization theorem	proved
Graph/anti-graph eigenspaces of unitary sheet exchange	proved
Bounded Cayley commutator residue and operator-valued seam memory	proved/typed
Connection curvature and curvature eigensector evolution	classical/proved in scope
Finite dagger algebra and Hilbert/dagger distinction	proved
Self-adjoint circle family and $\mathrm{sf} = \mathrm{Wind} = \mathrm{ind} = q$	proved
Reflection-compatible leakage ideal ladder	proved
Damped paired-prime family entire and trace-class valued	proved
Exact damped determinant divisor outside the closed strip	proved
Undamped bounded/compact and Schatten classification	proved
Local $\mathcal{S}_m$ damping-removal and explicit $\det_m$	proved and zero-free
Archimedean completion factor nonzero in the open strip	proved
Diagonal paired-prime carrier zero-blind in the critical strip	proved no-go theorem
Möbius pressure extraction $\ell * \mu = \Lambda$	classical/proved
Finite radical pushforward, adjoint, kernel, norm and coisometry	proved
Exact finite von Mangoldt image	proved
Prime-shell unitary realization and full defect pair	proved finite
Anti-seam current zero does not imply exact transfer	proved
One signed scalar witness may miss or cancel a nonzero current	proved finite
Exact order/endpoint/spectral-shadow blindness theory	separately published input
Observer-marked finite recovery and minimum lift counts	separately published input
Stable decoders, response residuals and completion-stable recovery	separately published input
Analytic Schatten determinant zero equals noninvertibility	classical/proved in scope
Nonvanishing-factor divisor promotion	proved
Multiplicative Haar carrier $\mathcal{D}_\times$ and inversion cut	proved
Euler-scale unitary group and generator $\overline{-ir\,\mathrm{d}/\mathrm{d}r}$	proved
Cut reverses Euler-scale flow and generator	proved
Generalized Euler characters $r^{i\xi}$	proved as distributional spectral characters
Mellin diagonalization of the Euler generator and flow	proved
Cosine/sine as inversion-even/odd spectral channels	proved
Logarithmic chart sends Euler characters to Fourier characters	proved
Hyperbolic–circular two-sheet character	proved
Native energy completion $\mathscr{X}_{\times,3}$	proved
Direct compact embedding $\mathscr{X}_{\times,3} \hookrightarrow L^2(\mathrm{d}^\times r)$	proved
Unitary logarithmic chart $\mathscr{X}_{\times,3} \simeq X_3$	proved
Intertwining of inversion, Euler flow and translation	proved
Native boundary–Gamma–prime overlay $\mathfrak{W}_\times$	constructed and proved bounded
Native completed-Weil representative $W_\times = W_\times^*$	proved
Native sign equals operator sign	proved
Native prime seam operator: compact dilation series and two-sheet symbol	proved
Classical explicit-formula zero-sum identification	cited classical input
Classical Weil positivity criterion	cited classical input
Cut-to-X_3 reflection extension and $X_3 = X_3^+ \oplus X_3^-$	proved/reproduced
Compact embedding $X_3 \hookrightarrow L^2(\mathbb{R})$	proved/reproduced
Bounded completed critical Weil representative W_3	proved/reproduced
Exact even split and essential zero threshold	proved/reproduced
Shifted compact even relative operator $K_{+,\eta}$	proved/reproduced

Statement	Status
Certified positive finite odd block $A_3 > 0$	imported certified data, proof reproduced
Odd Feshbach equivalence $W_3^- \geq 0 \iff F_3 \geq 0$	proved/reproduced
Global reduction $W_3 \geq 0 \iff W_3^+ \geq 0$ and $F_3 \geq 0$	proved
Positive odd reference $R_{3,\eta}^{\times;\sharp}$ for every $\eta > 0$	proved
Compact self-adjoint odd response $\mathbf{\Lambda}_{3,\eta}^{(1)}$	proved
Odd sign equivalence $F_{3,\eta}^{\times} \geq 0 \iff \sup \sigma(\mathbf{\Lambda}_{3,\eta}^{(1)}) \leq 1$	proved
Monotonicity $\mathrm{d}F_{3,\eta}^{\times}/\mathrm{d}\eta \geq I$	proved
Odd zero-charge anchor for $\eta \geq 14$	proved
Reduced carrier and bounded threshold anti-seam equivalence	proved
Global compact relative packet $\mathbb{K}_\eta$ for every $\eta > 0$	proved
Integer seam charge $\mathfrak{k}_\Sigma(\eta)$	proved finite and additive
Self-adjoint integer seam path $\mathbb{D}_{\eta,t} = I - t\mathbb{K}_\eta$	proved
Seam charge equals negative one-sided regularized spectral flow	proved
Threshold spectral zeta and value-at-zero seam-index formula	proved
Morse spectral zeta and congruence-safe value-at-zero formula	proved
Threshold spectral–zeta correspondence with completed Weil positivity	proved under declared controlled-sequence hypotheses
Euler–virial weighted trace identity and annihilation criterion	proved
Covariant removal of the prime phase derivative	proved
Complete-prime Arb nonvanishing and active-band derivative-gap certificate at $\eta = 10^{-2}$	certified local source theorem
Native strip-Hardy embedding and positive-measure boundary uniqueness	proved
Compatible active-band no-blindness	proved
Virial positivity of the pure Gamma–prime source at the declared fixed shift	proved
Explicit source floor S_(0.01,-) >= 0.008 I	proved
Inverse norm ||S_(0.01,-)^(-1)|| <= 125	proved
Final outward Hermitian five-by-five acceptance theorem	proved
Actual boundary/Feshbach source-resolvent columns	open / finite data
Actual outward five-by-five matrix value	open / finite data
Native channel reconstruction and compatible response-tail gate	proved
Universal target-faithful completion by the blind quotient	proved
Native observer–observed identity $\mathcal{C}_\times^*\mathcal{C}_\times = I$, $\mathcal{C}_\times\mathcal{C}_\times^* = P_{\mathrm{comp}}$	proved
Bilateral four-block cut reconstruction	proved
Cross seam zero if and only if the operator commutes with the cut	proved
Representation-complete framework closure	proved
Native seam-interval standing-wave frame	proved
Exact standing-wave Fourier–Laplace formula	proved
Pinned finite-grid Schur value at $\eta = 10^{-2}$	directed floating-point audit; not interval-certified
State-locked standing-wave top and residual	computed numerical shadow; not continuum-certified
Controlled global seam vanishing along $\eta \downarrow 0$	open
Direct zero-bearing determinant carrier replacing the excluded diagonal model	open
Fixed-fibre theorem for a direct critical-line coordinate	open
Completed no-leakage/admissibility theorem	open
Riemann hypothesis	`ABSTAIN`

Explicit exclusions

The paper does not use any of the following implications:

$$
\begin{aligned}
\text{quotient topology} &\Longrightarrow \text{differential geometry},\\
\text{recognition path defect} &\Longrightarrow \text{connection curvature},\\
\text{sheet transport} &\Longrightarrow \text{recognition agreement},\\
\text{area–rotation density} &\Longrightarrow \text{universal curvature law},\\
\text{dagger self-adjointness} &\Longrightarrow \text{Hilbert self-adjointness},\\
\text{Hilbert self-adjointness} &\Longrightarrow \text{empty threshold sector},\\
\text{observer equals observed} &\Longrightarrow \mathfrak{k}_{\Sigma}(\eta) = 0,\\
\text{acceptance of both cut sectors} &\Longrightarrow \text{cross seam } 0,\\
\text{framework representation closure} &\Longrightarrow \text{proof of RH},\\
\text{philosophical dissolution} &\Longrightarrow \text{operator certificate},\\
\text{circle index calibration} &\Longrightarrow \text{arithmetic index},\\
\text{damped trace-class family} &\Longrightarrow \text{completed } \xi \text{ determinant},\\
\text{higher regularization} &\Longrightarrow \text{missing interior zeros},\\
\text{archimedean factor} &\Longrightarrow \text{repair of a zero-blind carrier}.
\end{aligned}
$$

$$
\begin{aligned}
J_{\mathrm{def}} = 0 &\Longrightarrow UA = AD,\\
\text{one signed witness } 0 &\Longrightarrow \text{ full current } 0,\\
\text{reflection covariance} &\Longrightarrow \text{ critical-line localization},\\
\text{parameter equivariance} &\Longrightarrow \text{ fixed state in every critical-line fibre},\\
\text{finite determinant symmetry} &\Longrightarrow \text{ determinant–}\xi\text{ identity},\\
\text{Euler–harmonic derivation} &\Longrightarrow \text{ completed zeta explicit formula},\\
\text{native Weil overlay} &\Longrightarrow \text{ independent reproof of the classical zero sum},\\
\text{integer seam path defined} &\Longrightarrow \text{ integer seam charge vanishes},\\
\text{threshold or Morse zeta defined} &\Longrightarrow \text{ that zeta vanishes},\\
\text{Euler–harmonic derivation} &\Longrightarrow \text{ terminal no-crossing},\\
\text{positive finite odd block} &\Longrightarrow F_3 \geq 0,\\
\text{positive even finite section} &\Longrightarrow W_3^+ \geq 0,\\
\text{unitary relabelling} &\Longrightarrow \text{ repair of a blind kernel},\\
\text{one visible cut block} &\Longrightarrow \text{ the uncut operator},\\
\text{calibration matrix passes} &\Longrightarrow \text{ actual native certificate passes},\\
\text{floating-point finite-grid audit} &\Longrightarrow \text{ interval-certified finite sign},\\
\text{finite-grid no-crossing} &\Longrightarrow \text{ continuum no-crossing},\\
\text{directional Ritz residual} &\Longrightarrow \text{ global compatible tail},\\
\text{standing-wave shadow top} < 1 &\Longrightarrow \text{ native continuum sign},\\
\mathfrak{A}_{\mathrm{SS}} \text{ defined} &\Longrightarrow \mathfrak{A}_{\mathrm{SS}} = 0,\\
\text{principal holonomy } 1 &\Longrightarrow \text{ lifted seam charge } 0,\\
\mathfrak{k}_3(\eta) = 0 \text{ for one shift} &\Longrightarrow F_3 \geq 0,\\
\mathfrak{k}_+(\eta) = 0 &\Longrightarrow \mathfrak{k}_3(\eta) = 0,\\
\text{reformulation} &\Longrightarrow \text{ proof of RH.}
\end{aligned}
$$

B Developmental and realization ledger

This appendix records how the historical route is consumed by the final submission architecture. A historical stage is not promoted merely because a later theorem explains why it was useful.

Developmental stage	Durable mathematical contribution	Publication status
Original cut/index route	Recognition topology, labelled sheet transport, derived dagger, exact circle index calibration, finite arithmetic defects and the eventual integer seam obstruction.	Proved in declared settings; seam vanishing not inferred.
Direct reflected determinant route	Complete damped and undamped operator-ideal diagnostics and a structural zero-blindness theorem.	Proved no-go result; excluded as a direct RH carrier.

Developmental stage	Durable mathematical contribution	Publication status
Native numerical convergence route	Native frames, visible-transfer theorems, blind-kernel Schur bounds, compatible response tails, source closures and independent no-crossing certificate architecture.	Theorems proved; older finite values remain audits unless outwardly promoted.
Spectral-zeta route	Threshold and Morse zeta value-at-zero formulas, seam-index trinity and controlled-sequence correspondence.	Proved avatars and conditional terminal correspondence; vanishing open.
Euler–virial route	Weighted virial identity, covariant phase removal, complete-prime active-band source certificate and Hardy no-blindness.	Source positivity proved at the declared shift.
Bilateral closure	Universal blind-quotient observer completion, exact compatible analysis/synthesis and four-block uncut reconstruction.	Representation-complete framework theorem proved.
Finite terminal reduction	Rank-at-most-five Birman–Schwinger inertia formula for the remaining adverse source.	Exact reduction proved; actual outward matrix certificate open.

Realization hierarchy

Object	Role in the final paper	Status
Bilateral recognition / four-block reconstruction	Common representation-complete vessel accepting visible, blind and cross sectors.	Principal framework solution; proved.
Birman–Schwinger inertia	Preferred finite decision mechanism for the fixed-shift residual obstruction.	Exact theorem; actual certificate open.
Native operator enclosure	Independent complete-response verification route.	Method theorem proved; controlled actual sequence open.
Euler–virial positive commutator	Source-positivity proof engine.	Criterion and fixed-shift source theorem proved.
Hardy no-blindness	Faithfulness/admissibility theorem preventing hidden active-band states.	Proved.
Threshold and Morse spectral zetas	Cancellation-free coordinate systems for the seam integer.	Proved; no automatic vanishing.
Determinant winding	Conditional future avatar after Schatten and divisor licensing.	Open/conditional.

Submission boundary

The present paper claims

$$\boxed{\begin{array}{c}\text{representation-complete framework} + \text{ exact seam-integer reduction}\\ + \text{ rank-} \leq 5 \text{ fixed-shift obstruction reduction.}\end{array}}$$

It does not claim

$$\boxed{\mathfrak{k}_\Sigma(\eta_j) = 0 \text{ for one controlled sequence } \eta_j \downarrow 0,}$$

and therefore does not claim the Riemann hypothesis.

C Repository and publication source lineage

The manuscript reconstructs one mathematical journey from the historical MP RH branch stack and the earlier EMK–UGD development. Later operator results are no longer used as a black-box epilogue: their definitions and proofs are reproduced in the main text, while state-locked numerical packets remain traceable to their repository certificates.

EMK–UGD lineage

The earlier geometry supplies the mechanism preceding the operator stack:

V1. labelled sheet rotation rather than automatic quotient gluing;

V2. seam/residue coordinates and stable Cayley-step comparisons;

V3. a master connection and radial–tangential curvature sector;

V4. a bounded correction amplitude distinguished from its lifted memory;

V5. cocycle and recognition-closure tests.

Only statements retyped and proved under explicit hypotheses are consumed. Stronger universal-curvature, mock-modular, prime-torus and zero-closure claims remain outside the theorem package.

Published spectral-blindness companions

Two published papers supply completed calibrations for the information-loss and recovery language used here [Dab26b, Dab26a].

P1. *When Spectra Forget Order* proves exact determinant blindness for ordered matrix products, identifies finite order-sensitive channels, and constructs observer-marked stable endpoint recovery. Its word–endpoint–spectral-shadow hierarchy is the finite no-go model behind the present zero-blind carrier diagnosis.

P2. *Stable Recovery Beyond Spectral Blindness* develops target-relative blind kernels, minimal lifts, stable decoders, response-residual certificates, matched finite/complement bounds and completion-stable reconstruction. Its finite operator-cut and parity–Feshbach ledgers motivate the present native blind-kernel transfer, but the infinite-dimensional completed-Weil specialization is proved in this manuscript.

The current paper cites these results rather than reproducing their degree-five and degree-six combinatorial classifications.

Initial MP RH lineage

L1. Daggered phase-seam and circle index foundation: PR #50.
L2. Seam correspondence and leakage projections: PRs #62–#63.
L3. Finite reflected analytic candidates and carrier obstructions: PRs #64–#68.
L4. Lambda deformation and index bridge: PR #69.
L5. Kernel and relative-determinant candidates: PRs #70–#71.
L6. Arithmetic transfer and anti-seam current: PRs #72–#76.
L7. Primitive-topology correction and programme map: PR #77.
L8. Finite arithmetic representation and radical transfer: PRs #78–#82.
L9. Zero-identification and conditional promotion ledgers: PRs #89–#91.
L10. Recognition-seam topology as the mature front door: PR #94.

Completed-Weil and finite-block lineage

X1. Completed critical form and bounded two-channel envelope: PR #127.
X2. Boundary parity structure and Riesz bounds: PR #139.
X3. State-locked finite odd block and exact positive M_3 interval: PR #200.
X4. Parity and exact odd Feshbach reduction: PR #202.
X5. Typed proof graph and two terminal sector blockers: PR #205.
X6. Compact embedding, Gamma/prime defects and shifted even route: PRs #207–#208.
X7. Quantitative even packets and projection-consistent response theory: PRs #209–#212.

Compact response, native source and integer-path lineage

The present extension reproduces the mathematical content of the following packages.

O1. **Odd positive reference and compact response:** PRs #215–#217 construct the self-adjoint complement reference, compact relative operator and quantitative Gamma/prime tail architecture.

O2. **Large-shift anchor:** PR #218 proves strict odd no-crossing for $\eta \geq 14$.

O3. **State-locked runtime and response packets:** PRs #219–#220 restore the pinned semantic state and compute the first parity-correct response-Krylov packet.

O4. **Finite-grid Schur audit:** PR #221 records the directed floating-point value

$$0.9529631022229692$$

at $\eta = 10^{-2}$. Its full outward interval matrix enclosure remains open, so the packet is not consumed as a finite theorem.

O5. **Native source correction:** PRs #225–#227 derive the multiplicative Euler-scale carrier, prove unitary blindness invariance, and construct native blind-kernel and channel-tail gates.

O6. **Native seam-distance and visible-transfer pilots:** PRs #228–#229 identify the operator meaning of the finite generalized Feshbach matrix.

O7. **Box-tail correction:** PRs #230–#232 exclude an unresolved unwindowed Laguerre promotion and develop lawful windowed alternatives.

O8. **Standing-wave replacement:** PRs #233–#234 construct the exact native standing-wave frame and compute the current state-locked shadow top and residual.

O9. **Euler–harmonic concentration:** PR #235 packages the native flow, Mellin characters, sine/cosine cut parity, logarithmic Fourier chart and hyperbolic–circular sheet character into one theorem.

O10. Native Weil and integer-path concentration: the same publication revision separates the native boundary–Gamma–prime overlay from the classical zero-sum shadow and proves the compact integer seam path and its one-sided spectral-flow charge as the terminal operator architecture.

The paper reproduces the functional-analytic steps it consumes. It does not reproduce every backend table, and it does not promote a floating-point finite-grid or standing-wave shadow calculation to an interval-certified finite sign or a continuum sign theorem.

Historical versus logical order

Historically the work moved from sheet-rotation intuition through daggered topology, index experiments, diagonal determinant obstructions, completed-Weil finite packets and finally the native multiplicative response carrier. The manuscript's logical order is

$$\begin{aligned}&\text{cut} \to \text{ recognition} \to \text{ sheet transport} \to \text{ index calibration} \to \text{ carrier exclusion}\\ &\to \text{ multiplicative EMK carrier} \to \text{ Euler–harmonic derivation} \to \text{ native Weil overlay}\\ &\to \text{ bounded self-adjoint Weil operator} \to \text{ parity/Feshbach lift} \to \text{ compact relative packet}\\ &\to \text{ integer seam path} \to \text{ blind-kernel transfer} \to \text{ completed charge-vanishing problem.}\end{aligned}$$

The open content is the outward finite-matrix enclosure, continuum charge-vanishing sequence and independent even sign, not the existence of the native carrier, harmonic representation, Weil operator, compact response or integer path.

D Theorem dependency and nonpromotion graph

The unified manuscript has one foundational chain, one excluded direct determinant branch, and one native completed-Weil branch whose terminal sign remains open.

Foundational framework chain

$$\begin{aligned}\text{cut and recognition} &\longrightarrow X/R \longrightarrow \overline{C} \longrightarrow J_C \longrightarrow P_\pm\\ &\longrightarrow \text{typed sheet transport and target faithfulness}\\ &\longrightarrow \text{connection curvature and seam-memory readout}\\ &\longrightarrow \text{finite dagger and leakage calculus}\\ &\longrightarrow \text{circle spectral-flow/index calibration.}\end{aligned}$$

The closed foundational implications include

$$\begin{gathered}C \text{ recognition compatible and } C^2 \sim_R I \Longrightarrow \overline{C}^2 = I \Longrightarrow J_C^* = J_C,\ J_C^2 = I,\\ \Pi \text{ constant on quotient fibres} \Longleftrightarrow \Pi \text{ factors through the quotient},\\ U \text{ unitary} \Longrightarrow \operatorname{Ran} P_+ = \operatorname{graph}(U), \quad \operatorname{Ran} P_- = \operatorname{graph}(-U),\\ \operatorname{sf}(D_{\delta+tq};0) = \operatorname{Wind}(z^q) = \operatorname{ind}(\Pi_H M_{z^{-q}} \Pi_H) = q.\end{gathered}$$

Reflected-carrier diagnosis

The model ladder gives

$$A_a \in \mathcal{S}_q \Longleftrightarrow a \in \ell^q$$

for pure leakage profiles, and

$$JK_\lambda(s)J = K_\lambda(1-s), \quad K_\lambda(s)^* = K_\lambda(\overline{s}), \quad JK_\lambda(s)^*J = K_\lambda(1-\overline{s}).$$

For $\lambda > 0$,

$$K_\lambda(s) \in \mathcal{S}_1$$

locally uniformly and its determinant zeros lie only at

$$-\lambda \log p - \frac{2\pi i n}{\log p} \quad \text{and} \quad 1 + \lambda \log p + \frac{2\pi i n}{\log p}.$$

Hence the damped determinant is zero-free on the closed critical strip.

At $\lambda = 0$,

$$K_0(s) \in \mathcal{S}_q \iff \frac{1}{q} < \Re(s) < 1 - \frac{1}{q},$$

and, for $m \geq 3$,

$$\det_m(I - K_0(s)) = \exp\left[-\sum_{n=m}^{\infty} \frac{P_{\mathcal{P}}(ns) + P_{\mathcal{P}}(n(1-s))}{n}\right] \neq 0.$$

Thus

$$\boxed{\text{diagonal paired-prime carrier} \Longrightarrow Z_{\text{spectral}} = Z_{\det} = \varnothing \text{ in the critical strip.}}$$

This branch is excluded as a direct determinant realization of ξ with a nowhere-zero multiplier.

Finite arithmetic and defect chain

$$\ell * \mu = \Lambda \longrightarrow \operatorname{supp} \Lambda = \{p^k\} \longrightarrow \operatorname{rad}(p^k) = p \longrightarrow \mathcal{J}_N^{\text{ar}}(\Lambda) = 0.$$

More precisely,

$$T_N \Lambda = \sum_{p \leq N} \nu_p(N) \log(p) e_p.$$

The normalized radical map is a coisometry and its adjoint realizes the recognized carrier as fibre-constant functions.

For prime-shell transfer,

$$\Delta = UA - AD = S_{\text{def}} + J_{\text{def}},$$

with

$$\Delta^* \Delta = S_{\text{def}}^* S_{\text{def}} + J_{\text{def}}^* J_{\text{def}}.$$

Therefore

$$\boxed{\Delta = 0 \iff S_{\text{def}} = 0 \text{ and } J_{\text{def}} = 0.}$$

Neither $J_{\text{def}} = 0$ nor one scalar witness equal to zero is a complete transfer theorem.

Analytic coordinate and zero boundary

For a state involution J_X and coordinate map π,

$$\pi(J_X x) = 1 - \overline{\pi(x)} \Longrightarrow \pi(\operatorname{Fix} J_X) \subseteq \{s : \Re(s) = 1/2\}.$$

Equality requires a fixed-fibre theorem. It is not a consequence of equivariance alone.

For a holomorphic $\mathcal{S}_m$ family,

$$\det_m(I - K(s)) = 0 \iff 1 \in \operatorname{Spec} K(s),$$

so $Z_{\text{spectral}} = Z_{\det}$ for that declared event. The arrow to Z_ξ requires a nowhere-zero divisor identity and artifact control.

Native cut-to-Weil realization

The analytic core and reflection cut satisfy

$$\mathcal{D} = C_c^\infty(\mathbb{R}), \qquad (\kappa_0 f)(x) = f(-x),$$

$$\|\kappa_0 f\|_{X_3} = \|f\|_{X_3}.$$

Therefore

$$\kappa_0 \longrightarrow J_3 = J_3^* = J_3^{-1} \longrightarrow X_3 = X_3^+ \oplus X_3^-.$$

The completed Weil form extends boundedly and gives

$$\mathfrak{W}_\xi(f,h) = \langle f, W_3 h\rangle_{X_3}, \qquad [W_3, J_3] = 0.$$

The compact embedding $X_3 \hookrightarrow L^2(\mathbb{R})$ supplies the compactness mechanism for the native Gamma and prime defects.

The even branch gives

$$W_3^+ = A_0 - C_+, \qquad 0 \in \sigma_{\mathrm{ess}}(W_3^+),$$

so for $\eta > 0$,

$$W_3^+ + \eta I = A_{+,\eta}^{1/2}(I - K_{+,\eta})A_{+,\eta}^{1/2}.$$

The odd branch gives

$$A_3 > 0, \qquad F_3 = D_3 - C_3 A_3^{-1} C_3^*,$$

$$W_3^- \geq 0 \iff F_3 \geq 0.$$

Consequently,

$$\boxed{W_3 \geq 0 \iff W_3^+ \geq 0 \text{ and } F_3 \geq 0.}$$

Unified terminal obstruction

Define

$$\mathbb{W}_0 = W_3^+ \oplus F_3$$

and

$$\mathfrak{A}_{\mathrm{SS}} = \left[-\mathbb{W}_0(I + \mathbb{W}_0^2)^{-1/2}\right]_+.$$

Then

$$\boxed{\mathfrak{A}_{\mathrm{SS}} = 0 \iff \mathbb{W}_0 \geq 0 \iff W_3 \geq 0.}$$

Using the classical Weil criterion,

$$\mathrm{RH} \iff \mathfrak{A}_{\mathrm{SS}} = 0.$$

This is an exact reformulation, not a proof of vanishing.

The conditional compact-charge branch requires:

F1. a positive odd reference producing compact self-adjoint $K_{3,\eta}$;

F2. the global relative operator

$$\mathbb{K}_\eta = K_{+,\eta} \oplus K_{3,\eta};$$

F3. certified even and odd no-crossing bounds;

F4. zero charge along a sequence $\eta \downarrow 0$;

F5. completion stability;

F6. a no-leakage or admissibility theorem forcing the lifted charge to vanish.

Under the compact packet,

$$\mathfrak{k}_\Sigma(\eta) = \mathfrak{k}_+(\eta) + \mathfrak{k}_3(\eta)$$

and

$$\mathrm{sf}(I - t\mathbb{K}_\eta; 0) = -\mathfrak{k}_\Sigma(\eta)$$

away from endpoint spectrum, with one-sided regularization in general.

Nonpromotion table

Established object	Does not imply
recognition quotient	differential manifold or connection
quotient involution	critical-line representation
typed sheet action	recognition agreement
curvature eigensector law	actual zeta-zero obstruction
circle index identity	arithmetic or RH index
damped trace-class family	completed zeta determinant
higher regularized endpoint	nontrivial determinant zeros
radical zero current	absence of signed fibre cancellation
anti-seam current zero	exact route covariance
one signed witness zero	full current zero
parameter equivariance	fixed state in every critical-line fibre
finite positive odd block	positive odd complement
finite positive even section	global even sign
Surya anti-seam operator defined	Surya anti-seam operator vanishes
principal holonomy one	lifted seam charge zero
one shifted zero charge	unshifted positivity

The terminal decision remains

$$\boxed{\text{Riemann hypothesis: } \texttt{ABSTAIN}.}$$

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Revision Addendum: Quantitative Source Floor and Final Hermitian Five-by-Five Theorem

This addendum is part of the revised edition dated 29 July 2026. It supersedes earlier statements in the manuscript that the outward source floor at $\eta = 10^{-2}$ remained open. The Hermitian rank-at-most-five Birman–Schwinger reduction was already established; the new contribution is the explicit source floor and the resulting fully quantitative residual acceptance theorem.

A. Certified parent margins

Let

$$\eta = 0.01, \qquad \theta = 0.002.$$

The complete-prime Arb certificate at $\eta = 0.01$ supplies the directed margins

$$\Delta_{.01} = 0.10937358388065747,$$

for the covariant derivative gap on $0 < \xi < 7$,

$$L_{.01} = 0.012677731815045846,$$

for the lower source branch on $7 \leq |\xi| \leq 12$, and

$$F_{.01} = 0.09321765364422295$$

on the far ray $|\xi| \geq 12$. The complete prime mass satisfies

$$|q(\xi)| \leq D_0, \qquad D_0 = 0.5699896237284023.$$

Prime nonvanishing on $[-7, 7]$ is independent of the coercive shift.

Lemma 1 (Global Gamma derivative bound)**.** *For*

$$g(\xi) = \operatorname{Re}\psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log\pi,$$

one has

$$\boxed{0 \leq g'(\xi) < 9 \qquad (\xi \geq 0).}$$

Proof. Put $a_k = k + \frac{1}{4}$. The positive real-digamma series gives

$$g'(\xi) = \sum_{k \geq 0} \frac{a_k \xi}{2(a_k^2 + \xi^2/4)^2}.$$

With $y = \xi/(2a_k)$, the k-th summand is

$$\frac{y}{a_k^2(1 + y^2)^2} \leq \frac{1}{2a_k^2}.$$

Hence

$$g'(\xi) \leq \frac{1}{2}\psi_1\left(\frac{1}{4}\right) = \frac{1}{2}(\pi^2 + 8G_{\text{Cat}}) < 9,$$

where $G_{\text{Cat}} < 1$ is Catalan's constant. Positivity is termwise. $\square$

B. Shift transfer

For a shift α, write

$$\omega_\alpha(\xi) = g(\xi) + \alpha(1 + |g(\xi)|).$$

Where $g < 0$, decreasing α increases ω'_α. Where $g \geq 0$, decreasing the shift from $\eta = .01$ to $\theta = .002$ lowers the derivative by at most $(\eta - \theta)g'$. Therefore

$$\omega'_\theta - |(|q|)'| \geq \Delta_{.01} - (.01 - .002) \sup g' > 0.10937358388065747 - 0.008 \cdot 9 = 0.03737358388065747 > 0.$$

Thus the covariant derivative gap remains strictly positive on the active band.

On $[7, \infty)$, the recorded value $g(7) > 0$ and monotonicity give $g \geq 0$. Put

$$\ell_\alpha(\xi) = (1 + \alpha)g(\xi) + \alpha - |q(\xi)|.$$

Eliminating g between the two shifts yields

$$\ell_\theta = \frac{1 + \theta}{1 + \eta}\ell_\eta - \frac{\eta - \theta}{1 + \eta}(1 + |q|).$$

Consequently, on $7 \leq |\xi| \leq 12$,

$$\ell_\theta \geq \frac{(1 + .002)L_{.01} - (.01 - .002)(1 + D_0)}{1 + .01} = 0.00014175276123635545 > 0,$$

and on the far ray

$$\ell_\theta \geq \frac{(1 + .002)F_{.01} - (.01 - .002)(1 + D_0)}{1 + .01} = 0.08004373461552888 > 0.$$

Reality symmetry supplies the negative half-line.

Theorem 2 (Shift-transferred pure-source positivity)**.** *The complete odd Gamma–prime source at* $\theta = .002$ *satisfies*

$$\boxed{S_{0.002,-} \geq 0.}$$

Proof. The complete prime symbol remains nonzero on the active band; the covariant derivative gap remains strict there; the lower source branch remains positive outside; and native Hardy no-blindness excludes a compatible negative state invisible to the positive commutator. The same virial contradiction used at $\eta = .01$ therefore proves nonnegativity at $\theta = .002$. □

Corollary 3 (Explicit source floor)**.** *The pure source at the declared shift satisfies*

$$\boxed{S_{0.01,-} = S_{0.002,-} + 0.008I \geq 0.008I.}$$

In particular,

$$\boxed{\|S_{0.01,-}^{-1}\| \leq 125.}$$

Proof. The shifted pure source depends affinely on the coercive shift:

$$S_\eta = S_\theta + (\eta - \theta)I.$$

Apply the preceding theorem with $\eta = .01$ and $\theta = .002$. □

C. The already established Hermitian obstruction

At $\eta = .01$, the odd Feshbach complement has the exact form

$$F^{\times}_{3,\eta} = S_{\eta,-} - V_\eta V_\eta^*, \qquad \operatorname{rank} V_\eta \le 5.$$

Define

$$B_\eta = V_\eta^* S_{\eta,-}^{-1} V_\eta \in M_r(\mathbb{C}), \qquad r \le 5.$$

Then $B_\eta = B_\eta^*$ and the established inertia identity is

$$N_-(F^{\times}_{3,\eta}) = N_+(B_\eta - I), \qquad F^{\times}_{3,\eta} \ge 0 \Longleftrightarrow B_\eta \le I.$$

Thus the fixed-shift seam charge is

$$\mathfrak{k}_\Sigma(.01) = N_+(B_{.01} - I).$$

D. Final outward acceptance theorem

Let $\widetilde{X}$ approximate the five source-resolvent columns solving

$$S_{.01,-} X = V_{.01}.$$

Define

$$R = V_{.01} - S_{.01,-}\widetilde{X}, \qquad \widetilde{B}_{.01} = \frac{1}{2}\left(V_{.01}^*\widetilde{X} + \widetilde{X}^* V_{.01}\right).$$

Proposition 4 (Residual enclosure)**.** *One has*

$$\boxed{\|B_{.01} - \widetilde{B}_{.01}\| \le 125\, \|V_{.01}\|\, \|R\|.}$$

Proof. Since $X - \widetilde{X} = S_{.01,-}^{-1} R$,

$$\|V_{.01}^*(X - \widetilde{X})\| \le \|V_{.01}\|\, \|S_{.01,-}^{-1}\|\, \|R\| \le 125 \|V_{.01}\| \|R\|.$$

Symmetrization does not increase the bound. □

Theorem 5 (Final outward Hermitian five-by-five criterion)**.** *If outward arithmetic certifies*

$$\boxed{\lambda_{\max}(\widetilde{B}_{.01}) + 125\|V_{.01}\|\|R\| < 1,}$$

then

$$\boxed{B_{.01} < I, \qquad F^{\times}_{3,.01} > 0, \qquad \mathfrak{k}_\Sigma(.01) = 0.}$$

Proof. The residual enclosure and Weyl's inequality give $\lambda_{\max}(B_{.01}) < 1$. The exact Birman–Schwinger inertia identity then gives strict positivity of the odd shifted sign operator and zero odd threshold charge. Together with the already established favorable even fixed-shift source, the declared fixed-shift seam charge vanishes. □

E. Corrected claim boundary

Statement	Status
Pure-source positivity at $\theta = .002$	proved
Source floor $S_{.01,-} \geq .008I$	proved
Inverse norm $\|S_{.01,-}^{-1}\| \leq 125$	proved
Hermitian rank-at-most-five reduction	proved upstream
Final outward acceptance theorem	proved
Actual outward five-by-five value	open / finite data
Controlled cofinal $\eta_j \downarrow 0$ closure	open
Riemann hypothesis	`ABSTAIN`

A fixed-shift success alone does not imply unshifted positivity. The remaining global obligation is a controlled cofinal family of certificates or a direct uniform theorem at $\eta = 0$.